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Research study into the potential economic, societal and environmental impacts of mineral exploration and mining in Northern Ireland

Appendix A - Minerals Policy

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Appendix A

Work Package 4 - Minerals Governance and Sustainable Development Policy

This work package is condensed and now included in the final version of the main report as an adjunct to the first work package. The Appendix has been supplied in its original form due to the volume of relevant information contained within, and expands upon the main report.

Contents

4.1. Introduction

4.1.1. The Model Modern Mine

Globally, the vision for the modern model mine has been shifting in parallel with the evolving notion of sustainable development and changing expectations of mining sector stakeholders. In the 1990s, the vision focused on cleaner production and reducing negative impacts. In the new millennium, managing negative social impacts and enhancing the positive came into sharper focus. Now miners must concentrate on the credentials of their mineral products and the place of these products in supply chains geared to achieving the Sustainable Development Goals (SDGs) and Paris Agreement goals.

In this context, the model modern mine is one that is viable and adds value, producing valued mineral products and operating such that its host government and host community gain lasting value from the minerals extracted. The model mine will be closed in accordance with an agreed vision and so that the positive effects are self-perpetuating. Negative impacts that could not be eliminated during design, will be mitigated effectively in the operational phase. The mine will have the smallest possible footprint, both in terms of physical footprint and impacts on natural resources. When closed, the mine will not continue to be a source of impacts and the remaining landforms will be stable and blend into the landscape.

For existing mines, some new ideals are not achievable, especially where the mine already has a large footprint. Any mine is unlikely to come close to new ideals without an excellent understanding of the interests of all its stakeholders, assimilation of and alignment with new standards that capture new expectations and understanding of the mine's context.

Every mine is unique, in its design and its setting. The model modern mine will understand its setting and its impacts will be well defined. Effective stakeholder engagement and impact and risk assessments will inform strategy and planning. The mine will have been planned with closure vision from the start. Negative impacts on humans and natural resources will have been avoided as far as possible through design. Continued efforts will be made to enhance and demonstrate positive impacts.

The model mine will look at is impacts through a human rights lens and implement the UN Guiding Principles on Business and Human Rights. It will also have a good understanding of its place in the global response to the threat of climate change, be aware of how it can contribute to this and will be working to ensure that its mineral products are saleable in the low-carbon world.

The model mine will also disclose information on impacts and risks and how effectively these are being managed, it will do this to build and maintain trust and partnerships with stakeholders. Commitments made to stakeholders will be fulfilled and compliance obligations in permits and agreements will be observed. Closure objectives and completion criteria will be agreed and met.

The above is unlikely to be achieved without governance. The notions of minerals governance and environmental, social and governance (ESG) are explained in Sections 4.2.2 and 4.2.3.

4.1.2. The Reality

The reality is that most mines past and present do not fit the ideals described above and there are still some miners who assign little or no priority to these ideals. These miners and industry accidents, some of which have occurred under the auspices of the more responsible companies, have undermined trust in the mining sector. Recognising mining as very much part of the low-carbon sustainability-focussed future, the mining sector's many stakeholders are leveraging change and/or laying out their views on the ideals. Mining-sector sustainability standards have mushroomed. The overwhelming number of standards is seen to be problematic and so rationalisation and equivalency review processes are underway. The mining sector stakeholders having most influence on the ESG performance of mines and financiers and downstream manufacturers. This influence can exceed that of governments in many jurisdictions.

4.1.3. Purpose and Structure of this Chapter

This chapter acknowledges the impacts of mining and reviews governance in the mining sector, both globally and in Northern Ireland. The chapter aims to input to the Department for the Economy's (DfE) review of both minerals and petroleum licensing policies. It is focused on mining. The legislation surrounding minerals licensing regimes dates to the 1960s and the review is intended to ensure that the regimes remain fit for purpose. The DfE's review is intended to ensure that the licensing regimes continue to be effective in delivering key Programme for Government objectives, including that Northern Ireland's community prospers through a strong, competitive, regionally balanced economy and lives and work sustainably – protecting the environment. The review covers strategic issues of climate change and the setting of challenging decarbonisation targets for the UK and Northern Ireland, as well as the planned new energy strategy.

This chapter begins acknowledging the impacts of mining and introduces the notions of minerals governance and ESG (Section 4.2); it then looks at international policy (Section 4.3) and climate policy (Section 4.4); and then at minerals governance in Northern Ireland (Section 4.5). Thereafter the chapter covers specific topics worthy of further consideration in the review of Northern Ireland's minerals policy (Section 4.6).

4.2. Impacts and Governance

4.2.1. Overview of Impacts on Mining

The ultimate receptors of mining impacts are people and biodiversity and they are either affected directly or through impacts on the resources and ecosystems that they depend upon. While it is possible to generalise about the nature of impacts caused by mining, it is not possible to generalise about the significance of the impacts of mines. The significance of the impacts depends on the environmental and social setting of the mine and its proximity to sensitive receptors. It also depends on the design of the mine and the way it is constructed, operated and closed. Each mine is unique in this respect. Metrics used to compare the impacts and ESG performance of mines in different locations can be misleading if these are disconnected from setting.

Impacts of mines can be packaged in different ways as outlined in Table 4.1. Here, they are arranged in groups that are logical considering the cause of the impact and actions that can be taken to prevent or mitigate impacts in the design of a mine. As reflected in the impact controls column, many impacts can be avoided through design. The impact controls listed in this table are outlined very simply; there is a plethora of governance frameworks that elaborate on these controls, often in much detail.

4.2.2. Notion of Minerals Governance

Globally, the effectiveness of the environmental and social impact management at mines is heavily dependent on the mineral rights holders and mine operators, whether they are responsible and committed to effective management. Their commitment is also influenced by other actors in the mining sector, including host governments and business stakeholders.

The term "governance" has become widely used in the context of sustainable development, particularly since the 2030 Agenda for Sustainable Development was adopted by UN member states in 2015. The 17 SDGs are at the heart of the 2030 Agenda. It is recognised that the achievement of the SDGs requires governance, but the term "governance" used in this context has no universally agreed definition. However, it is accepted that governance is multi-dimensional and covers different actors, processes, structures and institutions involved in decision-making and implementation. [1]

An international think tank on sustainable development and governance, the Institute for Sustainable Development and International Relations, explains: "Sustainable development is the promise of a political response to a number of issues, and governance is the coordination of that response. Governance is therefore the way in which sustainable development is built. Its objective is to encourage the transformation of societies and to guarantee "well-being", according to the commitments made at the Rio Earth Summit in

1992 and reformulated in 2015 at the adoption of the 2030 Agenda for Sustainable Development and its 17 Sustainable Development Goals." [2]

Table 4.1 - Grouping of impacts considering impact nature and key controls.

Group	Nature of impacts	Impact controls (excluding compensation/ offsets)
Positive socio-economic impacts	Job creation Economic multiplier effects Payment of taxes and royalties Development of new infrastructure	**Impact enhancement** Investment in training and skills development Employment of nationals (and locals where possible) Procurement of goods and services in-country (and locally where possible) Participation in tax transparency initiatives Mandatory and voluntary community investments Providing community access to new infrastructure such as roads, dams, powerlines or pipelines.
Land transformation	Land take Habitat loss and fragmentation Land capability and use Land stability and drainage Landscape and visual impacts	**Impact avoidance** Decisions on location during project conception and planning (avoid sensitive areas and habitat fragmentation as far as possible). Design to minimise footprint Design to minimise landscape and visual impacts **Impact mitigation** Operational control of mine waste placement Rehabilitation
Water resource impacts	Water consumption (impact on availability) Water contamination (impact on quality)	**Impact avoidance** Decisions on location during project conception and planning. Design to minimise footprint. **Impact mitigation** Clean water diversion Dirty water collection, recycling and treatment Operation to prevent uncontrolled discharges
Carbon emissions	Greenhouse gas emissions	**Impact avoidance and mitigation** Design and operation to minimise carbon footprint, particularly Scope 1 and Scope 2 emissions
Air quality and noise	Windblown dust and vehicle entrained dust Vehicle, stack and fugitive emissions Vehicle, equipment and blasting noise	**Impact avoidance** Decisions on location during project conception and planning. Design and operation to minimise emissions **Impact mitigation** Operational control of mine waste placement
Hazards	Failure of mine waste and water holding facilities Spills of hazardous chemicals Blasting damage	**Impact avoidance** Decisions on location during project conception and planning. Design to prevent accidents **Impact mitigation** Operation of waste/water facilities to prevent impacts Safe closure of mine waste facilities

		Safe chemical transport Safe blasting practices **Emergency planning, preparedness and response**

4.2.3. Notion of Environmental, Social and Governance (ESG)

The term "environmental, social and governance (ESG)" has become popular in recent years, corresponding with the strengthening of the economic pillar of sustainability and the emergence of the green economy. It relates to the response of an institution to sustainability imperatives and is at the heart of responsible business, responsible finance and responsible sourcing.

There is no universally accepted definition of "ESG". Some definitions on the internet limit it to measurement and disclosure of ESG performance or relate ESG only to the direction of capital into sustainable investments. It is more than this; it also relates to the actions that an organisation takes to ensure and improve ESG performance – "walking-of-the-talk".

Many mining industry sustainability standards (Section 4.3.6) help to provide perspective on the ESG topics applicable to mineral exploration and mining. Figure 4.1 and Figure 4.2 aim to provide perspective in a nutshell. The latter figure links the topics to the International Council on Mining and Metal (ICMM) Mining Principles [3]. More background on ICMM and mining industry standards is provided in Section 4.3.6.

Figure 4.1 - ESG subjects relevant to mineral exploration and mining.

Corporate governance for sustainable development

Top level commitment (1.4, 6.3), stakeholder engagement (3.7,4.1, 4.4, 6.1,6.2, 9.3, 10.1)
Integrate sustainability principles into corporate strategy and decisions (investment, design, operation and closure) (2.1)
Compliance with law and permit conditions (1.1), respect human rights (3.1)
Functional management systems (1.1, 1.3, 4.3, 6.3), including risk-based controls (4.3, 6.3) emergency plans (4.4, 6.3)
Alignment with relevant leading principles and initiatives/ standards (1.3, 4.3), ESG disclosures (1.5, 6.2, 6.3, 10.3), assurance/ validation (6.3, 10.4)
Environmental and social impact, risk and opportunity assessment for new projects (4.1)

Environmental

- Apply the mitigation hierarchy (6.4)
- Responsible mine closure – plan & design & make financial provision for closure (6.1)
- Water stewardship (6.2)
- Extractive waste management (6.3)
- Reducing GHGs (6.5)
- Climate change adaptation (6.5)
- Biodiversity and ecosystems protection (7.2) – aim to achieve no net loss (7.2)
- Recover, re-use and recycle natural resources and materials (8.1)

Social

- Human rights due diligence (3.1, 3.3)
- Health and safety (5.1, 5.2)
- Workplace training & development
- Fundamental principles & rights at work (3.4) and support diversity in the workplace (3.5)
- Remuneration & working hours (legal limits or better) (3.5)
- Community engagement (4.1)
- Indigenous people (rights, culture, natural-resource based livelihoods and FPIC) (3.6 and 3.7)
- Avoid displacement, responsible resettlement (3.2)
- Social development (local procurement & employment and community development) (9.1, 9.2)
- Collaborate with government supporting improvements in ASM (9.4)
- Cultural heritage protection (3.7, 4.1 and 4.3)
- Social transition at closure (6.1)

Governance, other

- Payment of taxes & EITI (10.2)
- Responsible sourcing (promoting ESG in the value chain and supply chain due diligence) (2.2)
- Risk-based due diligence on conflict and human rights when operating in or sourcing from a conflict-affected or high-risk area (4.2)
- Anti-bribery and corruption (1.2)
- Responsible product labelling-hazard classification and labelling (8.2)
- Responsible political involvement

Figure 4.2 - ESG topics as they apply to mining, with cross references to the ICMM Principles that address these topics.

With the vast number of actors in finance and industry applying the term "ESG" there are cases where it has become disconnected with sustainability concepts and goals. However, there are ongoing drives to direct governance efforts towards achieving the SDGs. For example, the SDG Compass is a tool that was recently developed to help companies make progress in their contribution towards achievement of the SDGS and to report on this [4]. The SDG Compass was developed by the Global Reporting Initiative (GRI), the UN Global Compact, and the World Business Council on Sustainable Development (WBCSD). The core indicators were developed by the International Standards of Accounting and Reporting (ISAR).

4.3. International Policy

A wide range of actors participate in the mining industry and there is rapid evolution in their thinking on responsibilities for sustainable development and resource stewardship. This section outlines the many international governance frameworks that influence environmental and social management in the mining industry. It includes an outline of a global governance model proposed by the International Resource Panel (IRP).

4.3.1. Introduction

Minerals are inputs to almost every sector of the global economy and demand for mineral resources is expected to grow in the decades ahead. A key driver is the global transition towards the low-carbon economy, which requires metals-intensive energy-production technologies. Efforts to decouple economies from resource use and towards greater recycling will not meet the metals demand in the foreseeable future. Mineral resource extraction is expected to increase until necessary infrastructures are in place and resource circularity is effective globally [5].

The 2015 Sustainable Development Goals (SDGs) and the 2015 Paris Agreement have forged a consensus for a pathway to a universal just transition towards a low-carbon economy for all states and societies. Ethics underpinning the value of minerals and their role in realising common goals of global, national and local communities are changing fast [6].

Minerals governance is influenced by domestic law, international sustainable development initiatives and a plethora of standards. The standards have been generated by many different actors with an interest in the mining sector. The actors range from governments, mining companies and mining industry associations, through financial institutions (lenders, investors and insurers) and downstream manufacturers, to non-governmental organisations (NGOs). Motivation to apply these standards is increasing, with the aim of building trust in both mining companies and their products.

A recent detailed review of minerals law from an academic perspective [6], explains that the study of mining law has been transcribed to domestic and legal systems and is generally limited to national boundaries. It has not broadened systematically to encompass the international, transnational, regional, national, and local levels of normative orderings of social relations concerning mining and minerals. The review calls for a global perspective to the study of law and governance of mining and minerals as an academic discipline. It explains that it is necessary to engage with the expectations of all stakeholders and engage with sustainable development as an objective of the global community.

Concurring with this view, a model for global governance of mineral resources has been proposed by the International Resource Panel, which is supported by the United Nations Environmental Agency (UNEP). The IRP describes this model in a report on Minerals Governance in the 21st Century [5]. The model aims to improve the international governance architecture for mining to enhance its contribution towards sustainable development. It is envisaged that the model will supersede the complex array of governance frameworks in the mining sector.

4.3.2. Mining and the Sustainable Development Goals (SDG)

Mining can impact positively and negatively across the 17 Sustainable Development Goals (SDGs). It has most linkages to the goals relating to poverty eradication, decent work and economic growth, clean water and sanitation, life on land, sustainable and affordable energy, climate action, and industry and infrastructure.

Mining is needed for development as many of the minerals produced by mining are essential building blocks to technologies, infrastructure, energy and agriculture. Mining is not just valued for this but also because it can foster economic development. It provides opportunities for employment and can generate significant revenue streams for governments to invest in economic and social development. Furthermore, it can drive economic diversification through direct and indirect economic benefits.

The negative perspective is important to consider too. Historically, mining has contributed to many of the challenges that the SDGs are trying to address. These include environmental degradation, displacement of populations, worsening economic and social inequality, armed conflicts, gender-based violence, tax evasion and corruption, increased risk for many health problems, and the violation of human rights.

A study aimed at mapping of mining to the SDGs was undertaken in 2016 by the World Economic Forum, United Nations Development Programme (UNDP), Columbia Centre on Sustainable Investment, and the Sustainable Development Solutions Network [7]. The study is documented in a detailed report and a summary. The latter summary is given in Appendix A. This summary highlights the opportunity and potential for mining to positively contribute to all 17 SDGs.

A United Nations (UN) meeting on mining and the extractives industry was held on 25 May 2021. This was entitled "The Global Roundtable on Transforming Extractive Industries for Sustainable Development". In her closing remarks, the Deputy Secretary-General, Amina J. Mohammed, stated that the transformation of the sector will require coordinated efforts across all regions, and a shift in mindset away from short-term economic gains and towards long-term profits, and a focus on a just transition, in recognition of the intrinsic value of healthy societies and a healthy planet. Governments and companies, including private mining companies and multinational enterprises, need to create aligned national plans to achieve a net-zero economy and to commit to a global framework for extractive industries that aligns investments and operations with the Sustainable Development Goals and Paris Agreement [8].

At this UN meeting, Rohitesh Dhawan, the CEO of the International Council on Mining and Metals (ICMM), an association of mining companies representing a third of global mineral production, identified three ways mining companies can make positive contributions to the SDGs, while striving to minimise or eliminate negative impacts: partnering; performing and publishing [3]. He gave examples of ICMM partnering, including partnering with equipment manufacturers to accelerate the deployment of cleaner, safer vehicles at mine sites; with communities to build skills and support community resilience; and with global partners such as the United Nations Environmental Programme (UNEP) and the UN-backed Principles for Responsible Investment (PRI) to improve the safety of tailings facilities. Dhawan explained that performing involves consistently applying ICMM's Mining Principles, and other standards on responsible mining, and that publishing refers to disclosure of information on impacts – both positive and negative – in an open, transparent, and meaningful way. At the same time, he acknowledged that the mining industry needs to build trust between the industry and society – trust that to do the right thing and to be a responsible development partner.

4.3.3. Extractive Sector Segments and Governance

Globally, the extractives sector can be divided into several segments based on use, stakeholder interest and applicable governance frameworks. The segments are shown in Table 4.2. Governance of construction and industrial minerals generally receive much less attention than metal minerals. The primary reason for this is that metal minerals are generally exported into highly globalized value chains.

The IRP identify several differences between construction, industrial and metal minerals that are relevant to minerals governance [5]:

- Construction minerals producers are present in every country to supply local and regional construction and infrastructure projects. Industrial minerals are generally widespread, occurring in many countries, and products are not transported vast distances. Both construction and industrial minerals segments are commonly well integrated into the local economy and are less exposed to external shocks and price volatility. Also, these segments are less likely than metal minerals to be disrupted by automation due to their low value.
- Metal minerals are produced in a handful of countries, are widely traded and are commonly used in manufacturing processes remote from the locations where they are mined. Mining can only take place where geological conditions make it possible for economically recoverable mineral concentrations to exist. Not every country is geologically well endowed, and none can economically produce the diversity of metal minerals required by current manufacturing processes. Trade is supported by

seaborne bulk transport connecting countries hosting mines to countries with downstream manufacturing processes.

- Metal mineral value chains tend to have weak linkages to other economic sectors in host countries. The highly specialized and capitalized nature of metal minerals means that this sector lends itself to technological advances such as automation that can limit job creation and local procurement of goods and services.
- Metal minerals raise issues of supply security as disruption at either the supply source or the trading can disrupt economic activities at a global level. The high dependence of many countries on exports revenues and domestic resources means that they may become key issues within local politics.

Table 4.2 - Extractive sector segments

<table>
<tr><th colspan="2">Extractives sector segments</th><th>Minerals and metal</th><th>Notes</th></tr>
<tr><td colspan="2">Construction minerals</td><td>Sand, gravel, crushed rock, dimension stone (such as limestone, granite, syenite, marble), slate, lime, gypsum, clay (undifferentiated), cement</td><td>Data on production are patchy, and the role of development minerals in the economy therefore tends to be overlooked. .</td></tr>
<tr><td colspan="2">Industrial minerals</td><td>Asbestos, baryte, bentonite, boron minerals, bromine, diamond (industrial), diatomite, dolomite, feldspar, fluorspar, garnet, graphite, gypsum and anhydrite, helium, ilmenite, iodine, kaolin (china-clay), kyanite, lime, limestone, magnesia, magnesite, mica, nepheline syenite, olivine, perlite, phosphates (incl. guano), potash, quartz, salt, special clays, silica sand, sillimanite, soda ash, sodium sulphate, spinel, spodumene, sulphur, talc (incl. steatite and pyrophyllite), titanium oxides (rutile, anatase), vermiculite, wollastonite, zeolites, zircon</td><td>In 2016, production was recorded in 151 countries and its global value was estimated to be around USD 88 billion.
The key industrial materials are bauxite, gypsum and salt.</td></tr>
<tr><td rowspan="3">Metals</td><td>Iron and ferro-alloy metals</td><td>Iron, chromium, cobalt, manganese, molybdenum, nickel, niobium, tantalum, titanium, tungsten, vanadium</td><td rowspan="3">In 2016, the value of global production was estimated to be around USD 647 billion, calculated based on value contained in the mined ores.
The key metals by value are iron, copper, gold, manganese and chromium.</td></tr>
<tr><td>Non-ferrous metals</td><td>Aluminium (and bauxite, its ore), antimony, arsenic, bismuth, cadmium, copper, gallium, germanium, lead, lithium, mercury, rare earth metals, rhenium, selenium, tellurium, tin, zinc</td></tr>
<tr><td>Precious metals and minerals</td><td>Gold, platinum-group metals (iridium, osmium, palladium, platinum, rhodium, ruthenium), silver, gemstone diamonds, other precious and semi-precious minerals</td></tr>
<tr><td colspan="2">Mineral fuels</td><td>Steam coal (incl. anthracite and sub-bituminous coal), coking coal, lignite, natural gas, crude petroleum, oil sands, oil shales, thorium, uranium</td><td>In 2016, the value of its products was estimated to be around USD 2,187 billion,</td></tr>
</table>

4.3.4. Critical Minerals and Security of Supply

Many countries have assessed factors that may impact on availability of minerals and metals to support economic activity and have identified a subset of critical minerals. The minerals that a jurisdiction deems “critical” depends on the unique circumstances in that jurisdiction. Criticality does not just depend on geological scarcity and economic policies. Other important factors include geopolitical issues, trade policy, potential for substitution, evolution of specialised technology (including mining, recycling and low-carbon technology), regulatory oversight, and regional instabilities.

The IRP is of the view that criticality assessment methodologies need to be developed further, with more foresight, to enlighten public and industrial strategies, as well as strengthen minerals and metals governance [5]. The IRP suggests an international coordination body is formed for this. The body would have a similar role to that of the International Energy Agency in the energy sector.

The drivers of demand and supply are in constant flux and can interact and influence each other in ways that are counterintuitive. For example, higher demand and high prices can also lead to the development of new substitutes or new technologies to mine previously uneconomic resources that can result in abundance. A shift to a green economy and the new digital technologies are, on the one hand, creating demand for minerals and metals that were previously in more limited demand and, on the other hand, reducing demand through greater emphasis on recycling and other circular-economy related practices [5].

The UK currently does not have a critical minerals list or strategy. In the EU, it is recognised that the bulk of minerals and metals needed are imported. The minerals needed in the EU have been reviewed in three-year cycles since 2011. In 2017, the EU undertook a critical minerals assessment [9]. The resulting analysis identified 27 critical raw materials from a list of 54 candidate materials. They include: antimony, beryllium, borates, chromium, cobalt, coking, coal, fluorspar, gallium, germanium,indium, magnesite, magnesium, natural graphite, niobium, platinum group metals, phosphate rock, rare earth elements (light and heavy, silicon metal and tungsten. Canada announced its list of critical minerals March 2021

[10]. Australia's 2019 critical minerals strategy compares critical minerals lists prepared by other countries (EU, United States and Japan) and has been complemented with 2021 an update on the outlook for selected critical minerals [11].

4.3.5. Mining Industry Actors that Influence Sustainability Practices

International organisations and transnational corporations play a significant role in the mining industry, particularly in the metals segment. Many influence sustainability practices in the industry. International organisations with this influence are listed in Table 4.3. Most of these organisations have either generated or had an input to sustainability standards and guidelines that are applicable to the mining sector (Section 4.3.6). Transnational corporations having this influence include parent companies, financial institutions (lenders, equity investors) and downstream manufacturers.

The influence of parent companies on ESG in the mining sector is variable at present, but is radically changing as these companies are under pressure from other business stakeholders to improve their sustainability performance. Many parent companies, particularly those listed on stock exchanges, undertake annual sustainability reporting to inform business stakeholders about their ESG performance. The information disclosed in this way is deemed insufficient by modern business stakeholders and so there is a drive for much more detailed disclosures on ESG performance at an asset level (Section 4.3.9). There is a corresponding trend towards validation of the ESG performance of each mine site, which is putting mining companies under much pressure to ensure their performance measures up to the specific standards applied in the validation audits.

Mining is a capital-intensive industry and financiers, particularly lenders, have had an influence on environmental and social management practice in the mining industry in the last two decades. This influence has intensified since 2015, corresponding with the adoption of the 2030 Agenda for Sustainable Development. Responsible finance is discussed further in Section 4.3.8.

Downstream manufacturers' interest in the ESG performance in the mining industry has grown. One driver for this is increased consumer demand for sustainable products. Other drivers are competition, reputational risk and growing legal risks in supply chains. Impacts with significant potential to crystallize in legal liability are human rights impacts.

Table 4.3 - International organisations that influence sustainability practices in the mining sector.

Group	Organisation	Interest in sustainability in the mining sector
United Nations (UN) Agencies	United Nations Environment Programme (UNEP)	Wise use and sustainable development of the global environment.
	UNEP Finance Initiative	Works with banks, insurers, and investors to create a sustainable finance sector
	United Nations Development Programme (UNDP)	Helping countries achieve the SDGs
	International Labour Organization (ILO)	Promotes international labour rights
	International Atomic Energy Association (IAEA)	Promote the safe, secure and peaceful use of nuclear technologies
	International Finance Corporation (IFC)	Private sector arm of the World Bank, set the IFC Performance Standards on Environmental and Social Sustainability (IFC Performance Standards)
UN business engagement forums	UN Global Compact Initiative	Supports adoption of sustainable and socially responsible policies and reporting on their implementation. The initiative is the world's largest corporate sustainability initiative. It is a voluntary initiative based on chief executive officer commitments to implement universal sustainability principles and to take steps to support UN goals.
	Principles for Responsible (PRI) Investment Association	A network of investors that works to incorporate ESG into investment practices across asset classes. The PRI is an investor initiative that works in partnership with the UNEP Finance Initiative and the UN Global Compact Initiative. Supported the development of the Global Industry Standard on Tailings Management Has produced guidance on human rights in the extractives sector.
	Sustainable Stock Exchanges (SSE) Initiative	Promotes corporate investment in sustainable development. It is a UN co-organized by the United Nations Conference on Trade and Development (UNCTAD), the UN Global Compact , UNEP Finance Initiative and the PRI. Over 100 stock exchanges are part of the initiative.
Intergovernmental forums	Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development (IGF-MMMSD)	A voluntary initiative supporting more than 75 nations committed to leveraging mining for sustainable development to ensure negative impacts are limited and financial benefits are shared. The UK is a member of this forum.

	Organisation for Economic Co-operation and Development (OECD)	The OECD has 38 member countries and was founded in 1961 to stimulate economic progress and world trade. It seeks answers to common problems, identifies good practices and coordinates domestic and international policies of its members. The UK is a member of the OECD. The OECD operates the Global Forum on Transparency and Exchange of Information for Tax Purposes The OECD developed the OECD Due Diligence Guidance for Responsible Supply Chain Management of Minerals for Conflict Affected and High-Risk Areas
International panels of experts	International Resource Panel (IRP)	Resource management to improve use of resources worldwide. The IRP is a scientific panel of experts that aims to help nations use natural resources sustainably without compromising economic growth and human needs. The Secretariat of the panel is hosted by UNEP through its office in Paris, France
	Intergovernmental Panel on Climate Change (IPCC)	Produces assessment reports every few years summarising the scientific understanding of past, present and future climate change and suggesting international policy options for adaptation and mitigation.
Responsible finance associations and institutions	IFC – see above	See UN Agencies
	Equator Principles (EP) Association	Over 100 financial institutions countries are members of the association and apply the principles in decision making. This includes application of the IFC Performance Standards in countries not classed as “designated countries”. The UK is listed a designated country. (Designated Countries are those countries deemed to have robust environmental and social governance, legislation systems and institutional capacity designed to protect their people and the natural environment. The list of designated countries is updated regularly.) The signatories to the EPs are referred to as EPFIs. Some EPFIs will also apply all or some of the IFC Performance Standards in designated countries.
	Equator Principles Financial Institutions (EPFI)	
	PRI – see above	See UN business engagement forums
	PRI Institutions	Asset managers and owners who are signatory to the PRI More than 3,000 institutions are signatory (and their assets under management are worth more than USD 100 trillion)
Industry associations promoting responsible mining	International Council on Mining and Metals (ICMM)	Focused on strengthening environmental and social performance of the mining sector and enhancing mining’s contribution to society. It is an association of major mining and metals companies (28 companies) and over 35 other minerals associations. The company members represent one third of the mining industry with 650 sites in 50 countries. The member minerals associations included many country mining associations (Australia, Argentina, Canada, Chile, Colombia, Brazil, Ghana, Japan, Peru, South Africa, Southern Africa, United States), Euromines and Eurometaux, and mineral commodity associations (coal, cobalt, copper, gold, iron, lead, manganese, molybdenum, tin, zinc and zircon.

	World Gold Council	The World Gold Council is the market development organisation for the gold industry. There are 33 gold mining companies that are members. The Council aims to provide industry leadership. It Produced the Responsible Gold Mining Principles (RGMPs) in consultation with industry stakeholders. The principles set out what constitutes responsible gold mining. They are intended to inform the expectations of investors and the downstream gold supply chain, as well as create a framework for gold mining companies can provide confidence that their gold has been produced responsibly. The RGMPs recognise and consolidate existing standards and instruments under a single framework. Among the standards recognised are the United Nations Guiding Principles on Business and Human Rights, the OECD Due Diligence Guidance for Responsible Business Conduct and the Extractive Industries Transparency Initiative. Guidelines for Multinational Enterprises and the ICMM Performance Expectations.
	Canada Mining Association (CMA)	Developed the Towards Sustainable Mining (TSM) standard that is a globally recognized sustainability programme for mining companies. It was the first standard in the world to require site-level assessments and is mandatory for all companies that are members of implementing associations. Through TSM, ESG is evaluated, independently validated, and publicly reported against 30 performance indicators. Mining associations in Australia, Spain, Finland, Norway, Botswana, Argentina, Brazil, and the Philippines have adopted TSM.
	Prospectors and Developers Association of Canada (PDAC)	Produced the 3Plus guidelines for responsible exploration
	World Gold Council	The World Gold Council is the market development organisation for the gold industry. There are 33 gold mining companies that are members. It aims to provide industry leadership. It produced the Responsible Gold Mining Principles (RGMPs) in consultation with industry stakeholders. The principles set out what constitutes responsible gold mining. They are intended to inform the expectations of investors and the downstream gold supply chain, as well as create a framework for gold mining companies can provide confidence that their gold has been produced responsibly. The RGMPs recognise and consolidate existing standards and instruments under a single framework. Among the standards recognised are the United Nations Guiding Principles on Business and Human Rights, the OECD Due Diligence Guidance for Responsible Business Conduct and the Extractive Industries Transparency Initiative Guidelines for Multinational Enterprises and the ICMM Performance Expectations.

Responsible mining assurance/ responsible minerals sourcing	Initiative for Responsible Mining Assurance (IRMA)	Responsible mining assurance initiative supported by a coalition of mining industry stakeholders, including downstream manufacturers (including Microsoft and car and jewellery manufacturers), trade unions, NGOs and mining companies. Car manufacturers supporting this initiative include the BMW Group, the Daimler Group and Ford Motor Company.
	Responsible Minerals Initiative (RMI)	Supports responsible sourcing of minerals. Founded by the Responsible Business Alliance and the Global e-Sustainability Initiative. The RMI independent, third-party audits verify smelters and refiners can be verified as having systems in place to responsibly source minerals in line with current global standards.
	Copper Mark	Supports and assures responsible production of copper. The Copper Mark was founded by the International Copper Association. It is transitioning to become a more multi-stakeholder organization and it was incorporated as an independent organization in December 2019. The RMI and Copper Mark collaborated in the development of their assurance standards.
	Aluminium Stewardship Initiative (ASI)	An independent, third-party certification programme for responsible production and sourcing of aluminium
Conflict minerals support	European Partnership for Responsible Minerals (ERPM)	A multi-stakeholder partnership that aims to increase the proportion of responsibly produced minerals from conflict-affected and high-risk areas (CAHRAs) and to support socially responsible extraction of minerals that contributes to local development. It is an accompanying measure to the EU Conflict Minerals Regulation. The aim of this support is to enable more mines to comply with the standards required under the OECD Due Diligence Guidance. The EPRM focusses on tin, tantalum, tungsten and gold (3TG).
Metal exchanges	London Metals Exchange (LME)	Supports responsible sourcing of metals through various initiatives.
	London Bullion Market Association	Produced Responsible Gold Guidance (Version 9 was published in June 2021).
Non-profit associations established to promote transparency	Extractive Industry Transparency Initiative (EITI)	Good governance of oil, gas and mineral resources
	International Accounting Standards Board (IASB)	Global accounting standards, includes standards for the extractives sector
Other	EU Raw Materials Initiative	An association set up to ensure secure, sustainable, responsible and affordable supply of raw materials in the European Union (EU). Scope includes resource efficiency and supply of “secondary-raw materials” through recycling.
	World Economic Forum (WEF)	International organization for public-private cooperation. Platform for leaders from all stakeholder groups from around the world – business, government and civil society – to collaborate on multiple projects and initiatives. The World Economic Forum’s Responsible Mineral Development Initiative (RMDI) was launched in 2010 to explore the views, priorities and concerns of key stakeholders on mineral development.

	International Standard Organisation (ISO)	Several ISO standards influence management systems in the mining industry (eg ISO 14001:2015, ISO 45001:2018, and ISO 31000: 2018), through guidance and certification schemes. ISO is developing a standard on mine closure.
	International Women in Mining	Global organisation pursuing gender equality and promoting women's voices, access to opportunities and leadership in mining
	International Cyanide Management Code	A voluntary initiative for the gold and silver mining industries, and the producers and transporters of the cyanide used in mining. It is intended to complement an operation's existing regulatory requirements.
	Voluntary Principles on Security and Human Rights	A multi-stakeholder initiative and forum addresses security-related human rights abuses and violations. Members must promote and implement a set of principles that guide provision of security while respecting human rights. Members must also report on their implementation and actively participate in the forum.
	Responsible Mining Foundation (RMF)	A non-governmental organisation (NGO) that promotes continuous improvement in responsible extractives. It undertakes research, rates and ranks the performance of extractive companies using a Responsible Mining Index (RMI), and hosts many webinars. Note that this NGO is based in Switzerland, which is one of the three biggest commodity hubs in the world.

Some countries with well-established mining industries are playing a leading role influencing good practice in the sustainable development of minerals and metals. Canada is the most influential and this can probably be linked back to government strategy. In its Minerals and Metals Policy dated 2017, the Canadian federal government sets out its intention to continue to be an active, effective and influential partner on the international stage. A similar commitment was made in the preceding policy dated 1996. The Canadian Minerals and Metals Plan (Canadian Ministers, 2017) also envisions a global leadership role for Canada in the mining sector (Figure 4.3). The Mining Association of Canada's Towards Sustainable Mining (TSM) standard is a globally recognized sustainability model and has been adopted by mining associations in several other countries (Section 4.3.6).

There are no European countries with global prominence as leaders in best practice in the mining industry, probably because there is relatively little metal mining in Europe, including the European Union (EU) and the United Kingdom (UK) at present. Organisations in Europe do however influence best practice in the mining industry, these include mining companies with headquarters in Europe, financial institutions that provide capital for mining and, more recently, institutions in the downstream supply chain based in Europe.

Figure 4.3 - Extract from the 2017 Canadian Minerals and Metals Plan highlighting Canada's global leadership goals.

4.3.6. Explosion of Mining Industry Sustainability Standards

There has been an explosion of sustainability standards applying to the mining industry in recent years. The number is overwhelming; many mining companies are struggling to decide which standards they should align with. The image on the following page identifies some of the standards that are widely recognised by the mining industry and organises them into groups (Figure 4.4).

The phrase “explosion of standards” is not an exaggeration and is used in the IRP’s report Minerals Governance in the 21st Century [5]. The IRP presents this as a problem and as one of the motivations for the new governance framework for the extractive sector proposed by the IRP (Section 4.3.7). The IRP’s report has a chapter dedicated to this matter. It identifies over 80 international governance frameworks and initiatives applying to the extractives sector and organised these in various ways. Challenges with the existing instruments that are identified are as follows:

- They do not fully address the complex array of global commitments made through the SDGs and the Paris Agreement and they address the 2030 Global Agenda for Sustainable Development in a piecemeal way;
- They overlap in scope and they are not coordinated or cohesive, there is a lack of coordination between industry stakeholders;
- They can undermine the regulatory role of governments by claiming that voluntary self-regulation is more effective;
- Many are too sectorial and limited in dimension;
- Some have an exclusive focus on risk management and security of supply;
- Some have had unintended consequences; and
- Some are misaligned, distract from global commitments and can impede progress.

The IRP motivates for a manageable set of requirements to be used by decision makers involved in extractive sector governance. The framework should encompass governance institutions and mechanisms that act at the international, regional, national, local and project levels, and that are implemented by a range of actors (Section 4.3.7).

While the IRP has the above-mentioned criticisms of the many governance frameworks, it does identify a number as being commendable. The named examples include: the Africa Mining Vision, the Extractive Industry Transparency Initiative (EITI), the Dodd-Frank Act, the GRI, the Model Mining Development Agreement, the Natural Resource Charter, the

Initiative for Responsible Mining Assurance (IRMA), the wider work of the International Council on Mining and Metals (ICMM).

Some of the standards that are widely recognised by the mining industry (Figure 4.4) are worthy of further mention and are discussed below.

Overarching	Disclosure	Lender standards	Responsible mining/ sourcing	Issue specific
United Nations Initiatives UN Global Compact UN Sustainable Development Goals UN Guiding Principles on Business and Human Rights UN CEO Water Mandate	*Non-financial/ sustainability reporting initiatives* Global Reporting Initiative (GRI) Sustainability Accounting Standards Board (SASB) International Integrated Reporting Framework EU Corporate Sustainability Reporting Directive	*Equator Principle Financial Institutions* Equator Principles IFC Performance Standards World Bank Environmental, Health and Safety Guidelines	*Industry led initiatives* International Council for Mining and Metals (ICMM) Principles Towards Sustainable Mining (TSM)	Voluntary Principles on Security and Human Rights (VPSHR) Global Industry Standard on Tailings Management (GISTM) International Cyanide Management Code (ICMC) International Atomic Energy Agency (IAEA) Safety Standards IAEA Best Practice in Environmental Management of Uranium Mining World Health Organisation Standards Business and Biodiversity Offsets Programme Standard on Biodiversity Offsets
Other international bodies International Labour Organisation (ILO) Conventions Organisation for Economic Co-operation and Development (OECD) due diligence for responsible supply chains	*Climate specific disclosure* Taskforce on Climate-related Financial Disclosures (TCFD) Carbon Disclosure Project (CDP)	*Specific requirements of development banks* AfDB Environmental and Social Assessment Procedures EBRD Performance Requirements Asian Development Bank guidelines	*Supply led initiatives* Responsible Gold Mining Principles (RGMP) Aluminium Stewardship Initiative CopperMark BetterCoal Responsible Minerals Initiative	
International Standards Organisation ISO Standards (14000 environment, 45000 health & safety, and 31000 risk) and ISO Guidelines (26000 social responsibility)	*Other disclosure initiatives* Extractives Industry Transparency Initiative (EITI)		*Third party led initiatives* Initiative for Responsible Mining Assurance (IRMA) Responsible Mining Foundation	

Figure 4.4 - Organised view of some of the key standards relevant to the mining industry.

Lenders that finance mining projects often require conformance with international standards on environmental and social performance. The leading lender standards are the Equator Principles. Updated in 2020, the Equator Principles 4 requires that projects such as the mining projects (Category A projects) are subject to an assessment process to identify the environmental and social risks and impacts of the proposed project. Corresponding management plans must be developed and implemented in the context of an environmental and social management system (ESMS). The Equator Principles also require that projects comply with host country legislation. Where there are gaps in this legislation, the IFC Performance Standards (January 2012) and Environmental, Health and Safety (EHS) Guidelines (April 2007) should be applied. The latter guidelines are generally not applicable in countries like the UK as stricter requirements apply here, but some of the IFC Performance Standards could be considered applicable by lenders in specific situations.

ICMM plays a high-profile role in the mining industry [12]. Its company members represent one third of the mining industry (Table 4.3). The company members are required to observe ICMM's Mining Principles, which define the good practice environmental, social and governance requirements of company members through a comprehensive set of 38 Performance Expectations (the ICMM Performance Expectations) and eight related position statements on a number of critical industry challenges. Implementation of the Mining Principles is intended to support progress towards the global targets of the UN Sustainable Development Goals and the Paris Agreement on climate change.

Site-level validation of performance against the ICMM Performance Expectations is a new requirement for ICMM members and it is intended to increase trust. Reportedly, the various ICMM members are preparing for external validation audits and are currently undertaking internal audit programmes. ICMM members are also required to undertake sustainability reporting using the GRI framework and have credible assurance of corporate sustainability reports.

Wider work of ICMM that is valued by the mining industry includes the generation of guidelines on sustainability issues for the mining sector. These include guidelines on biodiversity, mine waste management, water stewardship, mine closure, social performance and health and safety.

Recently, ICMM has been helping with coherence of standards in the mining industry, by supporting equivalency reviews of performance validation standards against the ICMM Mining Principles. The outputs of equivalency reviews will be made available on the ICMM website [13].

The Mining Association of Canada's TSM standard is important to consider further. Mining associations in Spain, Finland, Norway, Botswana, Argentina, Brazil, Australia and the Philippines have recently adopted this. The standard is not widely implemented in these countries yet. TSM is acclaimed as being the first sustainability standard that required site-level assessment. The assessments are mandatory for all companies that are members of implementing associations. Through TSM, eight critical aspects of social and environmental performance are evaluated, independently validated, and publicly reported against 30 distinct performance indicators.

The IFC Performance Standards are strongly promoted in a draft report from the EU Horizon 2020 research programme on responsible sourcing of raw materials, which was released for public review in June 2021 [14]. The report identifies IFC standards as being the most comprehensive considering water, energy and material consumption and recycling efficiency. It recommends that the requirements provided by the IFC Performance Standards should also inform new regulations and policies and provide a common basis among EU member states. The report does undertake a detailed review of the standards and does not compare these standards with existing relevant law in the EU.

The IRMA standard is also strongly promoted in the above-mentioned report [14] as the only mining standard that has been developed based on a broad multi-stakeholder process. The report suggests that IRMA can therefore provide a common basis for mining regulations and a starting point for auditing of mining operations to achieve responsible mining. It does however note that there are other standards that need to be recognised such as the ICMM Performance Standards and the Copper Mark. The report states aspects missing from these standards should be the subject of supplementary audits, but it does not provide any insight into the perceived gaps in the latter standards. Again, it does not compare the standards with existing relevant law in the EU.

4.3.7. The IRP's Proposed New Governance Framework for the Minerals Sector

A new framework for global governance of mineral resources has been proposed by the UN-backed IRP [5]. Social licence to operate (SLO) is a term used extensively in the mining industry. The IRP is proposing that the concept is replaced with a new governance paradigm called "Sustainable Development License to Operate (SDLO)" that will promote actors in the mining industry to make decisions compatible with the 2030 Agenda's vision of sustainable development.

The term "SLO" refers to the right to exploit mineral resources. Mining companies recognize that community acceptance is as important as formal licences/ permits granted by the government. The term applies largely to local communities. Critiques of the SLO concept are that it focuses on business risk and it is limited to accommodating community demands to the minimum extent needed to avoid community opposition and associated conflict, reputational damage and costs [15] (Figure 4.5).

The proposed SDLO paradigm extends the SLO concept. It addresses a broader subject matter covering all concerns that fall within the remit of the SDGs and related targets; it is relevant to all actors in the mining sector across the public, private and civil society sectors; its implementation is a shared responsibility across nations and different actors along the minerals value chain. The IRP envisages that the SDLO framework creates a more coherent governance landscape across the whole extractive value chain through consolidation of existing relevant instruments, ensuring sustainable development is the overriding objective, and addressing gaps. The figure below illustrates the SDLO paradigm.

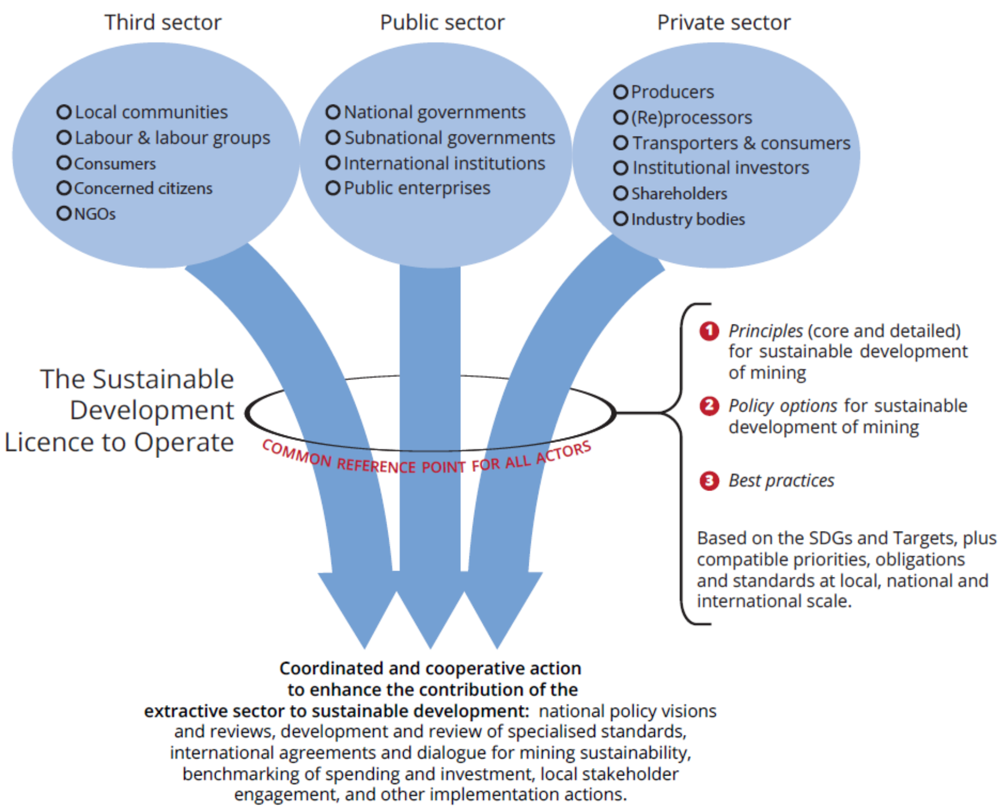

Figure 4.5 - Key components of the SDLO paradigm and illustrative implementation actions (IRP, 2020 – after Pedro, et al 2017).

The IRP explains that the new SDLO paradigm should address not only resource security, but also resource efficiency and decoupling of resource use from economic growth. It should also address all pillars of sustainable development – people, planet, prosperity, peace and partnership. It needs to be based on new metrics where success is measured against a quadruple bottom-line: on the strength of economic outcomes, sound environmental management, the respect of social values and aspirations and adherence to the highest standards of governance and transparency.

Options for operationalization of the SDLO that have been proposed by the IRP include a global international agreement that commits countries to a governance framework (much like the SDGs commit countries to sustainable development); a global platform for continued dialogue and advocacy on cross-cutting issues; and/ or regional platforms to engage host and home regions to reconcile issues of sustainable development and security of supply.

At the international level, the Panel discusses the creation of an International Minerals Agency, or the signing of an international agreement, to, inter alia, coordinate and share data on economic geology, mineral demand needs, and promote transparency on impacts and benefits. The IRP hopes that the UN Environment Assembly, the Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development, and wider ongoing UN processes focused on reviewing progress towards the 2030 Agenda for Sustainable Development could serve as fora for negotiating an international consensus regarding the specific policy options for the implementation of the proposed new global governance framework.

4.3.8. Responsible Finance

4.3.8.1. Overview

Pressure on the financial sector of the economy to direct capital into sustainable development has increased. This pressure is being transferred to the mining sector which is a capital-intensive sector. In some jurisdictions, financial institutions could be having more influence on the ESG performance of the mining sector than governments.

Pressures on the financial sector are reflected in the following:

- United Nations Secretary-General's Roadmap for Financing the 2030 Agenda for Sustainable Development, 2019 – 2021 (UN, 2019);
- The European Commission's Action Plan for Financing Sustainable Growth [16] and the associated Taxonomy Regulation [17] and Sustainable Finance Disclosure Regulation [18].
- The UK's appointment of experts, called the "Green Technical Advisory Group (GTAG)", to advise it on the development of "standards for green investment" to mirror the EU taxonomy [19].
- The growth in the number of signatories to the Principles for Responsible (PRI) Investment from 100 in 2006 to over 3,000 asset owners and asset managers.

Collectively, the signatories have assets under management worth more than USD 100 trillion [20].

- Over 100 stock exchanges are part of the Sustainable Stock Exchanges (SSE), 60 have written guidance on ESG reporting, 45 have markets covered by an ESG index, 39 have ESG bond segments, and 25 have mandatory ESG listing requirements (figures provided are correct as of 14 June 2021) [21].
- Nearly 120 financial institutions have adopted the Equator Principles [22], these institutions are known as Equator Principle Financial Institutions.
- Banks have started providing mining companies with sustainability-linked loans; the interest rates on the loans are linked to ESG performance [23].
- Public-facing rating agencies are rating the ESG performance of asset owners and asset managers. An example is the Asset Owners Disclosure Project [24].
- Financial institutions now have mandatory sustainability reporting requirements in the UK and EU, as outlined below.
- The new UK Stewardship Code 2020 [25] has been revised such that it has a strong focus on delivery of sustainable long-term investment by investors.

Mining projects and mines wanting to raise capital these days must embed ESG into the project design and operations. However, there are no clear-cut criteria that they must meet across the board. Mining companies need to consult with lenders and investors to understand their requirements and expectations. Disclosure of ESG information is seen to be important for this engagement.

4.3.8.2. Equator Principles Financial Institutions

One set of financial institutions does have clear-cut performance requirements. These are the Equator Principle Financial Institutions, which apply the Equator Principles and the IFC Performance Standards as outlined in Section 4.3.6.

4.3.8.3. Sustainability-linked loans

An example of a sustainability-linked loan is one obtained in 2019 by Polymetal International plc, a London-listed gold-and-silver producer with assets in Russia and

Kazakhstan. The loan is in the form of a credit facility of up to USD 75 million and it was agreed with Societe Generale [26]. The facility incorporates an adjustment mechanism linking interest rates to specific environmental and social indicators. The indicators relate to:

- Implementing a comprehensive climate management system;
- Ensuring tailings storage safety;
- Reducing fresh-water use;
- Maintaining occupational health and safety;
- Supporting and engaging local communities.

The above-mentioned loan is just one of several sustainability-linked loans secured by Polymetal. In May 2021, the company reported that it had agreed a USD 400 million of financing with interest rates linked to GHG emission intensity reduction targets. The financing consists of two loans obtained from AO Raiffeisenbank and UniCredit [27].

4.3.8.4. Ratings by Data Providers for Investors

In addition to data put into the public domain by mining companies, investors look at ESG ratings of mining companies by data providers such as Sustainalytics, MSCI, Bloomberg, Vigeo Eiris and Refinitiv. Inconsistency in the ratings given by the different data providers is a problem currently. This is attributed to poor availability of useful, material and validated sustainability information in the public domain. The situation will change as mining companies move towards making more detailed disclosures on ESG performance in the next couple of years, coupled with more assurance and validation of the information (Section 4.3.9).

4.3.8.5. Due Diligence Reviews

Many lenders and investors commission detailed due diligence studies before making a substantial investment in a mining project or company. The due diligence studies generally include an environmental and social review as outlined in Section 4.6.5.

4.3.8.6. The EU Taxonomy Regulation (and UK Equivalent under Consideration)

The EU Taxonomy Regulation will influence what investors based in the EU look for when financing mining projects and operations in future, however mining activities are not included in the EU taxonomy yet. The taxonomy concerns a unified classification system to distinguish economic activities that are environmentally sustainable and to deter greenwashing in the investment space (i.e., unsubstantiated or exaggerated claims that an investment is environmentally friendly.

The UK recently set up a new advisory group tasked with considering an equivalent to the EU Taxonomy Regulation. The UK Government says that the group's work will support investors, consumers and businesses to make green financial decisions [19].

It is envisaged that the EU Taxonomy will help financial institutions and other companies speak with a common language. The main users of the EU Taxonomy will be pension funds, private equity providers, banks and insurers offering financial products in the EU. As outlined below, these entities are obliged by the Sustainable Finance Disclosure Regulation (see below) to report on the extent to which their portfolios align with taxonomy objectives. Other companies are likely to use the EU Taxonomy voluntarily to facilitate the raising of finance.

The EU Taxonomy Regulation establishes six environmental objectives:

1. Climate change mitigation;
2. Climate change adaptation;
3. The sustainable use and protection of water and marine resources;
4. The transition to a circular economy;
5. Pollution prevention and control;
6. The protection and restoration of biodiversity and ecosystems.

There are four criteria under the Taxonomy Regulation that an economic activity must meet to qualify as "environmentally sustainable". One is that it contributes substantially to one or more of the above objectives. Another is that it does not harm any of the six objectives. The third is that it complies with minimum safeguards, which pertain to human rights (Section 1.7.1). The fourth is that it complies with Technical Screening Criteria (TSC).

The body responsible for defining the TSC is called the Technical Expert Group (TEG) on Sustainable Finance. So far, TSC has been defined for the first two objectives (climate change mitigation and climate change adaptation) in a delegated act that was adopted on 4 June 2021. A second delegated act presenting TSC for the remaining objectives will be published in 2022 [28].

Mining activities have been excluded from the published TSC to date. This is attributed to the complexity of the issues that must be considered for mining. It is expected mining will soon be included as it is recognised as having a central role in meeting some of the objectives; for example, metals and minerals are essential for manufacturing renewable energy.

4.3.8.7. The EU Sustainable Finance Disclosure Regulation

In the EU, the financial sector is now legally required to make disclosures on how ESG is integrated into decision making and will soon have to make disclosures on the sustainability credentials of financial products. The requirements have been introduced by the Sustainable Finance Disclosure Regulation (referred to here as the "Disclosure Regulation"). The regulation is in effect; it was adopted by the European Parliament and European Council on 27 November 2019 and substantive provisions of the regulation are effective from 10 March 2021. It is an unfinished regulation in the sense that subsidiary legislation that expands on some of the requirements is still incomplete.

Like the Taxonomy Regulation, the Disclosure Regulation falls under the EU Sustainable Finance Action Plan which promotes the direction of capital towards more sustainable businesses. Both regulations set precedents for other countries, they are 'first movers' among regulations of this kind.

In time, the Taxonomy Regulation and the Disclosure Regulation will be firmly linked. The linkages between the two frameworks will be further specified in the coming years.

The Disclosure Regulation applies to any firm that manages money and/or provides investment advice. These firms are described in the regulation as "financial market participants" and "financial advisers", respectively. The firms include banks, pension providers, insurance undertakings and investment firms. Under the Disclosure Regulations, these firms will need “entity-level” and “product-level” disclosures.

Entity-level disclosures are already required under the regulation, even though the regulation is “unfinished”. The entity-level disclosures must be made public (on firms’ websites) and must explain how ESG is investigated and incorporated into decisions and investment advice. Specifically, these must cover “sustainability risks” that could negatively affect the value of the investment. Larger firms (those with more than 500 employees) must explain considerations of “principal adverse impacts” from ESG risks in their investment decision-making and include a statement on how they do the due diligence to understand those risks. For smaller firms, the reporting requirement is “comply or explain”.

As subsidiary legislation is incomplete, disclosures made in March 2021 are only principles based. More detailed disclosures will be required under Regulatory Technical Standards (RTS) when these are finalised and in force (probably in 2022). Additional, more detailed periodic disclosure regimes are also set to come into place.

The draft RTS identify the following as indicators of adverse impacts are:

1. GHG emissions
2. Carbon footprint
3. GHG intensity of investee companies
4. Exposure to companies active in the fossil fuel sector
5. Share of non-renewable energy consumption and production
6. Energy consumption intensity per high impact climate sector
7. Activities negatively affecting biodiversity-sensitive areas
8. Emissions to water
9. Hazardous waste ratio
10. Violations of UN Global Compact principles and Organisation for Economic Cooperation and Development (OECD) Guidelines for Multinational Enterprises
11. Lack of processes and compliance mechanisms to monitor compliance with UN Global Compact principles and OECD Guidelines for Multinational Enterprises

12. Unadjusted gender pay gap
13. Board gender diversity
14. Exposure to controversial weapons (anti- personnel mines, cluster munitions, chemical weapons and biological weapons)
15. GHG intensity of investee countries
16. Investee countries subject to social violations (absolute number and relative number divided by all investee countries), as referred to in international treaties and conventions, UN principles and, where applicable, national law.
17. Exposure to fossil fuels through real estate assets (share of investments in real estate assets involved in the extraction, storage, transport or manufacture of fossil fuels)
18. Exposure to energy-inefficient real estate assets (share of investments in energy-inefficient real estate assets)

Product-level disclosure requirements apply to all financial products but increase for certain products. For all products, pre-contractual disclosures must set out how sustainability risks are factored into the investment or advice and include an assessment of the effect of sustainability risks on returns. Adverse impacts of products must be clearly communicated to the end-investor. For products promoting environmental or social characteristics" and "products promoting "sustainable investments" additional information must be disclosed, pre-contractually and on the firms' websites. Products that do not fall into these two categories are labelled as unsustainable. A firm that claims its products have sustainable characteristics or objectives will also need to make a product level disclosure in accordance with the technical standards.

In-scope financial products will include investment and mutual funds, insurance-based investment products, private and occupational pensions, and both insurance and investment advice.

The Disclosure Regulation will facilitate comparison of financial products, as well as the potential for consumers and advocacy groups to hold these to higher standards.

The mandatory obligations in the Disclosure Regulation come with the risk of regulatory action, including fines. The Regulation envisages national financial market authorities investigating non-compliance with the EU law in each individual member state. Variations in enforcement are foreseeable.

4.3.8.8. Shareholder Activism

Shareholder activism for sustainability is on the rise. More and more investors are using their equity stakes in companies to put pressure on management to address ESG issues or improve ESG performance. Shareholder activism can take any of several forms: management engagement, management negotiation, proxy voting, publicity campaigns, shareholder resolutions and litigation. Constructive activism is effectively being encouraged by an increased focus on engagement and stewardship in company law. The UK Companies Act 2006 (and its predecessors) contain numerous ways in which a shareholder can use even a relatively small shareholding to ensure that its voice is heard and engagement that is actively encouraged by the new UK Stewardship Code (De Falco, Jackson and Boney, 2020).

One high-profile example of shareholder activism is given in Section 4.8.6. The example relates to investors' response to the Brumadinho tailings storage facility failure in Brazil in 2019. Another example relates to the destruction of the Juukan Gorge caves in Australia in 2020 by Rio Tinto. The 46,000 year-old caves were of high conservation importance, particularly to the Puutu Kunti Kurrama and Pinikura aboriginal community. Australian superannuation funds [29] and London-listed pension funds [30] influenced the departure of three senior executives, including the global CEO, and subsequent actions that have been taken by the company.

4.3.9. Responsible Sourcing

4.3.9.1. Overview

The notion of responsible sourcing is relatively new in the mineral sector and is not formally defined. However, it is a major area of focus for mining companies because it presents significant business risks and opportunities. Figure 4.6 illustrates the actors in the minerals value chain with a role to play in responsible sourcing. Responsible sourcing puts increased emphasis on the ESG credentials of mineral products. Mining companies need to think about the provenance of their mineral products and the position of these products in supply chains geared to sustainable development and decarbonisation.

Figure 4.6 - Actors in the minerals value chain with a role to play in responsible sourcing. Figure adapted from Van den Brink et al (2019) [31].

There is little academic literature on responsible sourcing in the mining sector, but there is a notable study on this subject that was supported by the European Commission [31]. The study explored several different approaches to responsible sourcing, noted the diversity in the objectives applied and concluded that there are effectively two distinct approaches that are compatible.

The two responsible sourcing schemes are "supply chain due diligence" and "sourcing via sustainability schemes". The latter can be used in pursuit of the former. Together, these approaches can in principle enable companies to cover their complex supply chains from beginning to end. Supply chain due diligence aims to manage supply chains by mitigating risks, while sourcing via sustainability schemes seeks to guarantee that certain sustainability requirements are met in the production phase [31].

The above-mentioned responsible sourcing study noted that the British Standards Institution has developed a responsible sourcing sector certification scheme standard for construction products (BS 8902:2009). The BSI defines "responsible sourcing" as "the management of sustainable development in the provision or procurement of a product".

There is a study in the EU Horizon 2020 research programme on responsible sourcing of raw materials. A draft report from this study was released for public review in June 2021 [14]. The draft report presents a roadmap for actions for developing and implementing the notion of responsible sourcing, including actions for the mining sector. The actions are a mix of basic responsible mining recommendations coupled with recommendations on validation of the ESG performance of mines according to recognised standards (Section 4.3.6).

4.3.9.2. Conflict Minerals and Supply Chain Due Diligence

Supply chain due diligence in the context of conflict minerals has been a big focus in the mining sector [31]. The term "conflict minerals" is used for tin, tantalum, tungsten, and gold (in short, these are referred to as "3 TG"). After the end of the second Congolese war in 2003, violence and human rights abuses in the eastern parts of the Democratic Republic of Congo continued and media and non-governmental organisations reported on the role in the conflict of minerals in widespread use in our electronics and cars. Policies were developed to address the issue of conflict minerals. In 2010, the United States passed the Dodd Frank Act – Section 1502 of which is aimed at stopping army and rebel groups in the DRC from using profits from trade in conflict minerals to fund their violent operations. In 2017, the EU passed the Conflicts Minerals Regulation (EU Regulation 2017/821), which is designed to stem the trade in 3 TG as "conflict minerals" and came into force in January 2021.

The EU Conflict Minerals Regulation is mandatory for companies importing minerals or metals in the form of mineral ores, concentrates or processes. It sets out a due diligence process based on the five steps in the OECD Supply Chains Guidance (OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas, [32]). The steps are 1) establishing a management system; 2) identifying and assessing supply chain risk; 3) designing and implementing a strategy for responding to the identified risks; 4) independent auditing of supply chain due diligence; and 5) reporting on the due diligence.

The Conflict Minerals Regulation also requires that minerals must go through a chain of custody or a supply chain traceability system, supported by documentation, meaning that the minerals' origin can be ascertained regardless of how many commercial operators process, handle or transport it upstream prior to import into the EU. While the OECD Supply Chains Guidance contains specific recommendations for each actor in the supply chain for how to operate such a system, the EU Conflict Minerals Regulation is clear that the OECD Guidance is not the only way to comply. Other voluntary supply chain due diligence schemes may be approved by the European Commission [33].

A figure from the OECD Supply Chains Guidance is presented below (Figure 4.7). This shows focus areas for 3 TG supply chain risks. The OECD Supply Chains guidance addresses risks associated with illegal exploitation of minerals, the funding of conflict from mining, exploitation and abuse of local communities and mine workers and payment of taxes to the government.

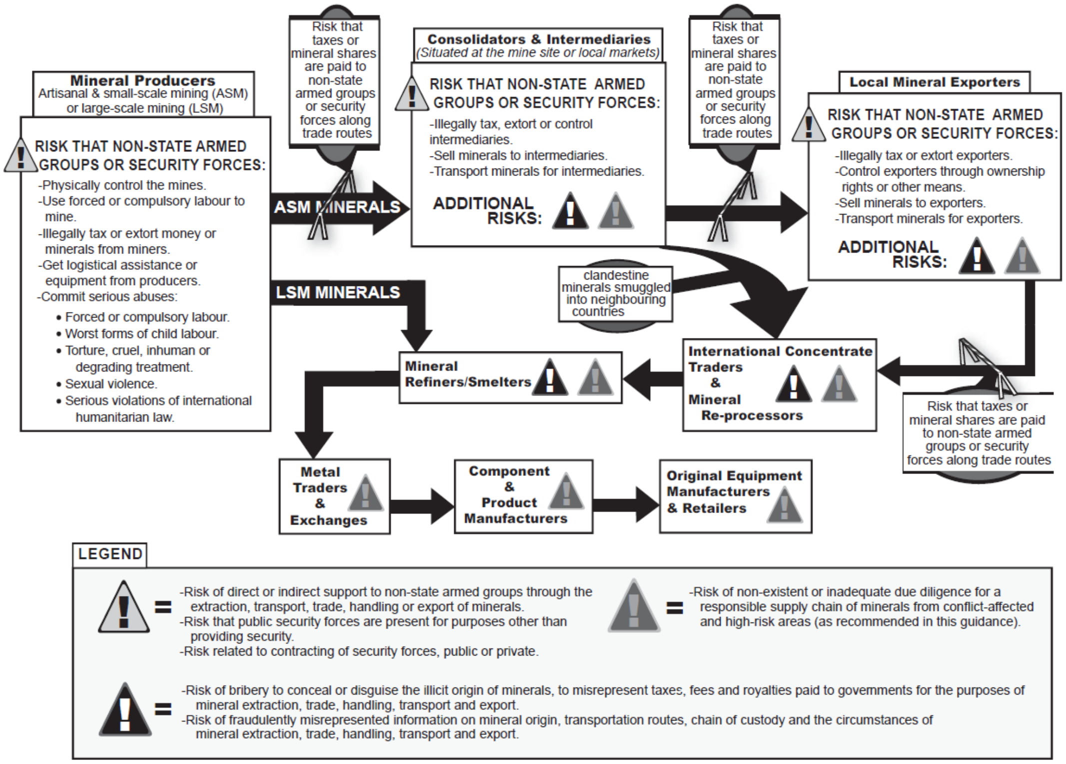

Figure 4.7 - The 3 TG supply chain risks covered by the OECD Supply Chains Guidance [32].

4.3.9.3. London Metal Exchange Initiatives

The London Metals Exchange (LME) has emerged as a leading proponent of responsible sourcing and has several initiatives that embrace both supply chain due diligence and sourcing via sustainability schemes. Metals traded on the LME are referred to as “brands”. The term applies to a specific metal, with a specific metallurgical composition, from a particular producer. The LME’s responsible sourcing initiatives include the following:

- Brands must comply with OECD Supply Chains Guidance (OECD Due Diligence Guidance for Responsible Supply Chains of Minerals from Conflict-Affected and High-Risk Areas).
- Producers must have certified management systems (ISO 14001 and OSHAS 18001).

- Brands must confirm alignment with a relevant internationally recognised sustainability standard (by 2022) and provide evidence of conformance with the chosen standard by 2023 (this does not stop trading on exchange).
- Brands are going to have electronic passports (named the LME passport) tracking their journey through the supply chain and minerals producers will be able add sustainability information to these passports on a voluntary basis.
- LME will look to provide comparability between disclosures related to carbon footprint and recycled content.
- New contracts designed to provide pricing and access to metals that support the electric vehicle transition (lithium hydroxide) and the circular economy (US scrap aluminium and regional steel scrap for India and Taiwan) will be launched in 2021.

The LME explains that the LME passport is a digital document register. It will store and maintain essential provenance documents for traded metal, incorporating both certificates of analysis and emerging certifications, standards and disclosures. The latter are to be registered in the system on a voluntary basis [34].

The LME's position on responsible sourcing is based on consultation with its stakeholders. The LME considers responsible sourcing to be both an ethical imperative and a commercial imperative. It explains: "The fundamental service of the LME is to price metals – and, by the nature of its market, the LME price will generally be the price of the least valuable brand in the brand lists. We must act to ensure that our price reflects the value of responsibly sourced metal and is not artificially depressed by metal which is not sourced in such a manner." [34]

4.3.10. Sustainability Reporting and other ESG Disclosures

Sustainability reporting practices emerged in the 1990s and gained impetus with the launch of the United Nations Global Compact in 2000, which encourages companies to apply its ten principles and report on their implementation. Over time many different reporting frameworks have emerged and now there is a drive to rationalise and improve these. There is also a trend towards sustainability reporting changing from voluntary to mandatory. Furthermore, there is a trend to make disclosures about ESG performance at an asset level linked to an increased interest in the provenance of mineral products.

4.3.10.1. Sustainability Reporting

Companies say that they are being asked to deliver the same sustainability information in too many different formats and are overwhelmed by requests for information from different stakeholders. They say they devote excessive effort and expense to answering numerous specialized requests. Investors complain that the information made available is not good enough for their investment decision making. Several reviews of reporting frameworks and practices have been undertaken in recent years. The reviews reveal the following shortcomings:

- Inconsistency, incomparability, or lack of alignment in standards;
- There are too many different reporting frameworks and there is inconsistency, incomparability, or lack of alignment in standards. Some reporting frameworks are misaligned with the SDGs.
- There is a demand for more sustainability disclosures that are material to financial performance from investors. Other stakeholders consider the focus on financial materiality excessive, there are matters that cannot be defined in financial terms presently that are important to disclose.
- The scope and depth of these disclosures made by companies differ considerably as a result of the subjective choices companies make about their approaches to sustainability reporting.
- Some companies are cherry picking the issues they report on, or expand on, and thereby manipulate the reporting process to portray an image of a socially and environmentally responsible company.
- Aggregated data presented by large companies with many different assets can disguise poor performance at specific assets.
- There is not enough assurance of the reliability of the data that is reported.

Many of the above shortcomings are being addressed by the reporting frameworks. Three leading sustainability reporting frameworks that have been used by mining companies are those of the GRI, the International Integrated Reporting Council (IIRC) and the Sustainability Accounting Standards Board (SASB). The GRI Reporting Standards are geared to a wider audience and disclosures that are of public interest, while IIRC and SASB focuses on financially material disclosures relevant to enterprise value. In the last year, the relationship between these frameworks has been resolved. In November 2020, the IIRC and SASB announced their intention to merge into a unified organization, the Value Reporting Foundation. The merger was recently completed. The newly formed Value Reporting Foundation explains that it will maintain the SASB Standards [35]. The SASB Standards and the GRI Standards are now seen to be complementary and there is a trend towards using them together. A guide to support this has been produced [36].

The GRI has recently produced tools to help companies embed the SDGs in their sustainability reporting [37].

The Value Reporting Foundation states that the SASB Standards have been developed with consideration of the existing metrics in use as much as possible. For example, the SASB Standards apply metrics used by the Carbon Disclosure Project (CDP). Wide consultation on the standards is undertaken with the aim of improving integration with other standards, reducing disclosure costs for companies and facilitating global corporate transparency efforts [38]. There are 77 sets of SASB Industry Standards, eight of these sets cover the extractive sector (including coal, oil and gas, and iron and steel production) and one set covers metals and mining specifically.

The SASB Metals and Mining Standards (SASB)[36] cover the following topics: greenhouse gas emissions; air quality; energy management; water management; waste and hazardous materials management; biodiversity impacts; security, human rights and rights of indigenous peoples; community relations; labour relations; workforce health and safety; and business ethics and transparency. The standards are complemented with booklets describing the basis for the conclusions and providing application guidance.

4.3.10.2. Mandatory Corporate Sustainability Reporting in the UK

Mandatory corporate sustainability reporting was introduced in the UK when the UK was part of the EU and by means of the EU Non-Financial Reporting Directive (Directive 2014/95/EU). In the UK, this directive was transposed into legislation under the Companies Act 2006, by means of the Companies, Partnerships and Groups (Accounts and Non-Financial Reporting) Regulations 2016. The regulations add to existing non-financial reporting requirements that were already in place under the Act.

Additional regulations under the Companies Act that extend ESG reporting requirements are the Companies (Miscellaneous Reporting) Regulations 2016; the Companies (Directors' Report) and Limited Liability Partnerships (Energy and Carbon Report) Regulations; and the Companies (Directors' Remuneration Policy and Directors' Remuneration Report) Regulations 2019.

Section 172 of the Companies Act 2006 relates to the notion of ESG. This section is clear that business leadership is not just about profit for shareholders. It encourages directors taking a stewardship approach to their business, their stakeholders and the environment. Company directors are under a statutory duty to promote the success of the company, which requires directors to consider (amongst other matters): (a) the likely long-term consequences of any decision made; (b) the interests of employees; (c) the need to foster the company's business relationships with suppliers, customers and other stakeholders; (d) the impact of the company's operations on the community and the environment; (e) the desirability of the company to maintain a reputation for high standards of business conduct; and (f) the need to act fairly as between the members of the company.

The Financial Reporting Council (FRC) provides guidance on the details to be provided in corporate sustainability reporting under Section 172 of the Companies Act 2006 and the above-mentioned regulations. The detail to be provided depends on the nature and size of the organisations. Companies that are medium-sized or larger must report on how they generate and preserve value, relevant risks and uncertainties and business performance. Companies with more than 250 employees must produce a Section 172 statement in their annual strategic report. This should explain how stakeholders have been engaged and how pertinent issues have influenced company decisions and strategy in the last year.

Companies with more than 500 employees, listed companies and financial institutions must report on various environmental, employee, social, community and human rights matters.

Mandatory corporate sustainability reporting requirements in the UK are expected to increase going forwards, especially following the FRC recommendation for the latter group of companies to use Task Force on Climate-Related Financial Disclosures (TCFD) and SASB and the publication of the Government's roadmap towards mandatory climate-related disclosures that will make TCFD mandatory across the economy by 2025 [39]. This recommendation has been further ratified by the G7 group of World Leaders in their final communiqé of the meeting held between 11 and 13 June 2021, in that the Leaders stated their support for moving towards mandatory TCFD reporting within their jurisdictions, for businesses and government, and look forward to the establishment of the Task Force on Nature-related Financial Disclosures (see Section 4.4.6).

4.3.10.3. Mandatory Corporate Sustainability Reporting in the EU

The EU is also elaborating on its corporate sustainability reporting requirements; it intends to amend the EU Non-Financial Reporting Directive (Directive 2014/95/EU). In this regard, on 21 April 2021, the Commission adopted a proposal for a Corporate Sustainability Reporting Directive. The proposal extends the scope to all large companies and all companies listed on regulated markets (except listed micro-enterprises). The proposal also introduces:

- A requirement for the audit (assurance) of reported information;
- More detailed reporting requirements, and a requirement to report according to mandatory EU sustainability reporting standards;
- A requirement for companies to digitally 'tag' the reported information, so it is machine readable and feeds into the European single access point envisaged in the EU Capital Markets Union Action Plan.

4.3.10.4. Responsible Mining - Sourcing and Validation

There is increasing recognition in the mining sector that sustainability reporting is not satisfying the disclosure expectations of stakeholders, including downstream manufacturers, and is not helping enough to build trust and transform the sector towards sustainable development. Linked to this there is a move to more detailed disclosure of performance at asset level.

A series of new reporting frameworks have developed to validate the ESG performance of mining operations at a site level. These are likely to be used as part of responsible sourcing initiatives (Section 4.3.9). Some have been adapted from existing governance frameworks, for example the ICMM Principles are now complemented with the ICMM Performance Expectations and a validation procedure. The validation will be applied at a site level and will be used to verify that mining assets conform to ICMM Performance Expectations. This validation would be undertaken by appropriately qualified technical specialists and is different from assurance of data presented in sustainability reports. The findings of the validation audits would be publicly disclosed [40].

Similar asset-level validation frameworks have been prepared by other organisations. Examples of these are listed under the headings “responsible mining” and “responsible sourcing” in Figure 4.4 showing a selection of mining sector standards in Section 4.3.6. The ICMM website hosts equivalency reviews of many of the standards, particularly those focused on the provenance of mineral products [41].

4.3.10.5. Climate-Related Disclosures

A specific review of climate-related disclosures that are relevant to the mining sector is presented in Section 4.4.6. This section considers existing disclosures that must be made under UK legislation and elaborates on TCFD disclosure requirements.

4.3.10.6. Mine-Waste Management Disclosures

New mine-waste management disclosures that investors are encouraging mine companies to make are outlined in Section 4.5.4.

4.3.10.7. Taxonomy and Sustainable Finance Disclosures

Financial institutions are required to make disclosures under the EU Taxonomy and Sustainable Finance Disclosure Regulations as outlined briefly in Section 4.3.8.

4.3.10.8. EITI Disclosures

Disclosures that must be made by holders of mineral rights and governments by means of the EITI are outlined in the following section.

4.3.11. Extractive Industry Transparency Initiative (EITI)

The UK is a member of the EITI. In the first years, EITI focused on regular publication of material oil, gas and mining payments by companies to governments ("payments") and material revenues received by governments from oil, gas and mining companies ("revenues") to a wide audience in a publicly accessible, comprehensive and comprehensible manner. The initiative has evolved so that there is a trend towards reporting of further information on governance of oil, gas and mineral resources [42].

The EITI now promotes disclosure of information on how rights are issued, how the resources are monetised, and how they benefit the citizens and the economy. It also now encourages countries to disclose "open data" online to enable users to make better use of EITI data to inform public debate about the extractive industries and to draw more from existing and emerging online sources than developing separate systems from collecting data for the EITI process.

The EITI explains: "As a multi-stakeholder organisation, the EITI builds trust between governments, companies and civil society. The EITI requires the disclosure of information along the extractive industry value chain, from licensing to extraction, to how revenue makes its way through to the government, to how it contributes to the economy and wider society. In doing so, the EITI strengthens public and corporate governance, promotes transparent and accountable natural resource management, and provides data that informs debate and reform in the extractive sector." [43] The image below illustrates how EITI works (Figure 4.8).

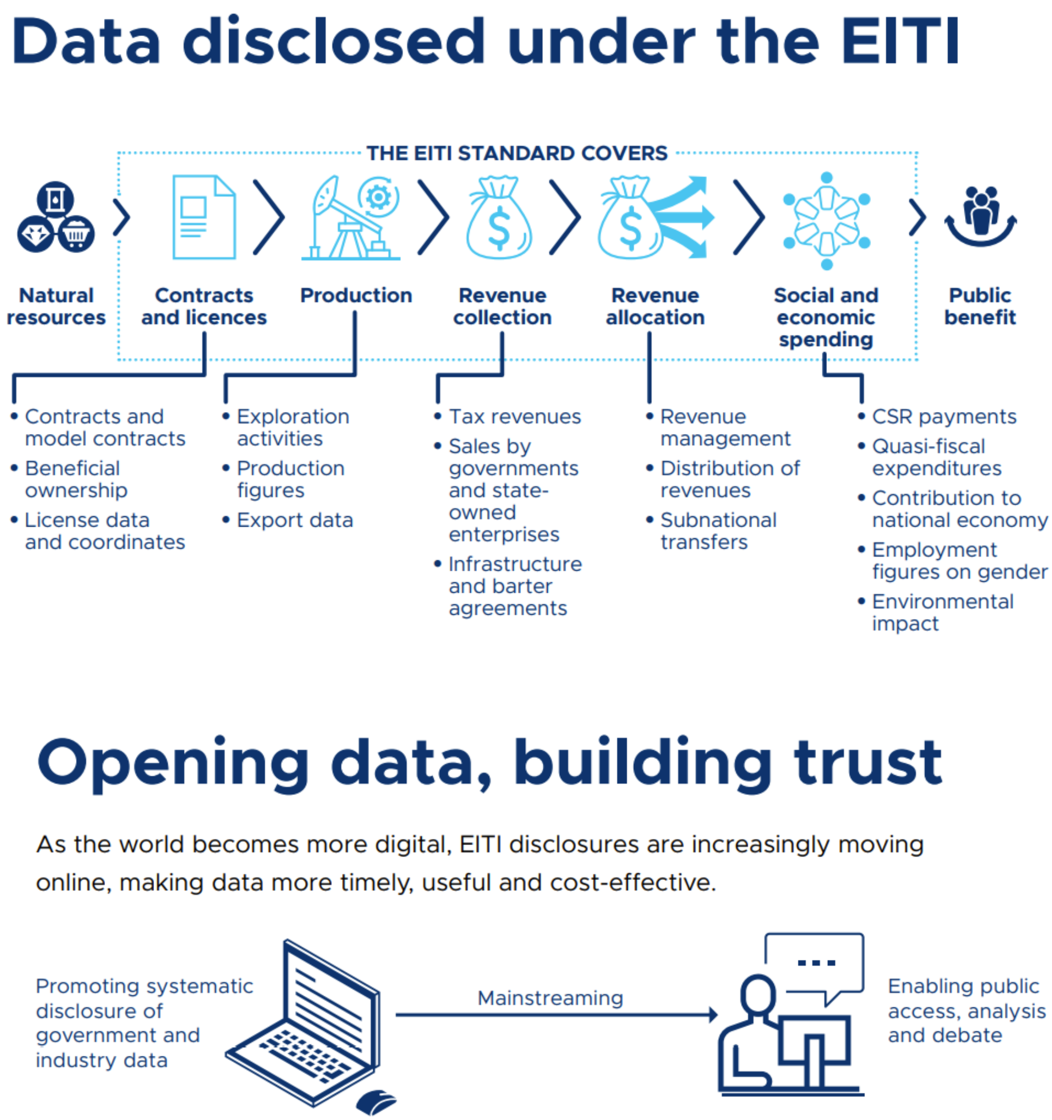

Figure 4.8: Figure illustrating how key EITI concepts (EITI Factsheet, 2020) [43]

The UK became an EITI member in 2014 as is classified as "making meaningful progress" towards meeting the 2016 EITI standard by the EITI Board, based on information from regular validation processes. The UK's EITI Secretariat published the fifth UK EITI report covering the year 2018 on 20 December 2019. This report includes a contextual report describing the activities of the extractive industries in the UK and provides information about the value and importance of the industry to the UK.

The notion of EITI has its origins in the 1990s and early 2000s, when there was much academic literature and campaigning about the potential benefits of the mining and oil and gas industries not being realised in many host countries. Principles to increase transparency of payments and revenues in the extractive sector were proposed in 2003. The UK Government was actively involved in the formulation of these first EITI Principles.

Initially it was expected that EITI would evolve as a voluntary corporate social responsibility initiative for companies, but it has developed into a disclosure standard implemented by countries. From 2004, financial assistance was offered to countries to implement EITI programmes in the form of the World Bank-administered Multi-Donor Trust Fund, which was then replaced by the Extractives Governance Programmatic Support facility in 2016.

An extensive review of EITI reporting requirements led to the development of an EITI Standard in 2013. The standard aimed at ensuring that EITI provided more intelligible, comprehensive and reliable information and linked with wider national dialogue about natural resources. The standard was also restructured to condense the requirements and policy notes into a shorter and more concise seven requirements.

The EITI Standard was revised further in 2016 and in 2019. The standard now encourages countries to build on their existing reporting systems and practices for EITI data collection and seeks to develop the EITI's growing status as a platform for progress that was bringing greater transparency and accountability to all aspects of natural resource management, including tax transparency, commodity trading and licensing. The 2016 Standard included disclosure requirements on beneficial ownership, ensuring that the identity of the owners of the oil, gas and mining companies operating in EITI countries would be public by 2020. The 2019 Standard is revised to include disclosure on contracts, gender, environment and commodity trading.

The 2016 Standard introduced a new validation system which aimed to better recognise efforts to meet and exceed the EITI Requirements and set out fairer consequences for EITI countries that had not yet achieved compliance with the EITI Requirements.

The EITI Standard is composed of two parts. Part I deals with the implementation of the standard and Part II deals with the governance and management of the international EITI. The standard now encourages countries to disclose "open data" online to enable users to make better use of EITI data to inform public debate about the extractive industries and to draw more from existing and emerging online sources than developing separate systems from collecting data for the EITI process.

4.3.12. Reporting on Mineral Resources and Reserves

The international practices for reporting on mineral resources and reserves are worth noting as mining companies apply these extensively to raise capital for mining projects and operations. The practices recognise that interests in the mining industry are transnational and aim to provide business stakeholders, particularly investors, with credible information on mining projects at different stages of development and mining operations. Based on standards, referred to as "codes", the practice addresses the challenge that, unlike many other industries, mining is based on depleting mineral assets, the knowledge of which is imperfect prior to the commencement of extraction.

Information reported in accordance with mineral resource and reserve codes is often readily available in the public domain. It includes some detail on identified ESG risks, in addition to detail on resource geology and on the mine plan and the financial model for the project/ operation, applied in the conversion of resources to reserves. The sections below identify key codes used in resource and reserve reporting and the types of ESG risks considered by these codes. Other detail required by the codes is not included here since the focus of this policy review is on ESG.

It is important to recognise that any permitting and sustainability information provided in public reports will be combed over by data vendors. Data providers to financial firms, traders, and investors scour through whatever ESG data they can get from companies to provide some assessment of ESG performance.

4.3.12.1. Mineral Resource and Reserve Reporting Codes

Mineral resource and reserve reporting codes support the mining industry to communicate risks consistently and transparently to earn trust. Information made public in accordance with the codes is used in decision making by many different mining industry stakeholders including investors and governments.

Guidelines on mineral resource and reserve reporting emerged in the 1900s and first became mandatory in Russia in the 1950s (GKZ reporting standards). In 1989 the Joint Ore Reserves Committee (JORC) of Australasian Institute of Mining and Metallurgy (AusIMM) released a code referred to as the "JORC Code" that was immediately incorporated into Australian Stock Exchange (ASX) listing rules. The JORC Code is widely applied internationally and has been updated several times, the 2012 version is currently valid and is being updated [44]. Since this time many mineral-rich counties have published codes.

Some of these codes are incorporated into minerals law (this is the case in Kazakhstan, for example). Many stock exchanges require that mining companies report resources and reserves in accordance with specific codes. Some stock exchanges recognise one code, while others recognise several codes.

In 1994, a Committee for Mineral Reserves International Reporting Standards (CRIRSCO) was created to contribute to maintaining trust by promoting high standards of reporting of mineral deposit estimates (mineral resources and mineral reserves) and of exploration progress (exploration results). Its current members represent Australia, Brazil, Canada, Chile, Colombia, Europe, India, Indonesia, Kazakhstan, Mongolia, Russia, South Africa, Turkey and the United States (Figure 4.9). CRIRSCO maintains the International Reporting Template, first published in 2012 and most recently updated in 2019, which represents the best of the existing codes [45].

The codes classify mineralized material into two major categories: mineral resources and mineral reserves. Every category has sub-categories, respectively measured, indicated and inferred mineral resources, and proved and probable mineral reserves, based on geological confidence and the potential for economic extraction.

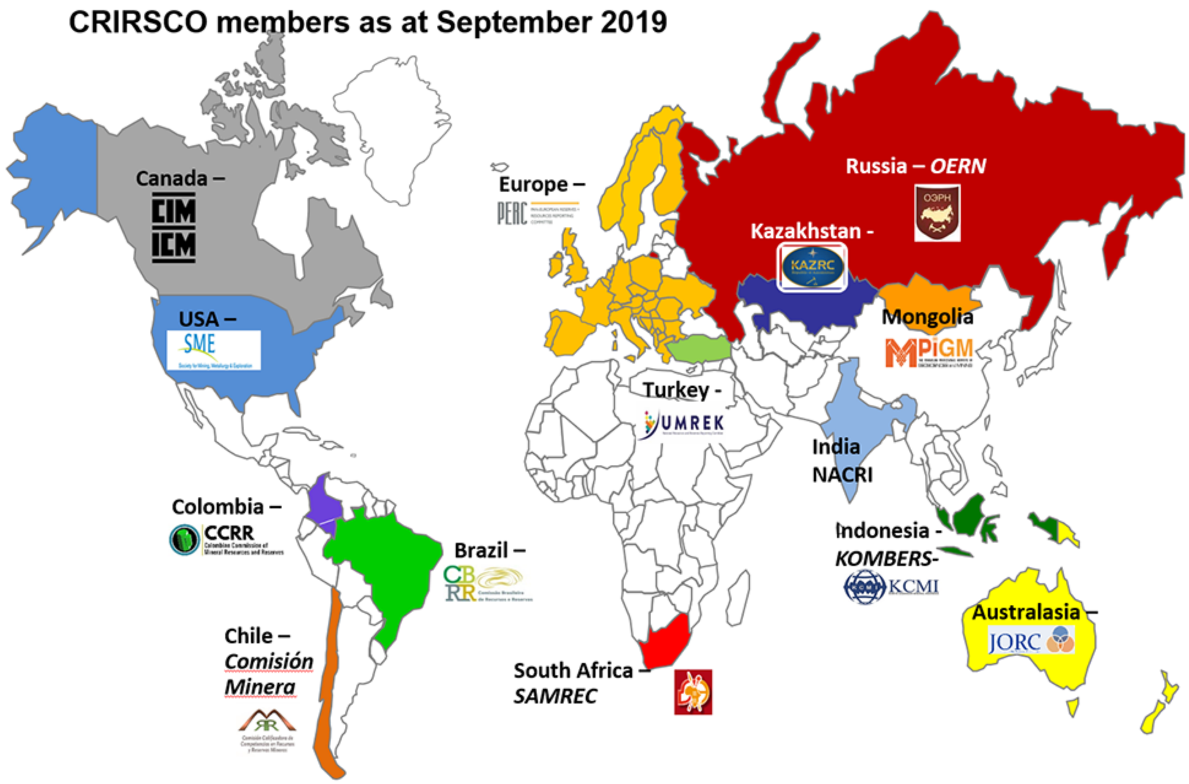

Figure 4.9: CRIRSCO members and their codes (Source CRIRSCO).

4.3.12.2. Permitting and Sustainability-Related Risks Considerations

The codes require that permitting and sustainability-related risks are considered in resource and reserve reporting, particularly in reserve reporting. The new CRIRSCO template (2019) reflects increasing emphasis on permitting and sustainability and the expectation that the subjects are considered from the earliest stages of project planning.

Most codes do not use the term “ESG” and few use the term “sustainability”. Instead, the codes refer to “environmental, social and governmental factors”, specifically as modifying factors that can affect the conversion of resources to reserves. Generally, these factors are interpreted as risks that can stop a project going ahead, cause an operation to be stopped or have significant costs that are material to the value of the asset.

The terminology point mentioned above could be taken that the various codes have not yet fully assimilated the increased level of interest of business stakeholders in the ESG performance of mining companies. However, reading the codes carefully it is quite clear that competent persons responsible for mineral resource and reserve reporting should be mindful of the expectations of mining industry stakeholders.

A few codes are more explicit than others about permitting and sustainability considerations. One is the new Indian Mineral Industry Code for Reporting Mineral Resources and Reserves in India (IMIC) that was published in July 2019 and prepared by the National Committee for Reporting Mineral Resources and Reserves in India (NACRI) with support from CRIRSCO. This aligns with the new CRIRSCO template.

A South African Guideline that supports the SAMREC Code, which is called the SAMESG Guideline, is more explicit than any other guideline on the level of attention that should be paid to ESG in resource and reserve reporting. This guideline is currently under revision to enhance its implementation. The guideline received an award from the United Nations Intergovernmental Working Group of Experts on International Standards of Accounting and Reporting (ISAR) in 2019 [46]. The award recognises "policy, institutional and capacity-building efforts of different stakeholders that encourage and assist enterprises to publish data on their contribution to the implementation of 2030 Agenda for Sustainable Development and facilitate dissemination of good practices in this area."

On permitting, the resource and reserve codes require disclosure on approvals required for mining and whether these can be obtained in a timely fashion and can be retained. Risks associated with approvals and legal rights must be disclosed.

The JORC Code (2012) has additional provisions relevant to permitting. These are:

- For a prefeasibility study (Section 39 of the code): Detailed assessments of environmental and socio-economic impacts and requirements will be well advanced

- For a feasibility study (Section 40 of the code): Social, environmental and governmental approvals, permits and agreements will be in place, or will be approaching finalisation within the expected development timeframe.

The new CRIRSCO template (2019) includes guidance that concurs with the above. It adds that primary permitting requirements should be identified in the scoping stage of mine planning (ahead of the prefeasibility and feasibility studies).

The South African SAMESG Guideline goes a step further than the above in that it promotes disclosure of findings of legal compliance audits. South African environmental and water legislation requires regular third-party audits of compliance with obligations in approvals and corresponding legally binding management plans.

The JORC Code (2012) (Table 1 of the code) requires that the status of agreements with key stakeholders and matters leading to the social licence to operate are disclosed. It also requires that the status of governmental agreements and approvals critical to the viability of the project are disclosed. It states that the report should highlight and discuss the materiality of any unresolved matter that is dependent on a third party on which extraction of the reserve is contingent.

The new CRIRSCO template (2019) states: "Public reports should discuss environmental, social, and health and safety impacts that are expected during development, operation and after closure. These impacts will affect employees, contractors, neighbouring communities and customers." The corresponding guidance given in the new template correlates with information disclosure specifications given in the JORC Code (2012) and the Canadian NI 43-101 Technical Report Template.

The various codes and the CRIRSCO template are not prescriptive about the material risks to be covered, recognising that these will differ from one mineral asset to another, depending on the unique environmental and social setting and the nature of the proposed operations. However, some codes do have specifications information to be disclosed as outlined below.

The widely-used JORC Code (2012) states reporting should cover:

- The status of studies of potential environmental impacts of the mining and processing operation;
- Details of waste rock characterisation and the consideration of potential sites, status of design options considered and, where applicable, the status of approvals for process residue storage and waste dumps should be reported;
- The status of agreements with key stakeholders and matters leading to the social licence to operate.

The Canadian Securities Administrators requires the following subjects are covered in mineral resource and reserve reporting as specified in the Canadian NI 43-101 Technical Report guideline:

- (4d) The nature and extent of the issuer's title to, or interest in, the property including surface rights, legal access, the obligations that must be met to retain the property, and the expiration date of claims, licences, or other property tenure rights.
- (4f) To the extent known, all environmental liabilities to which the property is subject.
- (4g) To the extent known, the permits that must be acquired to conduct the work proposed for the property, and if the permits have been obtained.
- (4h) To the extent known, any other significant factors and risks that may affect access, title, or the right or ability to perform work on the property.
- (20) Environmental studies, permitting, and social or community impact – considering: environmental studies; issues that could materially impact the issuer's ability to extract the mineral resources or mineral reserves; requirements and plans for waste and tailings disposal, site monitoring, and water management both during operations and post mine closure; project permitting requirements, the status of any permit applications, and any known requirements to post performance or reclamation bonds; any potential social or community related requirements and plans for the project and the status of any negotiations or agreements with local

communities; and mine closure (remediation & reclamation) requirements and costs.

4.4. Climate Policy Affecting the Mining Sector

4.4.1. Introduction

The global response to climate change can generally be categorised into two work streams termed “mitigation” and “adaptation”. Mitigation concerns efforts to limit, capture and/or store greenhouse gases (GHGs). Adaptation concerns efforts to learn to live with the changes that have already happened and may worsen in future.

While adaptation includes both future-proofing infrastructure and installations against the chronic and acute changes in weather patterns resulting from global warming, it also includes taking advantage of opportunities where they present themselves. For example, the UK agriculture industry may experience longer growing seasons and warm weather crops may become viable going forward.

Mitigation focuses on reducing emissions of greenhouse gases across the globe, dividing emissions into scopes. Scope 1 emissions are GHGs that are generated onsite, be it through combustion of fuels or from other processes. Scope 2 emissions refer to indirect emissions related to the use of energy at a facility, where the energy is generated at a third-party plant. Scope 3 emissions are effectively all other emissions throughout the value chain. Efforts in mitigation focus on emissions of carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), perfluorocarbons (PFCs), hydrochlorofluorocarbons (HCFCs) and sulphur hexafluoride (SF_6).

The previous chapter outlined international developments that promote governance of mining for sustainable development. This chapter focuses on climate-impact policy changes affecting the mining sector.

In 2015, both the agreement of the SDGs and the Paris Agreement galvanised climate action. Before then, attention to climate action in the global mining sector was a slow burn. This is reflected in the absence of compendiums of guidance on how to decarbonize mining. In contrast, there is abundant guidance for the sector in managing water, waste and social impacts. For close to two decades, international standards have promoted reporting on greenhouse gas emissions (GHGs), particularly Scope 1 and 2 emissions, but this requirement was not accompanied with significant pressures to reduce these emissions.

The EU and in particular the UK have been progressive in terms of the introduction of climate law and policy. The UK was the first country to introduce legally binding emissions reduction targets in the Climate Change Act 2008. Law and policies have created opportunities for development of low carbon mining companies, primarily because of the increasing availability of renewable energy. They have also driven decarbonisation through initiatives like the EU Emissions Trading Scheme, which applies to energy intensive operations. Prior to the UK's exit from the European Union a replacement scheme was implemented through the Greenhouse Gas Emissions Trading Scheme Order 2020. The monitoring and reporting requirements of the UK scheme are aligned with the EU scheme but separate.

Minerals processing activities like steel making, aluminium production and cement production fall under the above-mentioned EU and UK emissions trading schemes, but most mines do not meet the qualification criteria.

The financial sector is probably the biggest driver of decarbonization in the mining industry at present. Many investors are encouraging mining companies to reduce their carbon footprint at rates that exceed government targets. Rating agencies – public-facing and investor-facing agencies – are also pressurizing mining companies to improve their performance on this front. A well-known public-facing rating agency is the Carbon Disclosure Project [47].

Many larger mining companies, particularly ICMM members, have strategies to reduce their carbon emissions, which are defined in accordance with the TCFD framework (Section 1.4.6).

While data to enable mining companies to calculate their Scope 1 and 2 emissions is often readily available (as fuel and electricity are expensive commodities), calculating Scope 3 emissions, GHG emissions within the value chain presents a greater challenge. Similarly managing and reducing Scope 1 and 2 emissions is much more within the control of the organisation, whereas reducing Scope 3 emissions may involve extensive engagement with multiple levels of suppliers and customers. Most larger mining companies must still find a pathway to being carbon neutral by 2050 that includes Scope 3 emissions, presenting a challenge particularly for minerals companies with high Scope 3 emissions such as iron ore and coal.

This section looks at climate policy relevant to the mining sector in the UK and Northern Ireland, which focuses on decarbonisation, and briefly considers the EU Green Deal, which will set precedents for other nations. A notable feature of climate change legislation in the UK is that it is largely separate from the framework of environmental law.

4.4.2. Climate Change Legislation

4.4.2.1. UK Climate Framework

The UK Climate Change Act (2008), further amended in 2019, sets a legally binding GHG emissions target to achieve net-zero GHG emissions by 2050 for the UK and all operations, including mining operations within the UK. When first enacted in 2008, the target was initially set at an 80% reduction on 1990 levels by 2050. It was further increased to Net Zero by 2050 in 2019, reflecting the increased international ambition following the Paris Agreement of 2015.

The Climate Change Act requires a series of carbon budgets be established to achieve the 2050 target over set time periods and establish the Committee on Climate Change (CCC) to measure and report on them. While the Committee has voiced opposition to mining in the UK it has been constrained to coal mining, raising the concern that additional emissions resulting from coal use in steelmaking would further hamper the UK efforts to meet its commitments [48].

The UK is currently in the middle of the third carbon budget (2018-2022) having met the first two and is on course to meet the third and current budget although the CCC reported in 2019 that the UK was off course in delivering the fourth [49]. The UK has proposed a sixth carbon budget, which requires emissions reduction of 63% from 2019 to 2035 (78% relative to 1990), on the way to Net Zero by 2050.

The Climate Change Act originally required implementation of the Carbon Reduction Commitment Energy Efficiency Scheme (CRC) through the CRC Scheme Order 2010. The CRC ran until 2019 at which point it was ended by the UK Government. The scheme applied to mining in so far as the qualification requirements were based on the type of energy meter used and established an internal UK trading scheme for energy intensive organisations supplementary to the EU Emissions Trading Scheme. The scheme was un-officially replaced by disclosure requirements using organisational qualification criteria rather than metering.

The Climate Change Act also made provision for adaptation, requiring that the Secretary of State produce a report on risks from climate change every five years, and to produce a programme of adaptation to these risks.

Much of the existing energy and climate change legislation directly applicable to mining in Northern Ireland depends on qualification thresholds. These thresholds differ between regulations. For example, various requirements apply to “Quoted Companies”, companies listed in a European Economic Area country, the New York Stock Exchange or the Nasdaq stock exchange. Others apply to undertakings defined as micro, small, medium and large. The size of an undertaking is dependent on the turnover, balance sheet total and the number of employees.

Other qualification thresholds relate to the activity the company undertakes. Environmental permitting law in particular references industrial sectors and production thresholds to capture the larger industrial emissions within a permitting regime. It allows small and bijou entities to operate with proportionately lower environmental requirements and therefore costs.

The UK Government’s Ten Point Plan for a Green Industrial Revolution is outlined in Section 1.4.4.

Existing energy and carbon reduction schemes in the UK and their applicability to the mining sector are discussed in Section 1.4.3.

The CCC published a three-part report to the UK Parliament in June 2021 [49] that presents reviews of the UK Government's progress to date on reducing emissions and adapting to climate change, coupled with policy recommendations for every part of Government. The report does not consider mining specifically, but some summary recommendations relevant to the mining sector are:

- A Net Zero Test should be undertaken to check that all Government policy, including planning decisions, is compatible with UK climate targets.
- Delayed plans on surface transport, aviation and hydrogen must be delivered.
- Plans for the power sector, industrial decarbonisation, the North Sea, peat and energy from waste must be strengthened.
- The big cross-cutting challenges of public engagement, fair funding and local delivery must be tackled.

4.4.2.2. Climate Policy in the Devolved Administrations

The CCC [49] explains that delivering Net Zero by 2050 will require strong policy frameworks across all levels of Government, and collaboration between the governments of Wales, Scotland and Northern Ireland with Westminster to develop the required policies. There have been important developments in climate policy in the devolved administrations in recent years. An outlined is provided below:

- Both Scotland and Wales created ministerial portfolios that focus on Net Zero and decarbonisation following the May 2021 elections.
- Scotland has a Climate Change (Emissions Reduction Targets) (Scotland) Act 2019 that sets targets to reduce Scotland's emissions of all greenhouse gases to net-zero by 2045 at the latest. It has interim targets for reductions of at least 56% by 2020, 75% by 2030, 90% by 2040. The Act amends the Climate Change (Scotland) Act 2009. The Scottish Government recently updated its Climate Change Plan, which integrates the 2045 Net Zero target and its new interim targets into its delivery plan

for emissions reductions out to 2032. The CCC (2021) has advised the Scottish Government to scale up delivery across all sectors in line with the ambition set out in the recent Climate Change Plan Update.

- The Welsh Government has an Act that will help to mitigate for and adapt to the impacts of climate change, this is Environment (Wales) Act 2016. The Welsh Government recently increased its 2050 emissions target to Net Zero, from a 95% reduction on 1990 levels, by means of the Environment (Wales) Act 2016 (Amendment of 2050 Emissions Target) Regulations 2021. The CCC [49] has advised Wales to publish a new Net Zero Delivery Plan that sets out a long-term vision for meeting the Net Zero goal in 2050, with a particular focus on the Third Carbon Budget and beyond.
- The Northern Ireland Assembly is working towards legislating a Climate Change Bill before the next Assembly election in 2022 as outlined below.

4.4.2.3. Northern Ireland Climate Change Bill 2021

On the 22nd March 2021 the Northern Ireland Climate Change Bill was introduced to the Northern Ireland Executive as a Private Member's Bill by Ms Clare Bailey. The Bill passed its second reading on 10 May 2021.

The Bill aims to establish a binding net-zero greenhouse gas emissions target for Northern Ireland by 2045, as well as to provide for the establishment of a Climate Commissioner and a Northern Ireland Climate Office. On the day the Bill receives Royal Assent, it will automatically declare a state of emergency for the climate that will remain in place until verified evidence is provided that the goals laid out in the Paris Agreement of 2015 have been met.

The Bill requires the derivation and implementation of climate action plans to achieve a Net Zero carbon, climate resilient and environmentally sustainable economy by 2045. The plans must contain annual targets for, net GHG emissions, water quality, soil quality and biodiversity, and must include plans for the enhancement of storage of carbon.

The CCC [49] has advised that Northern Ireland legislates for a credible long-term emissions reduction target – the CCC previously advised that an 82% reduction on 1990 levels by 2050 is Northern Ireland's appropriate contribution to the Paris Agreement and the UK Net Zero goal. It has also advised that Northern Ireland publishes a final energy strategy that sets out how Northern Ireland will achieve a net-zero carbon energy system by 2050, in line with the pathways recommended in the CCC's December 2020 advice.

The Bill received criticism during its second reading from the then Minister for Agriculture, Environment and Rural Affairs, Edwin Poots, over concerns the Net Zero target is unachievable considering the economic structure of Northern Ireland. Since this criticism the political landscape including the Minister for Agriculture, Environment and Rural Affairs has changed, but no alternative Bill has yet been proposed to the Executive at time of writing.

4.4.3. Established Energy Saving and Carbon Reduction Schemes in the UK

4.4.3.1. Energy Savings Schemes in the UK

The Energy Savings Opportunity Scheme (ESOS) Regulations of 2014 (SI 1643) transposes the requirements of Article 8(4)-8(6) of the EU Energy Efficiency Directive (2012/12/EU). The scheme requires that large undertakings complete an energy assessment by a "Lead Energy Assessor", identify areas of significant energy consumption and savings opportunities, and produce a report that must be reviewed by the business top management.

The objective of the ESOS Scheme is to present top management with information on the potential value of implementing energy initiatives but does not mandate that energy use is either disclosed or that action on the identified energy savings measures is taken.

4.4.3.2. Carbon Reduction Schemes in the UK

Legislation promoting carbon reduction across the UK differs from environmental regulation that uses emission limit values and enforcement actions in response to exceedances of the limits. Instead, financial incentives and disincentives are used to drive GHG reduction and promote the uptake of renewable or energy efficient alternatives. The

CRC, ESOS and the emissions trading scheme do however include provisions for fines for non-conforming operators.

The UK has combined the “carrot and stick” through the CRC, through its involvement in the EU Emissions Trading Scheme from its inception in 2003 until the exit from the EU, and now in the UK Greenhouse Gas Emissions Trading Scheme. The trading schemes use a cap of permissible emissions distributed among qualifying installations within the jurisdiction. Whether or not an operation is included in the scheme depends on whether it is classified as a “Regulated Activity”.

Although mining itself is not considered a Regulated Activity, several processes that may be associated with certain mines such as sulphide metal ore sintering and smelting are Regulated Activities, as is fossil-fuelled power or heat generation. Thresholds apply to inclusion under these Regulated Activities. For example, in the case of power generation, the threshold is “at or above 20 MW capacity”. A mine operating below these thresholds is not required to participate.

If a mine does qualify for any reason it may be able to request free carbon allocation from the Government. The quantity of free allowances available depends partly on the exposure of that sector to carbon leakage. Should the mine emit more in a year than their allocation, they must purchase the difference. If they emit less than their allocation, they may sell the difference. This process does not apply to power generation; the energy sector is not eligible for free allowances and must purchase all its allowances each year.

Another cost focused climate change mitigation mechanism in the UK is the Climate Change Levy (CCL). The levy is paid on electricity, gas and solid fuels and was introduced as part of the Finance Act 2000. Despite the name, the CCL is also levied on energy generated by renewable technology and nuclear power. The levy is charged at £0.00541 per kWh on electricity, £0.00188 per kWh on natural gas, and £0.01429 per kWh on solid fuels.

Although energy intensive industries may claim relief on the CCL, energy intensive industry does not include mining, and may not necessarily include crushing and beneficiation, so a mine may be subject to the full CCL.

The UK has two incentive schemes to promote the installation of renewable energy for heating, one for domestic properties and the other for non-domestic properties, although the non-domestic scheme is no longer open to new entrants so new mines would not be able to take advantage of it.

4.4.4. Climate Change Consideration in Environmental Impact Assessment

When mining projects apply for planning permission in the UK, the EIA report must include an assessment of the greenhouse gas emissions from the project, the impact climate change may have on the project, and measures to reduce the risks to the environment and people. This requirement was introduced in 2017, in EIA legislation, as outlined below.

The EU Environmental Impact Assessment (EIA) Directive (2014/52/EU) established a requirement to consider climate change in EIA. Annex VI of the Directive, which outlines the information required in an EIA, requires that there is consideration of “climate (for example greenhouse gas emissions)” and “impacts relevant to adaptation”. In the Northern Ireland this requirement was transposed into law via the Planning (Environmental Impact Assessment) Regulations (Northern Ireland) 2017 (SR 83). Schedule 4 of the regulations outline the information expected in an EIA, and include in Section 5 “Description of the likely significant effects of the development resulting from, inter alia: … 5 (f) the impact of the development on climate (for example the nature and magnitude of greenhouse gas emissions), and the vulnerability of the development to climate change”.

4.4.5. Northern Ireland Climate Policy

Northern Ireland policies and commitments that may have an influence on climate action in the mining and minerals industry are discussed in this section. The policies and plans published for Northern Ireland considered here are:

- Regional Development Strategy 2035 (RDS), which was agreed in January 2012;
- The Strategic Planning Policy Statement for Northern Ireland – Planning for Sustainable Development (SPPS), dated September 2015;

- "A New Decade; a New Approach" published August 2020;
- "Draft Industrial Strategy: Economy 2030" published in 2017;
- "Rebuilding a Stronger Economy" published June 2020; and
- "The Draft Energy Strategy" published March 2021.
- "10x Economy" published May 2021

4.4.5.1. Regional Development Strategy 2035

"Take action to reduce our carbon footprint and facilitate adaptation to climate change" is one of the eight aims of the RDS 2035 (as expressed in paragraph 2.10). The corresponding regional guidance in the strategy is not specifically relevant to the mining sector, except for the intention to increase the use of renewable energies, which is expressed in Paragraph 3.26.

4.4.5.2. Strategic Planning Policy Statement for Northern Ireland (SPPS)

The policy explains that sustainable development is at the heart of the SPPS and the planning system (Paragraph 3.1) and recognises that a central challenge in furthering sustainable development is mitigating and adapting to climate change (paragraph 3.10). It states in paragraph 3.10 that the planning system should help to mitigate and adapt to climate change by various means, including the following which are relevant to mining developments:

- Requiring the siting, design and layout of all new development to limit likely greenhouse gas emissions and minimise resource and energy requirements;
- Promoting the use of energy efficient, micro-generating and decentralised renewable energy systems.

4.4.5.3. A New Decade, A New Approach

This agreement outlines the establishment of the Northern Ireland Executive following the hiatus from 2017. The document includes a commitment to tackling climate change "head-on" in line with the Paris Agreement. It also includes a commitment to producing an Energy Strategy with targets and actions for a fair transition to a low carbon economy.

The agreement also suggests that a Northern Ireland specific Climate Change Act should be brought forward to give targets a legal underpinning comparable with commitments in other UK devolved administrations. It also commits to the establishment of an Environmental Protection Agency to ensure targets are met and also includes a commitment to closing the controversial Renewable Heat Incentive Scheme and replacing it with an alternative.

4.4.5.4. The Draft Industrial Strategy; Economy 2020

The strategy introduces the Energy Policy as a core economic and social activity and reiterates the commitment to develop an Energy Strategy for publication at the end of 2021.

4.4.5.5. Rebuilding a Stronger Economy

The white paper has limited implications for impacts on the mining sector but discusses the post COVID-19 recovery and looks to build capacity in the green employment sector.

4.4.5.6. The Draft Energy Strategy

The ambition of the Draft Energy Strategy [50] is for Northern Ireland to make a fair contribution to the UK's commitment to a target of net-zero GHG emissions by 2050, including an overall goal of achieving net-zero carbon energy by 2050, while ensuring that energy remains affordable. For Northern Ireland this equates to an 82% reduction in total GHG emissions. The key principles of the strategy are:

- Protecting consumers through the energy transition;
- Growing a green economy;
- Setting clear targets, standards and regulations to drive energy efficiency;
- Replacing fossil fuels with indigenous renewables; and
- Creating a flexible and integrated energy system.

The greatest implications of the Draft Energy Strategy on the minerals industry in Northern Ireland are likely to fall on fossil fuel extraction and use, which are not within the scope of this study. However, there may an influence on future mine design, such as:

- Emphasis on avoidance of emissions through designing out emissions early in the mining lifecycle;
- A push to install low or zero carbon solutions at existing and future mine sites such as wind power generation or ground source heat pumps; and
- Minimum energy efficiency standards.

SRK notes that the Draft Energy Strategy itself would not directly drive a material increase in demand for minerals required in the construction of renewable or energy storage infrastructure. Manufacturing of such infrastructure within Northern Ireland might however motivate development of supply chains from local mines, if feasible.

4.4.5.7. The 10x Economy

Published in May 2021 the 10x Economy is a vision for a decade of innovation in Northern Ireland. The ambition with the vision is to make economic growth innovation led, inclusive and green and use a new set of metrics to measure progress. Details of these metrics or actions to be applied to achieve green growth have not yet been developed.

4.4.6. UK Government Ten Point Plan for a Green Industrial Revolution

The UK Government published a Ten Point Plan for a Green Industrial Revolution in November 2020. Although the plan is ambitious, there are no mentions of mining or indication of any change to the regulatory framework for mining. The investment in renewable technologies and the critical minerals they will require worldwide will have a significant effect on the value of certain minerals globally, but it is not expected this plan will have any significant effect on mining in Northern Ireland.

The Ten Point Plan centres on leading in various areas of climate change mitigation. The investment aims of the plan are as follows:

1. £160 million in advancing offshore wind;
2. £240 million net zero hydrogen fund for driving the growth of low carbon hydrogen production;
3. £385 million in an Advanced Nuclear Fund and £215 million into Small Modular Reactors to deliver new and advanced nuclear power;
4. £1 billion to support the development of electric vehicles and £1.3 billion for charging points to accelerate the shift to zero emission vehicles;
5. Tens of billions on rail enhancements, £4.2 billion in city public transport and £5 billion on buses, cycling and walking;
6. £15 million in "FlyZero, a 12-month study to investigate "Jet Zero" and £20 million into the Green Maritime Demonstration Programme;
7. 600,000 heat pumps installed in buildings by 2028 and extension of the Green Homes Grant;
8. £1 billion in a Carbon Capture Storage and Utilisation Infrastructure fund;
9. £40 million in the second round of the Green Recovery Challenge, the creation of new national parks and Areas of Outstanding Natural Beauty (AONB), and £5.2 billion in a 6-year flood and coastal defence programme; and
10. £1 billion Net Zero Innovation portfolio, £100 million in greenhouse gas removals technology and a further £100 million for energy storage innovation challenges, as well as £222 million in fusion power to push innovation, as well as the UK's first Sovereign Green Bond and introduce mandatory climate risk disclosures (Section 1.4.6) to facilitate green finance.

4.4.7. European Green Deal

The European Green Deal will affect Britain's markets, trade negotiations and stance in global climate action, even though the UK has now left the EU. Miners looking for capital from Europe, to mine in Europe and/or to supply products to the EU will have an interest in both the European Green Deal and EU Sustainable Finance Action Plan.

The European Green Deal is a set of initiatives underpinning the European Commission's aim to make Europe climate neutral by 2050. It moves climate-change concerns to the heart of EU policy making [51]. The European Climate Law is the cornerstone of the Europe Green Deal. The goal of no net emissions of greenhouse gases in 2050 is to be enshrined in a European Climate Law and is coupled with a more-immediate target to cut the EU's emissions by at least 55% by 2030. The European Climate Law enters into force soon and requires alignment of existing EU laws with the climate neutrality goal and corresponding strong coherence across EU policies.

The Europe Green Deal proposes measures to ensure European industry does not experience unfair competition if countries outside of Europe are less ambitious on climate. These measures include a tax on certain imports; specifically, a Carbon Border Adjustment Mechanism (CBAM) is proposed to ensure that the price of imports reflects more accurately their carbon content.

The Europe Green Deal includes clean energy initiatives that aim to develop an integrated power sector based largely on renewable sources. A regulation on batteries and waste-batteries is envisaged to strengthen the sustainability of supply chains and improve the recycling of industrial, automotive, electric vehicle and portable batteries in the EU. Carbon footprint reporting and supply chain due diligence are expected to be mandatory.

EU strategies relating to sustainable industry include the New Industrial Strategy for Europe and the Circular Economy Action Plan. These aim to develop markets for climate neutral products that are also more durable, re-usable, repairable and have increased recyclable contents. The industrial strategy aims to ensure that industries are equipped to keep pace with and drive the digital and green transformations of the economy. The strategy will support carbon-intensive industries in implementing the transition towards low-carbon products and help adversely affected regions in dealing with the negative impacts of the transition.

The European Strategy for Data will facilitate the development of a 'single market for data' and develop electronic product passports that can improve the availability of information of products sold in the EU to tackle false green claims [52].

Financing of the Europe Green Deal is covered in a Europe Green Deal Investment Plan that intends to deliver EUR 1 trillion through several mechanisms including direct provisions from the EU budget and EU Emissions Trading Scheme. The EU Sustainable Finance Action Plan is complementary to the European Green Deal. This plan aims to reorientate capital flows towards sustainability, mainstream such aspects into risk management and foster transparency and long-termism [53]. The EU Taxonomy and Sustainable Finance Disclosure Regulations come under this plan. The requirements of these regulations and the implications for the mining sector are outlined in Section 1.3.8.

4.4.8. Climate-Related Disclosures Required by Government and Financial Institutions

Disclosure of company information into the public domain is increasingly required by governments and financial institutions (Section 1.3.10). This section focuses on disclosures on climate action, particularly disclosures related to carbon footprint and decarbonisation.

Public disclosure of GHG and energy information allows comparison between competitors, hypothetically driving businesses to improve their energy and carbon performance to remain competitive for financiers and consumers. Disclosure also facilitates partnerships between different sectors of the economy aimed at addressing decarbonisation imperatives. For example, partnerships between the mining sector and the financial sector or between mining, downstream manufacturers and retailers.

Mining operations within Europe generally emit comparatively less GHG than other industrial sectors and as such often fall below the regulatory radar. They do not however, fall below the radar of lenders and investors.

4.4.8.1. Energy and Climate Change Disclosures Required in the UK

Since the implementation of the Companies Act in 2006, quoted companies in the UK have had to disclose annual GHGs within their annual Directors' Report. In 2018 the scope of this disclosure was expanded by the Companies (Directors' Report) and Limited Liability Partnerships (Energy and Carbon Report) Regulations. This expanded disclosure is known as "Streamlined Energy and Carbon Reporting" (SECR), which replaced the CRC.

SECR requires disclosures from both large and medium sized companies. Disclosures must include direct emissions from a site resulting from the combustion of fuels and process emissions from the facility, and must also include Scope 2 GHG emissions from the purchasing of energy, the methodology used to calculate emissions, and an intensity metric such as tonnes of product per tonne of carbon dioxide equivalent (CO_2e).

Traded companies with more than 500 employees must also produce a separate non-financial report in line with the Companies, Partnerships and Groups (Accounts and Non-financial Reporting) Regulations 2016. The regulations transpose the EU Non-Financial Reporting Directive (NFRD) 2014/95/EU and require disclosure of ESG information, in addition to the existing carbon disclosure requirements (Section 4.3.10).

In the UK, Her Majesty's Treasury and the Financial Reporting Council are proposing to make TCFD compliant disclosures mandatory across the UK economy through their Roadmap Towards Mandatory Climate-Related Disclosures [54]. The Roadmap aims to ensure that by 2025 comprehensive information on both transition and physical risks relating to climate change are available through the value chain. An outline of TCFD requirements is provided in the following section.

4.4.8.2. Climate Change Disclosures Required by Financial Institutions

Banks that provide loans for industrial developments often have minimum ESG standards that must be met by the minerals companies they lend to, for example the IFC has "Performance Standards" and the European Bank for Reconstruction and Development has "Performance Expectations". These standards include requirements relating to greenhouse gas (GHG) emissions from the mineral operators.

The IFC Performance Standard 3 requires that GHG are avoided and reduced in the design and operation of developments. It requires that developers implement technically and financially feasible and cost-effective measures for improving efficiency in energy consumption and compare the energy efficiency with benchmarking data (PS 3.6). It also requires that developers consider alternatives and implement technically and financially feasible and cost-effective options to reduce GHG emissions during the design and operation of the project (PS 3.7). For mines, these options may include, but are not limited to, alternative project locations, adoption of renewable or low carbon energy sources.

IFC Performance Standard 3 also requires reporting on Scope 1 and 2 emissions for sources with annual emissions greater than 25 kCO_2e. Quantification of GHG emissions must be conducted by the client annually in accordance with internationally recognized methodologies and good practice.

The Equator Principles require certain categories of facilities to also consider physical risks that may result from climate change, such as the risk to infrastructure from increased rainfall or permafrost thawing. The Equator Principles also require that all Category A projects (most mining projects are Category A projects) and projects emitting more than 100,000 tCO_2e from Scope 1 and 2 have evidence of an analysis of alternative technologies that may emit less GHGs. The latter projects must also consider transition risks, the risks that may be posed to the operation by the transition away from fossil fuels and into a low carbon economy.

Many institutional investors expected mining companies to make disclosures using the TCFD framework. Within the TCFD framework, transition risks include:

Policy and legal risk including jurisdictions developing carbon taxation or trading mechanisms, legislative constraints on water use or requirements for energy efficient or low carbon technology to be implemented. Legal risk also includes the potential for claims to be brought against organisations for inaccurate or misleading disclosures.

Technology risk including the extent to which new low carbon or energy efficient technology can displace old technology or fossil fuel assets.

Market risk, including the response of the market to supply and demand of commodities linked to technological change, or away from fossil fuel powered energy generation.

Reputational risk, the risk of being identified as a large emitter, or being associated with industries with lower response to the transition to a low carbon economy.

The TCFD considers physical risks to include:

- Acute risk, event driven risk such as that associated with severe storms or extreme weather; and
- Chronic risks, e.g. risk associated with long term shifts in climate, such as permafrost melting destabilising high latitude infrastructure or seal level rise threatening ports.

The framework acknowledges climate related disclosures may include opportunities such as:

- Improvements in resource efficiency through the reduction of energy and fuel usage;
- Renewable energy uptake, improving security of supply and developments in energy storage technologies;
- Innovations in development of low carbon products and services;
- New market opportunities through collaboration with development banks, governments, small-scale entrepreneurs; and
- Increased resilience to the changing climate and to the changing economic landscape.

The disclosure of these risks and opportunities should be in the categories in Table 4.4.

The most-commonly used scenarios within the sphere of climate science are representative concentration pathways (RCPs). These can be used to model the climate's reaction to various hypothetical concentrations of greenhouse gases in the atmosphere, including potential climatic changes at regional or even mine site resolutions. This allows an approximation of the potential risks to be established, facilitating adaptation strategies.

The TCFD also recommends scenarios also be conducted in the context of transition risks. Energy and power company ConocoPhillips for example uses scenario analysis of investment trends, moderate and accelerated transition and global carbon price to inform its decisions [55]. With high exposure both to changes in financial markets and changes in weather patterns, mining will likely have to move towards increased use of scenarios for identifying risks and opportunities.

The TCFD recommends that the approach organisations take to addressing climate-related risks is based on scenario analysis. Scenarios are a tool useful in estimating how the climate and the economy will alter due to climate change.

Table 4.4. - TCFD recommended disclosures

Governance	Disclose the organisation's governance around climate related risks and opportunities It is recommended the organisation: Describe the Board's oversight of climate related risks and opportunities a) Describe management's role in assessing and managing climate related risks and opportunities
Strategy	Disclose the actual and potential impacts of climate-related risks and opportunities on the organisation's businesses, strategy and financial planning where such information is material. It is recommended the organisation: a) Describe the climate related risks and opportunities the organisation has identified over the short, medium and long term b) Describe the impact of climate-related risks and opportunities on the organisation's business, strategy and financial planning c) Describe the resilient of the organisation's strategy, taking into consideration of different climate-related scenarios, including the 2°C or lower scenario
Risk Management	Disclose how the organisation identifies, assesses and manages climate-related risks. It is recommended the organisation: a) Describe the organisation's processes for identifying and assessing climate-related risks. b) Describe the organisation's processes for managing climate-related risks. c) Describe how processes for identifying, assessing and managing climate-related risks are integrated into the organisation's overall risk management
Metrics and Targets	Disclose the metrics and targets used to assess and manage relevant climate-related risks and opportunities where such information is available. It is recommended the organisation: a) Disclose the metrics used by the organisation to assess climate-related risks and opportunities in line with its strategy and risk management process b) Disclose Scope 1, Scope 2 and if appropriate Scope 3 greenhouse gas (GHG) emissions and the related risks c) Describe the targets used by the organisation to manage climate-related risks and opportunities and performance against targets

4.5. Review of Relevant Northern Ireland Policy and Legislation

This section outlines legislation and planning policy relevant to the development of mines in Northern Ireland and presents some recommendations for consideration in the review of this legislation and policy.

4.5.1. Mineral Rights and Granting of Exploration and Mining Licences

Since 1998, Northern Ireland has had a devolved government within the UK, presided over by the Northern Ireland Assembly and a cross-community government (the Northern Ireland Executive). The UK Government and UK Parliament are responsible for reserved and excepted matters. In Northern Ireland, under the Mineral Development (Northern Ireland) Act 1969, with certain exceptions (exceptions include gold and silver) mineral rights are vested with the Department for the Economy (DfE). (Responsibility for gold and silver rests with the Crown Estate Commissioners.) Exploration and mining licences are granted under the Mineral Development (Northern Ireland) Act 1969 by the DfE.

Provisions relevant to sustainable development and ESG are absent from the Mineral Development (Northern Ireland) Act 1969. However, this does not mean that sustainability matters are neglected in the granting of mineral licences. Instead these matters are provided for under other legislation. The DfE cannot take decisions on licences without certain planning and environmental approvals being in place (Sections 4.5.3 to 4.5.5). Legislation relevant to these approvals is given in (Table 4.5).

Guidance on applications for minerals licences is available from the DfE [56]. Applications for mining licences must include the following information:

- Relevant planning permissions;

- Surface rights held or surface rights agreements;
- The mine plan showing storage, extraction, remediation works and facilities, and work to protect the environment;
- The relevant experience and technical qualifications/ expertise of the applicant;
- The financial capital available to develop the project;
- Statutory insurances and public liability cover to indemnify the DfI.

Table 4.5 - Planning and Environmental Legislation Relevant to Mining in Northern Ireland

Topic	Legislation
Planning permission	• Planning Act (Northern Ireland) 2011 (Chapter 25) • Planning (Development Management) Regulations (Northern Ireland) 2015 (SR 71) • The Planning (General Development Procedure) Order (Northern Ireland) 2015, No 72
EIA	• Planning (Environmental Impact Assessment) Regulations (Northern Ireland) 2017 (SR 83)
Mine waste management	• Planning (Management of Wastes from Extractives Industries) Regulations (Northern Ireland) 2015 (SR 85)
Nature conservation	• Wildlife (Northern Ireland) Order 1985, No. 171 (NI 2) • Nature Conservation and Amenity Lands (Northern Ireland) Order 1985, No. 170 (NI 1) • Nature Conservation and Amenity Lands (Amendment) (Northern Ireland) Order 1989, No. 492 (NI 3) • Environment (Northern Ireland) Order 2002, No. 3153 (NI 7) (including amendments) • Wildlife and Natural Environment Act (Northern Ireland) 2011 • Conservation (Natural Habitats, etc.) Regulations (Northern Ireland) 1995 (SR 380) as amended by SR 2007/ 245; SR 2009/8; SR 2011/216; SR 2012/368 and SR 2015/182 • Planning (Trees) Regulations (Northern Ireland) 2015 (SR 84)
Holding of hazardous substances	• Planning (Hazardous Substances) (Amendment) Regulations (Northern Ireland) 2014 (SR 190). • Planning (Hazardous Substances) (No. 2) Regulations (Northern Ireland) 2015 (SR 344) • Planning (Hazardous Substances) (No.2) (Amendment) Regulations (Northern Ireland) 2016
Major hazards	• Control of Major Accident Hazards Regulations (Northern Ireland) 2015 (SR 325) (COMAH)
Water abstraction, discharges and prevention of pollution	• Water (Northern Ireland) Order 1999 SI 662 (including amendments) • Water Environment (Water Framework Directive) Regulations (Northern Ireland) 2017 (SR 81) • Water Framework Directive (Classification, Priority Substances and Shellfish Waters) Regulations (Northern Ireland) 2015 (SR 351) • Groundwater Regulations (Northern Ireland) 2009 (SR 254), as amended by SR 2009/ 359; SR 2011/211, SR 2014/ 208 and SR 2016/ 119 • Water Abstraction and Impoundment (Licensing) Regulations (Northern Ireland) 2006 (SR 482), as amended by SR 2007/122
Alteration of a watercourse	• Drainage (Northern Ireland) Order 1973 No. 69 (NI 1)
Flood risk assessment	• Water Environment (Floods Directive) Regulations (Northern Ireland) 2009 (SR 376) • Reservoirs Act (Northern Ireland) 2015
Drinking water quality	• Water Supply (Water Quality) Regulations (Northern Ireland) 2007 (SR 147) as amended by SR 2009/246, SR 2010/218 and SR 2015/363 • Private Water Supplies Regulations (Northern Ireland) 2009 (SR 413) as amended by SR 2010/131
Industrial emissions	• Pollution Prevention and Control (Industrial Emissions) Regulations (Northern Ireland) 2013 (SR 160), as amended by SR 2014/304
Air quality	• The Air Quality Standards Regulations (Northern Ireland) 2010 (SR 188) as amended by SR 2017/2
Additional noise legislation	• Environmental Noise Regulations (Northern Ireland) 2006 (SR 387)

	• Control of Noise (Codes of Practice for Construction and Open Sites) Order (Northern Ireland) 2002 (No 303)
Additional waste legislation	• Waste (Northern Ireland) Regulations 2011 (SR 127) • Waste Management Licensing Regulations (Northern Ireland) 2003 (SR 493) as amended by SR 2009/76; SR 2011/403; SR 2014/137; SR 2014/137. SR 2014/253 and SR 2015/ 301 • Hazardous Waste Regulations (Northern Ireland) 2005 (SR 300) as amended by SR 238/2015 and SR 288/ 2015. • Waste and Contaminated Land (Northern Ireland) Order 1997 (SI 1997/2778(NI 19)) • Waste and Contaminated Land (Amendment) Act (Northern Ireland) 2011
Naturally occurring radioactive materials (NORM)	• Radioactive Substances Act 1993 (RSA 93), which deals with the control of radioactive material and disposal of radioactive waste in the UK • Radioactive Substances Act 1993 (Amendment) Regulations (NI) 2011 (SR 290) • Radioactive Substances Exemption (NI) Order 2011 (SR 289) • Ionising Radiation Regulations (NI) 2000 (SR 375)
Environmental Liability	• Environmental Liability (Prevention and Remediation) Regulations (Northern Ireland) 2009 (SR 252) as amended by SR 361/ 2009 and SR 210/ 2011
Climate Change	• Climate Change Act 2008
Archaeology	• Historic Monuments and Archaeological Objects (Northern Ireland) Order 1995 No 1625 (N.I. 9)

The small print of the mining licence application requires detail on the reserve calculations and the geological evidence that they are based on. However, there is no specification of the code (Section 4.3.12) that applies to the reserve calculations.

4.5.2. Planning Permission

Planning permission must be obtained by a project proponent before a mining licence can be granted. For most mining developments, completion of an environmental impact assessment (EIA) process and submission of an EIA report and a mine waste management plan are prerequisites to the grant of planning permission, as outlined further in Sections 4.5.3 and 4.5.4. Although not specifically required by planning legislation, Northern Ireland regulatory authorities do also consider other important environmental approvals that might be required for a project before taking a planning decision (Section 4.5.5).

The regulatory authority responsible for determination of planning permissions in Northern Ireland is the local authority. Under the Planning Act (Northern Ireland) 2011 ("the 2011 Act"), which came into force in April 2015, most planning functions were transferred from central government to district councils. However, the Department for Infrastructure is responsible for processing planning applications deemed to be of regional significance. Section 26 of the 2011 Act defines a development of regional significance as one that is of significance to the whole, or a substantial part, of Northern Ireland or involves a substantial departure from the local development plan for the area to which it relates.

Planning policy relevant to mining developments includes the following:

- Regional Development Strategy 2030 (RDS), which is the spatial strategy for Northern Ireland and was agreed in January 2012;
- The SPPS for Northern Ireland – Planning for Sustainable Development (SPPS), dated September 2015;
- Provisions of the Planning Strategy for Rural Northern Ireland (PSRNI) that are currently retained under the transitional arrangements of the SPPS;
- Planning Policy Statements on specific subjects that are currently retained under the transitional arrangements of the SPPS;
- Local development plans, which are prepared for local council areas, by local councils, in consultation with the local community, and considering the RDS.

The RDS promotes the new developments to sustain rural communities and employment opportunities, which respect local, social and environmental circumstances and are integrated into the rural landscape (see Paragraphs 3.94 and 3.96). Policy provisions for mining and minerals in the SPPS (see Paragraphs 6.148 to 6.166) and PSRNI (MIN 1 to Min 8) are similar in that these:

- Recognise that mineral development can bring economic value;
- Are not in favour of mineral development in areas of natural beauty, scientific interest, archaeological and historical importance, cultural-heritage importance, nature-conservation importance and good quality agricultural land – particularly in designated areas of importance and where features of importance would be affected;
- Do not completely prohibit mineral development in the above-mentioned areas, but require careful assessment of the need for and consequences of mineral development in the areas;
- Require that measures to protect water resources are included in applications for mineral development;

- Require that measures are taken to avoid and mitigate visual intrusion and impacts on landscape quality;
- Consider mining operations to be bad neighbours of housing because of their continuous and disruptive nature and require a degree of separation between the mineral workings and other developments, particularly where operations involve blasting;
- Require consideration of the safety and amenity of people living in close proximity to the development, including along the roads that will be used;
- Require consideration of the adequacy of roads for traffic associated with the development, improvement where necessary and particularly consideration of safety and convenience of other road users.
- Require proposals for closure of the mine that include progressive restoration of the site during the life of the mine;
- Where mineral reserves are known and proven to acceptable standards, surface development that would prejudice future exploitation of the reserves, particularly where the reserves are of value to the economy.

The latter point is interesting. It comes from MIN 5 of PSRNI and it uses the language of modern mineral resource and reserve reporting, but there is no reference to recognised resource and reserve reporting codes (Section 4.3.12). Such codes define a level of work that must be undertaken before resources and reserves can be declared. The SPPS Regional Strategic Policy on Minerals makes a similar point in SPS Paragraph 6.155 but does not use language that aligns with international codes.

It is also noted that MIN 5 of PSRNI mentions lignite (brown coal) as an example of a mineral that may be important. PSRNI dates to 1993. Now, with modern views on climate action, this is not an ideal example.

There is commendable emphasis on visual and landscape implications of development proposals in the Northern Ireland Minerals planning policy. Mine infrastructure needs to be located to avoid impacts as much as possible, skylines need to be preserved and existing landforms and features need to be considered in the design of infrastructure. Attention to visual and landscape qualities of the environment is generally a gap in international standards on mining and sustainable development.

The SPPS sets out the factors which should be taken into consideration in the determination of mineral development applications.

- Paragraph 6.163 – “In decision-taking the factors to be considered on a case by case basis will depend on the scale of the proposed minerals development and its local context. In all circumstances proposals will be assessed in accordance with normal planning criteria, including such considerations as: soil quality (where this is particularly suitable for agriculture), water quality (of rivers, lakes, reservoirs and groundwater), tree and vegetation cover, wildlife habitats, natural features of interest in the landscape and sites of archaeological and historic interest.”
- Paragraph 6.164 – “Visual intrusion is often the most significant environmental impact associated with mineral workings and where permission is granted, landscape quality should be protected by attaching conditions designed to avoid or mitigate any adverse impacts. Regard should be paid to the preservation of skylines and to the proposed location of plant, stockpiles and overburden/waste within the development. Additional scope for the satisfactory mitigation of proposals impacting upon the environment may include, where applicable, measures designed to prevent pollution of water bodies, watercourses and ground water. Such measures should be included in applications for mineral extraction and processing plants, including settlement ponds. The provision of 36 reliable protective measures will be an important factor in assessing the acceptability of the proposal.”
- Paragraph 6.165 – “When assessing proposals, it is also important to have particular regard to the safety and amenity of the occupants of developments in proximity to mineral workings. Proposals impacting upon the residential amenity of occupiers of nearby development may be satisfactorily mitigated by requiring sufficient separation between mineral operations and housing development, particularly where such operations involve blasting. The distance required will vary according to a number of factors including the nature of operations, intervening topography and the layout / design of housing development.”

- Paragraph 6.166 – "In line with the objective to secure the sustainable restoration, including the appropriate re-use of mineral sites, planning applications should be required to provide adequate details demonstrating the satisfactory restoration of sites subsequent to the completion of operations. Such provisions must be underpinned by appropriate conditions attached to any grant of planning permission."

4.5.3. Environmental Impact Assessment Input to Planning Decisions

In Northern Ireland, an EIA report is not subject to an EIA approval decision directly as is the case in many other countries. Instead, it is a supporting document that informs the decision to grant planning permission. The EIA report is a product of an EIA process.

The Planning (Environmental Impact Assessment) Regulations (Northern Ireland) 2017 define what projects should be subject to an EIA process; the information to be included in the EIA report; who should be consulted as part of the EIA process; and the procedures for submitting and advertising an EIA report. Most metal mining projects are likely to be classified as EIA developments, especially if they involve underground mining.

The planning authority must take account of the likely environmental effects of a proposed EIA development before making a planning decision for the development. The information to be included in an EIA report includes:

- A description of the development, including its physical characteristics, production processes and residues and emissions;
- Assessment of alternatives considered, and reasons for the choice;
- Consideration of disaster risks;
- A description of elements of the environment likely to be significantly affected – including population and human health, biodiversity and use of natural resources, land, soil, water, air, climate change (mitigation and adaptation), material assets, cultural heritage, the landscape, and risks of major accidents;

- A description of the likely significant effects and cumulative effects of the development;
- A description of mitigation measures and monitoring plans;
- A non-technical summary; and
- An indication of any uncertainties in the compiled information.

The EIA Regulations include requirements for stakeholder engagement, including optional opportunities for engagement prior to EIA report submission and mandatory engagement activities following submission. Once the EIA report is submitted, the project proponent must make it available for review by the public. The planning authority will publish notice of the planning application by local advertisement and will allow the public to provide representations within four weeks. The notice will include details on where the EIA report can be viewed or obtained. The planning authority will also consult the district council and other authorities providing them at least four weeks to provide representations.

The requirements of the EU EIA Directives (Directive 2011/92/EU and 2014/52/EU) are implemented in the legislation in Northern Ireland as the legislation pre-dates Brexit. There are useful EIA guidelines that were published by the European Commission in 2017 [57]. It is unclear whether these are still considered to be valid references in Northern Ireland.

4.5.4. Mine Waste Management

The Planning (Management of Wastes from Extractives Industries) Regulations (Northern Ireland) 2015 (2015 MWR, SR 85) transposed the Directive 2006/21/EC ("the Mine Waste Directive"). The aim of the Mine Waste Directive is to prevent, or reduce as far as possible, the production of waste and the potential harmful effects of such waste on the environment and human health. Under 2015 MWR, the grant of planning permission is prohibited (by Regulation 4) without prior approval of the mine waste plan.

Information to be included in mine waste plans is defined in Regulations 6, 9, 12 and 13 of the 2015 MWR. For each waste stream, the plan must include details on:

- The source of the waste;
- Intentions to minimise, treat and/ or recover waste;

- Waste characterisation;
- Waste transport and deposition methods; and
- Potential impacts on environment and human health.

The plan must also include details on the design and management of the waste disposal facility and activities including:

- Detail of control and monitoring measures to secure the stability of the waste and to prevent pollution of air, soil, surface water and groundwater; and
- A plan for closure.

Where there are Category A facilities at a mine site they need to be clearly identified. Waste disposal facilities are classified as Category A facilities if: failure or incorrect operation could give rise to a major accident; any contained waste is classified as hazardous; and/ or it contains any substances classified as dangerous. Where Category A facilities are to be developed, the waste management plan must include a major accident prevention policy, safety management system and internal emergency plan.

The planning authority can only approve the waste management plan after it has been subject to public review. The planning authority is required to publish notice of the waste management plan by local advertisement and to allow the public to provide representations within two weeks. The notice will include details on where the mine waste management plan can be viewed or obtained. The authority must take comments made by the public into account when taking the decision whether to approve the plan.

Regulation 8 of the mine waste regulations requires that there is a financial guarantee in place for the closure of the mine waste facility prior to commencement of mine waste management operations. Regulations 9(1)(e) and (f) require that the planning authority ensures that there are suitable arrangements in place for closure of the facility and Regulation 15 requires the planning authority to determine the quantum and form of financial guarantee.

The 2015 MWR refers to several other EU Regulations, Directives and Decisions as listed below.

- Characterisation of waste is required in accordance with Commission Decision 2000/532/EC which is subsequently amended by 2014/955/EU (Commission Decision of 18 December 2014 amending Decision 2000/532/EC on the list of waste pursuant to Directive 2008/98/EC of the European Parliament and of the Council).
- Schedule 2 of the 2015/85 Regulations requires characterisation based on the hazardous characteristics of the waste. Annex III of Directive 2008/98/EC defines properties of waste which render it hazardous. This annex has been replaced by Commission Regulation 1357/2014 of 18 December 2014 and Council Regulation 2017/997 of 8 June 2017.
- Schedule 3 of the 2015/85 Regulations details the criteria for Category A facilities. Criterion include the implications of failure, waste materials classified as hazardous according to Directive 2008/98/EC (replaced by Commission Regulation 1357/2014 and Council Regulation 2017/997) and waste containing substances classified as dangerous according to Regulation (EC) No 1272/2008 of 16 December 2008 on classification, labelling and packaging of substances and mixtures.
- 2009/337/EC (Commission Decision of 20 April 2009 on the definition of the criteria for the classification of waste facilities in accordance with Annex III of Directive 2006/21/EC of the European Parliament and of the Council concerning the management of waste from extractive industries), is not explicitly mentioned in the 2015/85 Regulations but is referred to in terms of the waste facility classification.
- Design of the facilities to ensure that there is no water pollution and conformance with the Water Framework Directive and daughter directives. The regulations specifically refer to the Water Framework Directive (Directive 2000/60/EC of the European Parliament and of the Council establishing a framework for community action in the field of water policy), the Groundwater Directive (Directive 2006/118/EC of the European Parliament and of the Council of 12 December 2006 on the protection of groundwater against pollution and deterioration) and Directive 2006/11/EC (Directive 2006/11/EC of the European Parliament and of the Council of 15 February 2006 on pollution caused by certain dangerous substances discharged into the aquatic environment of the Community). Directive 2006/11/EC is no longer in force and the WFD (2000/60/EC) has been amended by Decision No.

2455/2001/EC establishing the list of priority substances and Directive 2008/105/EC on environmental quality standards in the field of water policy (subsequently amended by Directive 2013/39/EU).

Mine waste management guidance documents that should be used in conjunction with the 2015 WMR are listed below:

- European Commission Reference Document on Best Available Techniques for Management of Waste from the Extractive Industries, in accordance with Directive 2006/21/EC 2018 [16].
- Other EU guidance documents, which include:
- CEN/TR 16363:2012 Characterisation of Waste – Kinetic testing for assessing acid generation potential of sulphidic waste from extractive industries;
- CEN/TR 16376:2012 Characterisation of Waste – Overall guidance document for characterisation of waste from extractive industries;
- CEN/TS 14429:2005 Characterisation of waste – leaching behaviour tests – Influence of pH on leaching with initial acid/base addition;
- CEN/TS 14997:2006 Characterisation of waste – leaching behaviour tests – Influence of pH on leaching with continuous pH-control; and
- EN 12457 – 1/2/3/4 Characterisation of waste – leaching compliance tests of granular waste materials and sludges.

The following UK and international guidance documents:

- Guidance on the Classification and Assessment of Waste (1st Edition v1.1, May 2018). Technical Guidance WM3. Northern Ireland Environment Agency (NIEA).
- BRE Report "An engineering guide to seismic risk to dams in the UK";
- BRE "An engineering guide to the safety of embankment dams in the UK";
- BS5930/EN - BS 5930:2015 "Code of practice for ground investigations" came into effect on 31 July 2015 (supersedes BS 5930:1999+A2:2010, which is withdrawn);
- ICOLD Bulletins; and
- Eurocode 7: Geotechnical Design HSE Guidance.

Other Northern Ireland legislation that is directly relevant to waste management plans includes:

- The Quarries Regulations (Northern Ireland) 2006 (SR 205) – the geotechnical assessment for waste dumps must be in accordance with Part V of the regulations;
- The Mines Regulations (Northern Ireland) 2016 (SR 2015 No. 427);
- Water Environment (Floods Directive) Regulations (Northern Ireland) 2009 (SR 376);
- Reservoirs Act (Northern Ireland) 2015; It is noted that "A pond within an extractive waste site or waste facility" is not a Controlled Reservoir as defined under the Reservoir Act (Northern Ireland), 2015, refer to Section 5 (2) (f) of the Act.
- Health and safety legislation (https://www.hseni.gov.uk/topic/legislation).

4.5.5. Other Environmental Approvals

Additional approvals that are likely to be required for a mining development include:

- A Pollution Prevention & Control (PPC) permit;
- Consent to discharge water and alter watercourses;
- Water abstraction licences;
- A licence for the storage of explosives;
- An excavation licence for disturbance of cultural heritage sites and conservation of the information held within these sites;
- Derogation licences for relocation of identified protected species;
- A hazardous substance consent.

4.5.6. Influence of UK's EU Exit on Environmental Legislation

4.5.6.1. EU Regulations and EU Directives

The current environment and social statute book in Northern Ireland has been shaped by UK legislation and EU legislation. Prior to the UK's exit from the EU, Regulations generated by the EU were directly applicable in the form published by the Commission throughout the

European Community. EU Directives were required to be transposed into the UK statute book by the Treaties to the European Union. At present, the core of environment and social law across the EU and UK is similar.

For the UK to complete its exit from the EU in January 2020 and retain the environmental standards from EU regulations, Section 2 of the European Union (Withdrawal) Act 2018 (Amended 2020) required that any EU derived domestic legislation in force before exit day, continued to have effect on and after exit day. Additionally, Section 16 required that the environmental principles enshrined in the EU Treaties be replicated in UK law, including:

- The precautionary principle;
- The principle of preventative action to avert environmental damage;
- The polluter pays principle;
- The principle of sustainable development;
- The principle of integrating environmental requirements into policies and activities;
- Public access to information;
- Public participation in environmental decision making; and
- Access to justice in relation to environmental matters.

4.5.6.2. The Environment Bill 2021

At time of writing, the UK Parliament House of Lords is considering the Environment Bill. If enacted, the Bill will be used to implement the above environmental principles, and will perform two functions, firstly to require objectives be established for environmental improvement, and secondly to establish the Office for Environmental Protection (OEP), a body to oversee progress on the objectives. The Office will also fulfil some regulatory roles required following the UK's exit from the EU.

The aim of the Bill is to make provisions for:

- Targets, plans and policies for improving the natural environment;

- Statements and reports about environmental protection;
- The Office for Environmental Protection;
- Waste and resource efficiency;
- Air quality;
- Recall of products that fail to meet environmental standards;
- Water;
- Nature and biodiversity;
- Conservation of covenants;
- Regulation of chemicals; and
- Connected purposes.

The Bill also makes some amendments to other acts and regulations, including empowering devolved administrations such as the Northern Ireland Executive to implement producer responsibility schemes, extended powers to revoke water abstraction licenses, cementing an obligation to achieve net biodiversity gain across public authorities, and provisions enabling the continuation of chemicals registration legislation.

The Bill does not discuss climate change however, as similar requirements for plans, targets and reporting on targets are contained in the Climate Change Act 2008 (Section 1.4.2). The OEP also should not overlap with the CCC (Section 1.4.2) as they are equivalent bodies.

4.5.7. Considerations for Updates to Legislation and Planning Policy

The preceding sections (Sections 1.5.1 and 1.5.5) show that there is a substantial body of law and planning policy relevant to responsible mining development in Northern Ireland. Updates to the legislation and planning policy should consider the following:

- International trends and governance frameworks (Section 4.3);
- Climate policy (Section 4.4);
- The stages in the development and the life of a mine and the need for ESG in all stages (Section 4.6.1);

- The strong role of financiers in mining project development and their expectations (Sections 4.3.8, 4.4.8 and 4.6.2);
- Impact assessment practices (Section 4.6.3) and impact management weaknesses in the mining industry (Section 4.6.4);
- The new overt focus on human rights (Section 4.6.5);
- Stakeholder engagement good practices (Section 4.6.6);
- New disclosures on biodiversity impacts that might be required in future and the application of the mitigation hierarchy to limit biodiversity impacts (Section 4.6.7);
- Available good practice guidance on mine closure (Section 4.6.8);
- Perspectives on enhancing the positive impacts of mining (Section 4.6.9);
- Special features of mine water management (Section 4.6.10);
- New international standards on mine waste management and associated public disclosure requirements (Section 4.6.11);
- Options for decarbonisation of mines (Section 4.6.12);
- Guidance on good exploration practices (Section 4.6.13).

4.6. ESG Considerations for Updates to Minerals Governance Framework Northern Ireland

In addition to international policy, it is recommended that several matters specific to the mining industry are considered when revising minerals policy in Northern Ireland (Section 4.5.7). This section elaborates on these matters.

4.6.1. Stages in the Development and Life of a Mine

Minerals policy should take account of the stages in the development and the life of a mine and recognise that, to avoid and minimise negative environmental and social and enhance positive social impacts, ESG needs to be in place in all stages (Figure 4.10). Opportunities to avoid and mitigate negative impacts of mining are the greatest in the earliest stages of project planning (Figure 4.11).

Mining projects are conceptualised when exploration establishes that a mineral resource can be defined (scoping stage). The projects are then evaluated, by means of pre-feasibility and feasibility studies to define reserves (economically mineable parts of the resource). Decisions to finance mining projects and to bring projects into production are based on understanding of resources and reserves. Internationally recognised codes are applied when reporting publicly on resources and reserves (Section 4.3.12).

Construction and operation of mines can only commence after permissions for this have been obtained. The bulk of the approvals to be obtained relate to ESG. Usually an EIA is undertaken in the pre-feasibility and feasibility stages to provide input to the project design and permitting decisions. The potential to avoid adverse impacts is highest at the earliest stages of project planning when the mining project is conceptualised; many project alternatives are considered during conceptualisation.

Many decisions on alternatives are taken before the official EIA process commences, with the submission of a screening application to regulatory authorities. It is important that there is ESG input to those decisions and this is documented so that they can be revisited if necessary. During the EIA and at the permitting stage, after large investments in exploration and evaluation have already been made, stakeholders can be expected to ask questions about the decisions on alternatives. These can even extend to the deposits targeted for exploration and development; these days some stakeholders are unconvinced that deposits in highly sensitive settings should be developed and don't accept arguments that there is no alternative deposit that could have been considered.

Another depiction of stages in the life of a mine is given Figure 4.12. This figure prepared by the Ontario Mining Association [58] gives perspective on the financial investment in each stage of development and how long each stage takes. It also highlights that closure costs are substantial and can be highly variable.

Good practice mine planning starts with closure vision and builds closure into the mine design and the life-of-mine (LOM) plan, which defines activities in the construction, operational and decommissioning phases. Best practice envisages a positive legacy post closure.

The Ontario Mining Association's figure is presented on its website with a statement that provides a useful insight on the probability of bringing mines into production. It explains: "It takes a great deal of will, effort and time to discover a viable ore deposit and bring it into production. There is no way of predicting where profitable ore deposits will be found. Each prospector, explorationist and investor may fervently hope for the next 'big find' but only about one mineral exploration project in ten is taken to the drill stage, and one drill program in 1000 finds a viable mineral deposit; hence, less than one project in 10,000 becomes a mine. It typically takes ten to fifteen years of exploration, data analysis, planning and financing to bring a mine into production."

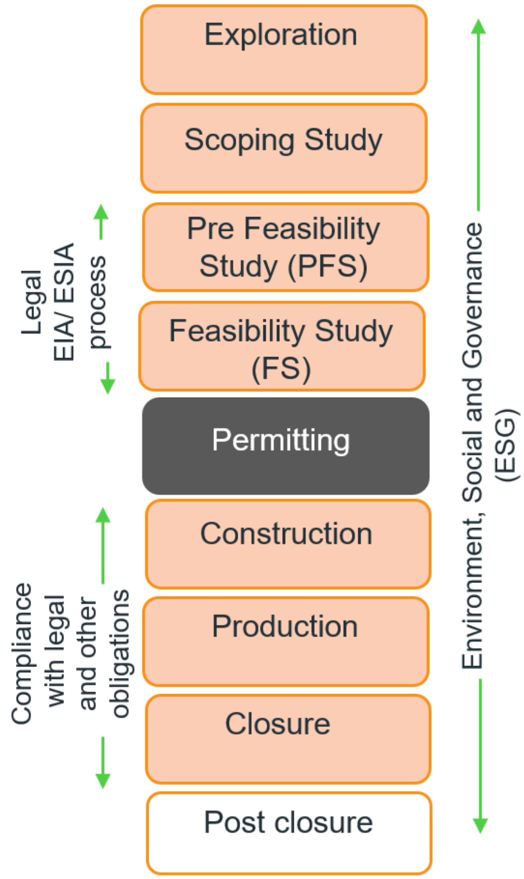

Figure 4.10 - Mining stages and ESG.

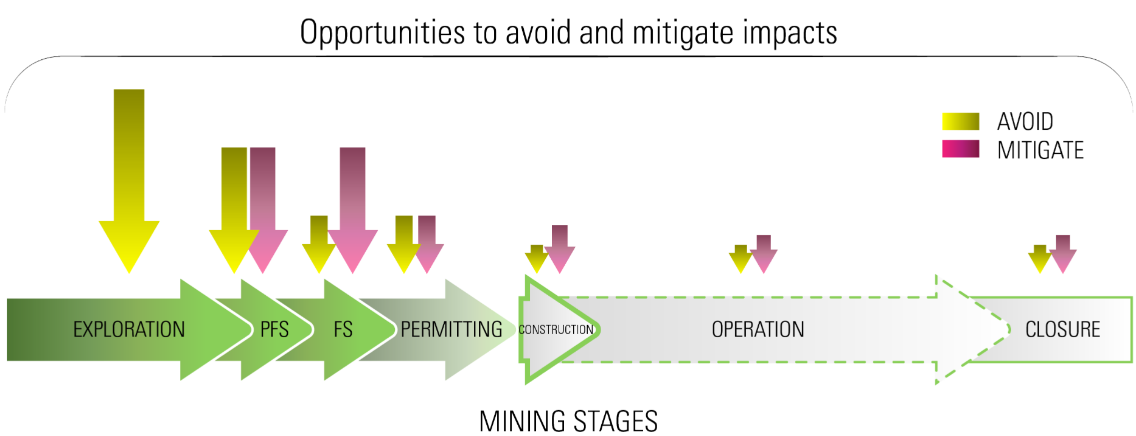

Figure 4.11 - Opportunities to avoid negative impacts are greatest in the mine project conception and evaluation.

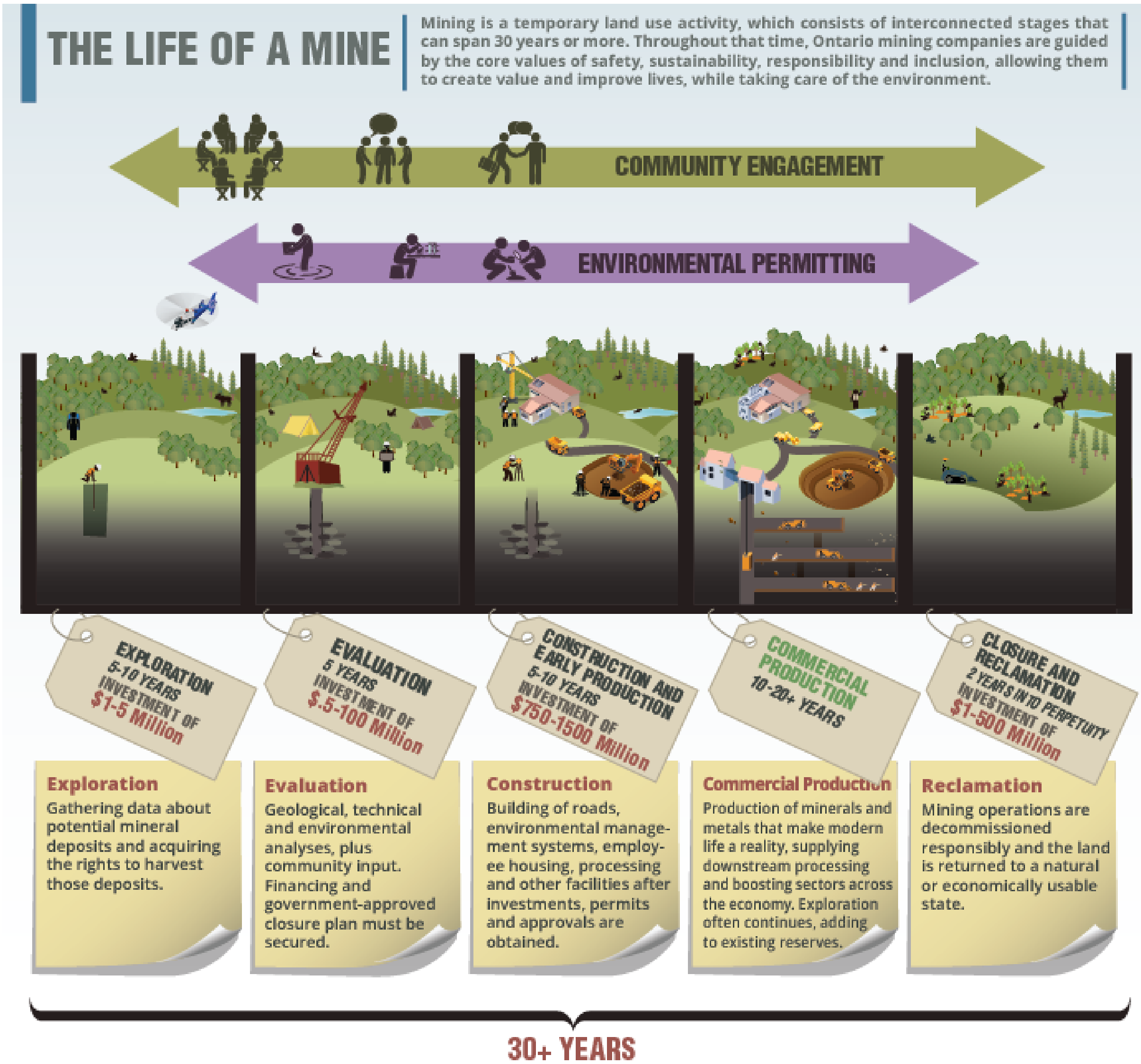

Figure 4.12 - Mining stages explained by the Ontario Mining Association

For very large mining developments, the costs of each stage of mining can be even greater than those given in Figure 4.10. A similar diagram in the Minerals Governance in the 21st Century report [5] states that exploration costs can be USD 1 million to USD 1.75 million per year, evaluation costs can be in the order of USD 5 million year in the scoping and PFS stages, while FS costs can be 2 to 5% of capital expenditure for the project. The same diagram suggests closure costs can be more than USD 1 billion for large mines. The source of the diagram is Canada's Queen University's Mine Design Project Wiki.

4.6.2. Evaluation of Mining Projects by Financiers

Normally project financiers (equity investors and lenders) commission a due diligence study when they are interested in large-scale financing of a mining project. An environmental and social review is undertaken as part of the due diligence. Until recently, the reviews were focused heavily on risks to the investor but now there is a trend to look at sustainable development outcomes too. The due diligence usually begins with a desktop study using information available in the public domain, as well as information made available by the developer. Where the financier is interested in providing a large loan for a project or taking a large stake in the mining company, the due diligence often includes site visits to verify the available information.

The environmental and social matters that financiers always have an interest in because they are a risk to the financier are:

- Matters that can stop a project going ahead such as inability to obtain permits on time or failure to obtain a social licence to operate;
- Matters that could stop operations, such as suspensions as a result of pollution incidents or failure to observe permit conditions or blockades by unhappy community members;
- Matters that can affect the value of the asset, such as cases management measures that are material to the value of the asset, examples are very high water treatment costs or a need to repair or replace mine waste disposal facilities; and
- Closure liabilities.

Increasingly, financiers are also interested in reputational risks, particularly when viewed through the human rights lens (Section 1.6.5). Modern financiers are also more and more interested in how mineral products fit in supply chains geared to decarbonisation and the ESG credentials of mineral products.

The topics that are typically covered by sustainability consultants when undertaking environmental and social reviews are:

- Environmental and social approvals for the project and existing operations – the status of approval processes, approvals obtained and compliance with conditions of approval;
- Environmental and social agreements for the project and existing operations – status of any agreements with government/ communities that have to be reached/ honoured;
- Management systems for environmental and social management and health and safety, including the status of the management system, including the adequacy and appropriateness of the impact assessment, management plans, monitoring and auditing systems and capacity to implement environmental and social management;
- Stakeholder engagement – how constructive relationships with stakeholders are established and maintained;
- Technical issues – environmental and social matters that present risks to the project/ operation;
- Closure plan/s and costs for the project and existing operations – adequacy of the plan/s and cost estimate/s, funding mechanism/s to be used to provide for closure.

4.6.3. Environmental and Social Impact Assessment

It is worth considering some features of impact assessment practice generally applied in the mining industry, particularly the metals sector, when revising Northern Ireland minerals policy.

4.6.3.1. General impact assessment practice in the metal mining industry

It is possible to generalise impact assessment practice in the metal mining industry across the globe because there is much commonality. The need to raise much capital for mining projects is a key force influencing this. International agencies, such as UN agencies, have supported many governments in amendment of mining law, to both help attract investment and promote sustainable development, and have influenced understanding of acceptable impact assessment practices at the same time. International standards that are applied by financial institutions in the evaluation of mining projects, such as the Equator

Principles and IFC Performance Standards, have also influenced internationally accepted good practice.

The term “environmental impact assessment (EIA)” is used in law in Northern Ireland and in many other countries, but international standards applying to the mining industry use the term “environmental and social impact assessment (ESIA)”. There is no difference between an EIA and an ESIA, but the term “ESIA” emphasises that impacts on people need attention in the impact assessment process.

An EIA/ ESIA is a process that produces an EIA/ ESIA report (also called an environmental statement in some countries). The report records the process and defines the impacts of the project. In most (if not all) countries, there is a legal requirement to undertake an EIA/ESIA for a mining development. The EIA/ ESIA report must be approved before other approvals for the project can be issued. In most countries, a licence to proceed with mining operations will not be issued until the EIA/ ESIA report has been approved and there is a record of the approval decision in place (a decision notice or certificate).

In Northern Ireland, and other parts of the UK, there is no such thing as EIA approval. Instead, the EIA process and EIA report inform the decision on planning permission (Section 4.5.3).

To align with international standards, a mine ESIA report must include a management plan that explains how the identified impacts will be managed. This is usually called an environmental and social management plan (ESMP). The ESMP can be one plan, an umbrella plan underpinned by many other plans or a package of plans. In the case of mining projects, the ESMP that complements an ESIA report is usually accompanied by a preliminary closure plan.

In many jurisdictions, the ESMP becomes legally binding when the ESIA is approved. Obligations to comply with the ESMP commitments are included in the conditions of the ESIA approval and exploitation licences.

In many jurisdictions, the application for an exploitation licence may require submission of additional environmental or social documentation over and above the ESIA and ESMP. The additional documentation can include plans to enhance socio-economic impacts (Section 1.6.9).

The EIA/ ESIA report is used for decision making as follows:

- It is used by regulatory authorities to determine whether they can grant environmental approval for the project;
- It is used by project financiers (equity investors and lenders) to understand the environmental and social risks associated with financing of the project.

As well as informing the above decision making, the report is usually also used to inform stakeholders of how the project may influence them so that they can provide meaningful feedback to these decision makers as part of the on-going ESIA process.

After an EIA/ ESIA report has been approved and the official approval to proceed with the project is obtained, the report is used as a base for secondary approvals such as: permits for water abstraction, use and discharge; permits for release of emissions; and permits for disturbance of archaeological sites.

Where there is physical displacement of people, the EIA/ ESIA is usually the base for the resettlement action plan that needs to be developed when a decision has been taken to proceed with the project.

4.6.3.2. How does the ESIA align with project planning?

Often the legally-required EIA/ ESIA process only commences during the feasibility stage. This is because there is a high level of information sharing on the project with government and other stakeholders required during the official EIA/ ESIA process. Developers are often only ready to undertake the level of information sharing required at the feasibility stage. For example, developers may be reluctant to share information on the preferred sites of infrastructure until they have secured options to purchase land. However, ESG input into project planning is required from project conceptualisation (Section 4.6.1).

An EIA/ ESIA should be undertaken concurrently with the pre-feasibility and feasibility stages of mine project planning. The EIA/ ESIA team and project engineers should work together so that impacts are avoided and/or mitigated in the project design.

The EIA/ ESIA team generally provides the project engineers with information on sensitive features in the setting of the project and design criteria. They also provide input to decisions on alternatives. Furthermore, through the stakeholder engagement exercises that are integral to the EIA/ESIA process, the ESIA facilitates communication between stakeholders (regulatory authorities, local communities and other interest groups) and the project engineers.

4.6.3.3. How long does it take to complete a mine ESIA?

As mentioned above, an EIA/ ESIA is usually on the critical path for project development. It normally takes 18 months or more to complete an EIA/ ESIA for a mine. One reason for this is that some baseline studies take at least a year to complete. Usually 12 months of data is required for water studies (air quality/climate, surface water hydrology, groundwater and water quality) and data for all seasons of the year is required for biodiversity studies.

4.6.3.4. How long does it take to obtain ESIA approval?

Globally, approval periods for ESIA approval range between 6 months and four years to obtain ESIA approval. The former is typically the case in less developed countries and countries like Russia and Kazakhstan. The latter is the case in more developed countries. There are examples in the United States and Europe where it has taken ten or more years to obtain key environmental and mining approvals. The Rosemont Project in Arizona in one example [59]. Long approval timeframes negatively impact on investor interest in financing projects.

4.6.3.5. How much does an ESIA cost?

A good quality EIA/ ESIA for a mine that defines project impacts and provides meaningful input in the project design often costs more than GBP 1 million. The cost depends on the complexity of the issues to be investigated. Studies that contribute substantially to the costs are geochemistry and water studies. The costs are higher when detailed water studies, especially groundwater or marine studies, and biodiversity studies are required. They will also be high where people will be physically or economically displaced by the project and detailed social studies are required.

4.6.3.6. Stakeholder engagement in the EIA/ ESIA process

Stakeholder engagement is an integral part of an impact assessment process, but engagement during this process is not sufficient to build constructive relationships with stakeholders. More background on good practice stakeholder engagement is provided in Section 4.6.6.

4.6.4. Impact Management Weaknesses

Impact management weaknesses that are often encountered in the mining industry globally summarised in Table 4.6. This table is based on SRK's experience of reviewing mines in many different jurisdictions and our general knowledge of the industry. Where possible, these should be addressed in government policy.

Table 4.6 - Impact management weaknesses often encountered in the mining industry

Manageme nt tools	**Weaknesses**
Impact assessment	• Attention to the ultimate receptors of impacts is often lacking, specialists will use technical standards and limit values to define impacts without seeing the people, biodiversity and ecosystems affected. • Impact receptors are often not clearly mapped and therefore are not properly considered in impact assessments – specifically, local communities and land uses are not mapped, downstream water users are not identified and mapped, and habitats are not delineated. • Impact receptors are not adequately engaged during project planning, impact assessments and the operational phase. • Positive impacts receive little attention during project planning and are not defined in terms of social and economic outcomes. • Specialist inputs into the impact assessment are not integrated. • Water and waste studies are limited, with geochemistry and groundwater studies being particularly weak (this could be related to the high costs of these studies). • Material environmental and social impacts are not technically defined – the impact assessments are too generic. • Cumulative impacts on people and biodiversity are not well defined.

	• Impact studies do not strongly influence the design of projects. • Impact studies do not directly influence management plans. • Impact studies do not link to management systems.
Project design	• Decisions on project alternatives are taken before the impact assessment commences and/or without adequate consideration of environmental and social matters. • Positive impacts receive little attention during project planning. • Not enough effort is invested in designing out/ avoiding negative impacts.
Impact managemen t plans	• The plans are generic and superficial so there is a weak basis for implementation of effective management, as well as setting conditions of regulatory authority approvals, ongoing monitoring, audits and enforcement. • The plans are not recognised as commitments made to stakeholder in permitting processes and are not implemented because they are forgotten or replaced with updated plans. • Management plans do not link to management systems.
Monitoring of impacts	• Monitoring programmes are disjointed, they are not adequately connected to original baseline data collection programmes and monitoring points are changed too frequently during the life of mine, reducing the usefulness of the data for tracking of trends. • Positive impacts are not measured and monitored in terms of social and economic outcomes.
Tracking of compliance	• Commitments made during project planning, the EIA and permitting processes, and other stakeholder engagement processes are often not rigorously recorded and/or translated into compliance obligations. • The numerous compliance obligations incurred in permitting processes, as conditions of approvals or actions in legally binding management plans, are not systematically tracked and implemented.
Planning for closure	• Closure is not integrated into mine planning. • Closure plans are superficial. • Closure cost estimates are too low and financial provisions made for closure are negligible compared to the real closure costs.

4.6.5. Human Impacts and the Human Rights Lens

There is an increased trend to look at impacts of people through a human rights lens. This has been given impetus by the United Nations Guiding Principles on Business and Human Rights. Impacts (UNGPs), which were endorsed in June 2011 by the UN Human Rights Council. The UNGPs have since influenced many of the international standards applying to the mining industry. New standards incorporate these principles and older standards have been updated to ensure there is explicit reference to the UNGPs.

The mining industry's business stakeholders, particularly institutional investors, generally require that the UNGPs are applied. In addition, the requirement to observe the UNGPs is built into the EU Taxonomy Regulation (which is mentioned in Sections 4.3.8, 4.3.10 and 4.4.5).

Mines can impact human rights in the planning of projects and in management of operations. They can do this through their impacts listed in Section 4.2.1 (Table 4.1). The UN-backed PRI has produced a document that explains investor's interest in mining and human rights called "Digging Deeper: Human Rights and the Extractives Sector" [60]. This gives numerous examples of ways in which mines could impact human rights. The examples are not limited to environmental impacts and land acquisition issues, they extend to employee and labour relations, supplier and business partner engagement, provision of security, artisanal mining, impacts on indigenous people, human trafficking and sexual exploitation and use mining revenues being used to fund conflict and corruption.

The UNGPs identify what governments and businesses need to do to embed respect for human rights in a business context. The UNGPs are based around a 'Protect, Respect and Remedy' framework as shown in Figure 4.13. Pillar 2 of the UNGPs requires businesses to know how negative impacts linked to their business activities and relationships might infringe upon the rights of people. Businesses must take concrete steps to improve these. At the core of Pillar 2, is the implementation of human rights due diligence. This is the continuous process of assessing, addressing, tracking and communicating the negative impacts on people.

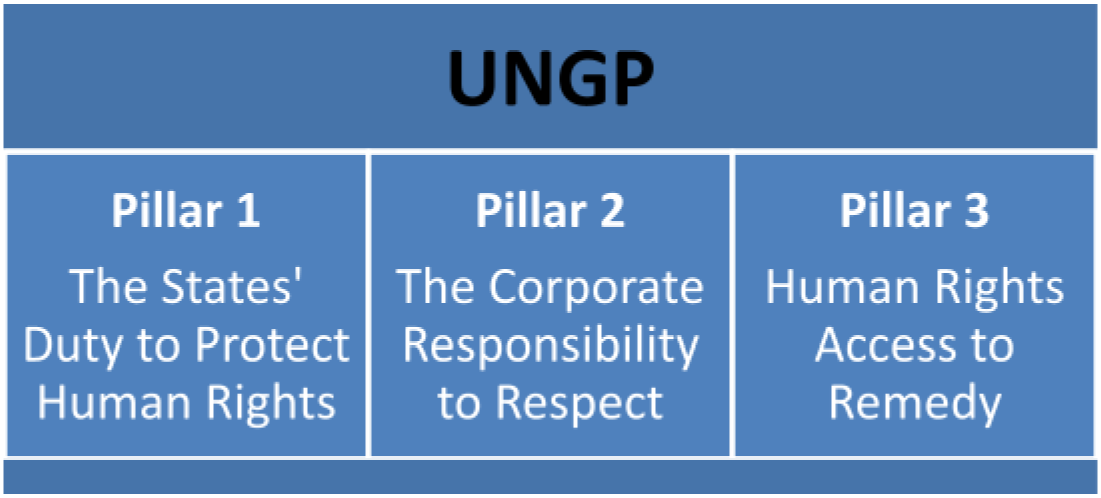

Figure 4.13 - UN Guiding Principles on Business and Human Rights

Companies in the UK are not legally required to undertake human rights due diligence. However, this is promoted by a UK guideline published in 2016 [61]. The guideline is called the “Business and human rights: A five-step guide for company boards.” It was produced by the UK Equality and Human Rights Commission. The guide is not applicable in Northern Ireland; it is applicable in England, Wales and Scotland. The guide explains “Human rights due diligence focuses on risks to people, not risks to the business. It is an ongoing process through which a company understands when, where and how it could impact on human rights, prioritises these risks for action, takes steps to address them, tracks the effectiveness of its efforts, and communicates with internal and external stakeholders.”

The European Commission has committed to tabling an EU-wide human rights due diligence law in 2021 [62]. Furthermore, the minimum safeguards in the EU Taxonomy Regulation (Sections 4.3.8, 4.3.10, and 4.4.5) pertain to human rights. Article 18 of this regulation requires an economic activity to comply with minimum safeguards that relate to the UNGPs and OECD Guidelines for Multinational Enterprises, as well as the core international human rights treaties and International Labour Organisation instruments referred to in the UNGPs and OECD Guidelines [63].

The UK Government’s guidance on its involvement in international projects includes a useful summary of human rights initiatives that should be applied to projects and how these fit together [64]. It also depicts practical tools to assist with management and due diligence of project-related human rights risks and impacts for business and state actors. These include: the International Labour Organisation (ILO) standards and core conventions; the OECD Common Approaches; the Equator Principles; and the IFC Performance Standards.

The above-mentioned UK guidance on international projects also highlights the fact that human rights are embedded in the IFC Performance Standards, stating: "The IFC Performance Standards address the majority of rights recognised under the International Bill of Human Rights, with a focus on those most relevant at a project level." The guidance maps project-related human rights against each of the IFC Performance Standards.

Basic actions that companies need to take to conform to the UNGPs are:

- Issue a policy statement from senior management committing to the responsibility to respect human rights;
- Reflect that commitment in their operational policies and procedures;
- Establish human rights due diligence processes to identify, prevent, mitigate and account for how they address their adverse human rights impacts, and integrate the results of such assessments into their operations;
- Communicate how they address human rights impacts;

Act to remedy adverse impacts, including by establishing or participating in effective operational level grievance mechanisms, and track the effectiveness of such actions. An example of a mining industry standard that addresses human rights is the ICMM Principle 3 on Human Rights [65]. The topics covered by this standard are as follows:

- 3.1: Respect human rights;
- 3.2: Avoid involuntary resettlement;
- 3.3: Manage security while protecting human rights;
- 3.4: Respect the rights of workers;
- 3.5: Provide fair pay and working hours;
- 3.6: Respect indigenous peoples;
- 3.7: Work to obtain free, prior and informed consent; and

- 3.8: Promote workplace diversity.

4.6.6. Stakeholder Engagement

Effective engagement is essential for building constructive relationships with stakeholders and for addressing human rights risks. Most mining companies have an ongoing programme of stakeholder engagement that extends to all stakeholders, including employees, business stakeholders, government stakeholders and local communities. Effective stakeholder engagement requires involvement of top-level management. Relationships of trust between mining companies and their stakeholders grow when concerns raised are addressed and there is follow through on commitments made.

Usually engagement with local communities is undertaken by a team of community liaison officers, who are often appointed from the local communities, under the supervision of an appropriately qualified stakeholder engagement manager. Where there is a high potential for opposition to the project or unrealistically high expectations of the project, a substantial investment in stakeholder engagement is required.

A leading standard on engagement with local communities is IFC Performance Standard 1, which relates to impact and risk management. The standard requires:

- Careful planning of engagement based on a good understanding of the communities, and including measures to allow effective participation of disadvantaged or vulnerable people;
- Disclosure of information ahead of engagements. The information needs to be relevant, transparent, objective, meaningful and easily accessible information. It also needs to be culturally appropriate, in local language(s) and understandable to affected communities;
- The consultation needs to be a two-way process that is free of external manipulation, interference or coercion and intimidation; enables meaningful participation; and is documented;

- A procedure for external communications, which should include sustainability reporting and regular reporting to local communities on the project/ operation and how their concerns are being addressed;
- A grievance mechanism to receive and facilitate resolution of affected communities concerns This mechanism should be culturally appropriate and readily accessible at no cost and without retribution. It must not impede access to judicial or administrative remedies. Communities should be informed about the mechanism as part of the stakeholder engagement process.

4.6.7. Biodiversity and Ecosystem Impacts

In the UK, including Northern Ireland, law and national biodiversity strategies and action plans establish detailed frameworks for protection of biodiversity. Valuable and threatened species, habitats and ecosystems are identified and EIAs are undertaken to inform project design and the determination of planning approvals. While law and practice do provide for protection of biodiversity in the planning of new developments in the UK, this is not the case in many parts of the world.

Impacts on biodiversity affect a range of ecosystem services, which translate into implications for human well-being (livelihoods, safety, security and health). The notion of managing ecosystems impacts is not well developed in law in the UK or elsewhere yet.

From 2023 onwards, mining companies are likely to be required by financial institutions to make public disclosures on impacts on nature in accordance with a new framework. This is to be developed by the Taskforce on Nature-related Financial Disclosures (TNFD) that was launched June 2021. The TNFD will complement the TCFD, which is popular amongst corporates and financial markets in outlining the economic risks associated with the climate crisis. The TNFD aims to guide companies to prepare a complete picture of environmental risks, associated disclosure and action, and a shift in global financial flows away from nature-negative outcomes and toward nature-positive outcomes.

The extent of land used up by mining activities, current and past, is small when compared to other human footprints, such as agriculture. In the EU, current and past mining sites are part of the “Artificial land” category of the land cover census published by Eurostat. In 2015, artificial land, of which mining is only a small part, represented 4.4 % of the EU land cover, as compared to 21.5 % for croplands [5].

Detailed study is required to define the impacts of a specific mine on biodiversity. The impacts at each site are unique; they depend on the area and type of biodiversity and the type, scale and extent of mining. Opencast, underground and alluvial mining are examples of different types of mining that have different levels of impact on biodiversity.

Mine impacts on biodiversity can largely be attributed to land take, vegetation clearing, alteration of landforms and drainage, and water pollution. Emissions from modern mines generally do not have biodiversity impacts, unless poorly managed. However, dust blowing from mine waste facilities and entrained by vehicles on unsurfaced roads can be an issue. Biodiversity impacts are generally greatest when mines are developed in relatively undisturbed areas and/or the biodiversity impact receptors are of high conservation importance. The impacts can be exacerbated if mining opens access to remote locations and causes a population influx.

Many mineral deposits are in high diversity zones. Increased demand for minerals and the depletion of easily accessible reserves are pushing exploration and mining into previously inaccessible and/or fragile areas.

With sensitive and appropriate design, the adverse impact of mining developments on biodiversity can be limited. Implementation of the mitigation hierarchy is key to avoiding and minimising biodiversity impacts (Fig. 4.14). For new projects and extensions to existing projects, the hierarchy should be implemented with the ambition of no net loss. These days efforts to avoid sensitive habitats have increased correlating with a decline in acceptance of economic justifications for significant biodiversity impacts.

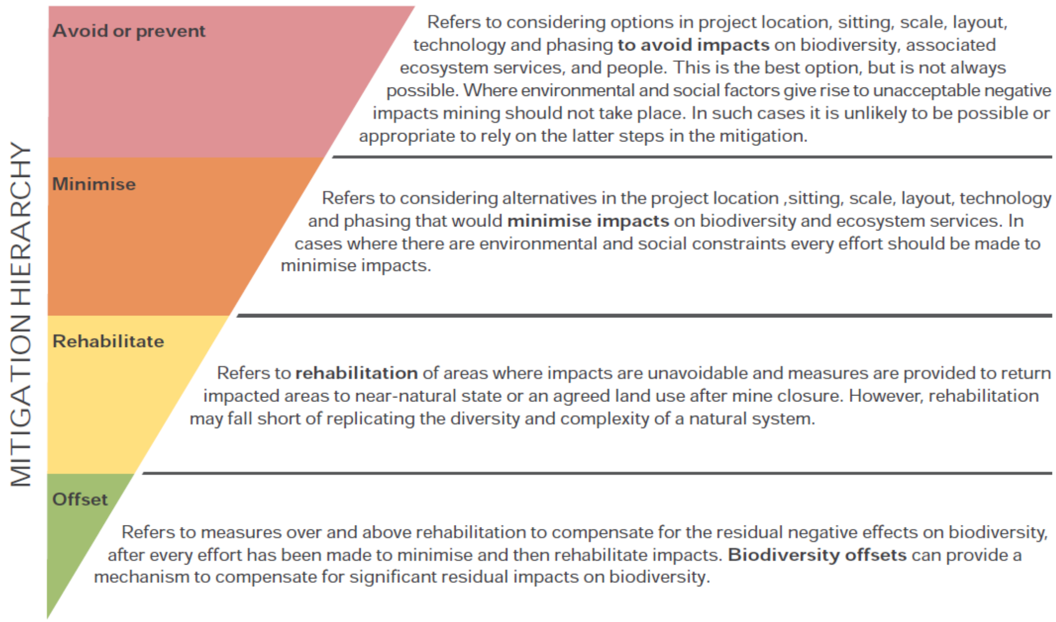

Figure 4.14 - Application of the mitigation hierarchy to mining projects. Source: Departments of Environmental Affairs and Mineral Resources, South Africa (2013) [66].

In countries where frameworks for protection of biodiversity are lacking, financiers often expect application of the International Finance Corporation (IFC) Performance Standard 6 on Biodiversity Conservation and Sustainable Management of Living Natural Resources (2012). The standard requires projects to achieve no net loss of biodiversity in areas of natural habitat, and a net gain in areas of critical habitat through adoption of the mitigation hierarchy.

Good practice guidance on mining and biodiversity is available from ICCM [67] and the Mining Association of Canada's Towards Sustainable Mining initiative [68].

4.6.8. Mine Closure

Although closure and post-closure occur at the end of a mine's productive life, the best closure outcomes are planned well before any mining begins. Ideally mine planning should start with closure vision and build closure into the mine design and the life-of-mine (LOM) plan, which defines activities in the construction, operational and decommissioning phases. Best practice also envisages a positive legacy post closure.

The closure stage of mining can be subdivided into rehabilitation during the operational phase, decommissioning and then reaching agreement that closure is complete, followed by post-closure. Decommissioning and closure typically lasts for between two to five years depending on the size of the mine. Post-closure has no end point and can extend in perpetuity.

The UK has a great number of mines that could be classed as being in post-closure, including coal fields in the north east of England and south Wales, metal mines in Cornwall and across mid-Wales, and clay and slate mines in Wales and the south west of England.

If mine closure is not considered at the design of the mine, the impact of the mine closure can be very costly. Historically, fixing this fell to the public purse. This is true in the UK today. This problem is reflected in a recent Welsh Government consultation paper that concerns over 2000 disused coal tips in Wales and the effects of climate change on the stability of the tips [69]. Some coal-tip landslips occurred in Wales in February 2020 following Storms Ciara and Dennis. These structural failures heightened awareness of the potential risks that disused coal tips present to communities and to the environment. In response, the Welsh Government has established a Coal Tip Safety Task Force to deliver a programme of work to address the safety of coal tips in Wales. The programme involves responding to immediate safety concerns and developing a new long-term policy approach to the legacy of disused coal tips. There is a Law Commission project on a new regulatory framework for coal tips in Wales that complements this work. The consultation paper is specifically about the regulatory framework. It points out the gap in legislation applying to disused coal tips and the complicated ownership of the tips, many are on privately owned land.

Some mines close when they are no longer profitable and their operators are strapped for funds and facing a variety of other challenges [5]. Many mining jurisdictions now have detailed legal requirements on closure provisions that must be established based on engineering-driven closure cost analysis. The provisions Nevada, United States, is recognised as leading authority on closure cost estimation [70].

The EU has recently published a useful guide on mine closure planning, cost estimation, financial provision for closure [71]. The guide includes unit rates for items in closure cost estimates. The guide focuses on environmental impacts, it does not consider social impacts. It was developed by consultants based in Germany.

Within the Northern Ireland framework, there is a requirement to have a financial guarantee in place for the closure of the mine waste facility prior to commencement of mine waste management operations under the Planning (Management of Waste from Extractive Industries) Regulations (Northern Ireland) 2015. This is only applicable to mine waste facilities. Northern Ireland Planning Policy requires that applications for mineral extraction include satisfactory restoration proposals and that planning permission includes conditions for their implementation (SPPS Paragraphs 6.161 and 6.166). The policy does not refer to financial provisions that need to be made for mine closure.

There are different types of mine closure cost estimates used by various mining industry stakeholders for different purposes (Figure 4.15). This can be confusing. The ICMM has produced a useful guide on these. The guide aims to enhance understanding of the various types of estimates, enabling consistent understanding and communication across the industry, between industry disciplines and with external stakeholders. Without this understanding, it is difficult to compare cost estimates produced by mines. The document defines and contextualises the key concepts related to closure costing, accounting and reporting requirements and the purpose of each.

- Examples of negative environmental impacts covered in closure plans are:
- Acid rock drainage from mine-waste facilities contaminating watercourses;

- Failures of mine-waste facilities resulting in releases of waste and water to the environment and endangering health and safety; and
- Exposed shafts or unsafe pits into which people and animals can fall.

If done well however, the environmental liability of a mine can be greatly reduced during rehabilitation, decommissioning and closure of mines. Environmental impacts during these phases may include short term noise, light and dust nuisance from earthworks, which can be managed using water sprays and protocols for not operating during unsociable times of day.

Mine closure plans need to take account of climate-change predictions and there is an increasing expectation that the plans account for impacts and changes in the stability of the facilities remaining post closure that could arise centuries later. Most engineering standards for storm events and seismic events include return periods of 100 years or less. In theory, post-closure is forever. In Nevada, United States, the storm event that must be considered for closure was reportedly increased to the 500-year/ 24-hour event. New mine closure legislation in Kazakhstan says that the design service life of structures that will remain post closure should be about a thousand years.

Socio-economic transitioning at the end of a mine life is also a critical part of closure. All mines have a finite life span, and an operator's responsibility to ensure the impacts of closure on the social setting are minimised extends beyond the workforce that will not be employed in mining after closure. Modern expectations of mines are that they will start work on ensuring positive outcomes for project affected persons years in advance of closure, be that preparing some of the workforce to move to be employed at another site or to up-skill the people that will remain in the area in other professions, such as agriculture or engineering. Within the UK there are also many examples of closed mines being used for tourism, such as the Eden Project in Cornwall (previously a clay mine), coal museums in the South Wales Valleys, or extreme sports centres such as in north-Wales' historical slate mines.

Key mine closure cost estimate types – high level summary

Closure cost estimate types	LoA (or LoM) cost estimate	Financial liability cost estimate	Sudden closure cost estimate	Regulator cost estimate (financial assurance)
	Costs that the operator expects to incur in the context of the current mine plan at the end of the mine life	Estimated liability based on applicable accounting requirements	Cost to close the operation in its current state	Costs that form the basis of a guarantee provided to a regulatory body
Closure and rehabilitation earthworks	√	√	√	√
Long term water management costs	√	X*	X*	X*
Decommissioning, decontamination and demolition	√	√	√	√
Project (owners) management costs	√	√	√	√
Post closure monitoring and maintenance costs	√	√	√	√
Socio-economic costs	√	X	√	X
Employee retrenchment costs	√	X	√	X
Land holding costs	√	X	√	X
Contingency	√	X	√	X

* Included in the case of a constructive obligation

Figure 4.15 - High-level comparison between the different types of mine closure cost estimates (Source ICMM, 2019)

The IRP 2020 has several recommendations on mine closure for governments. These include:

- The mine closure plan should be submitted to the government at the feasibility stage and must include plans for decommissioning and closure of each component of the mining area with cost estimates.
- An appropriate funding mechanism is essential to ensure sufficient funds are available for mine closure activities.
- The plan must be reviewed periodically.
- There should be continual testing of assumptions and recommendations to match evolving social, economic and environmental conditions and expectations.

- It is advisable to have an independent mine closure law that establishes a single agency to implement the law. This model gives the business community assurance that one agency will take the lead on closure problems and will not have to answer to many differing opinions on how closure implementation and success will be measured. The model also allows the public to go to a single space for information on mining.
- Legislation governing mine closure should be modernized.
- Appropriate technological alternatives for closure should be considered.
- Governments should consider the interest and opinions of civil society, especially those communities directly affected by mining enterprises.
- The experiences of countries with well-developed closure policies should be considered.
- Policies should be designed to encourage or provide incentives for technological innovation in mine closure, to reduce costs of compliance (economic incentives tend to provide greater incentives for innovation than technology or performance standards).

The above recommendations are interesting considering that the new regulatory framework for coal tips in Wales that was recently proposed identifies the need for a single supervisory authority with a duty to supervise the management of all disused tips.

There is very little guidance available on how to deal with social impacts at closure and to preserve a mine's positive legacy post closure. The ICMM is one good source of guidance.

The ICMM's suite of closure guidance for mining companies [72] is widely recognised, it includes training materials and the following documents:

- ICMM's 'Integrated Mine Closure: Good practice guide' into training materials (2020);
- ICMM Closure Maturity Framework (2020);
- ICCM Key Performance Indicators: Tool for closure (2020); and
- ICCM Financial Concepts for Mine Closure (2019).

4.6.9. Enhancing Positive Impacts

4.6.9.1. Opportunities

The mining sector has potential to play a positive role in promoting broad-based development and support the SDGs by stimulating the larger economy and thus driving economic transformation. The path to prosperity of many countries has been largely based on the exploitation of extractive resources and is being repeated in some developing countries. Botswana is an example of the latter case, having judiciously used diamond resources to promote broad-based development [5].

In addition to generating vast government revenue through taxes, royalties and other levies, mines can foster the emergence of small and medium-scale enterprises that supply goods and services to the industry. They can become development nuclei in remote regions with limited economic options and can also open access to modern infrastructure and leverage it to support a wider range of development objectives and boost productivity in other sectors. Furthermore, mines can facilitate the transfer of technologies and know-how, strengthening local human capital formation, which is the key to structural transformation [5].

The potential of mines to promote broad-based development and structural transformation of economies is not routinely realized in mineral-rich developing countries. The primary reason is that mines do not develop strong linkages to the local economy. Another reason is that national administrations are often not equipped to deal with the unevenly distributed and finite nature of mineral deposits, the volatility of commodity prices, and large and volatile inflows of foreign capital. This is exacerbated by information asymmetries and technical complexities of large-scale mining, and weak agreements negotiated with large multinational companies. Mining can cause conflict and lead to political dynamics that compromise governance where the mineral-derived value and benefits are in question.

Governance frameworks aiming to strengthen linkages of mining to the local economy promote local content and local participation, especially in the provision of goods and services. This depends on having the right conditions for developing competitive industries that will turn minerals and metals into a wider range of added value goods and services. Among these conditions are a well-trained and diversified workforce.

In many mining operations, procurement in the form of operating and capital expenditures constitutes between 50 and 65% of the production value of mining. The percentage of operational expenditure that goes to local suppliers of goods can range from 100% to very little due to the variable national capacity to supply mining industries with the wide range of skilled human resources, services and goods required by the industry. The same can be said for capital expenditure [5].

Equipment needed for mining, ore processing and/or metallurgical plants is mostly provided by a limited number of suppliers from developed countries. Modern equipment needed to ensure resource-efficient and economically competitive operations tends to be technologically complex. There are only a few suppliers providing specific equipment such as haulage trucks, loaders or drilling rigs and machines. These companies have the capacity to provide very rapid worldwide assistance to their clients.

4.6.9.2. Related initiatives: African Mining Vision and minerals law in Africa

The African Mining Vision [73] has been lauded as a policy option for broad-based sustainable development. Key features of the AMV is that it has a core emphasis on creating local value and maximizing the benefits that can be derived from mineral exploitation. Interventions promoted by the African Mining Vision are:

- Improving the quality of geological data to promote fairer deals and more equitable returns on mineral sector investments. This includes building capacity for geological mapping and comprehensive geoscientific databases, including mining cadastral systems and self-adjusting, progressive tax regimes that change with increasing profitability (or the rate of return) and allow the state to earn windfall rents during resource booms.

- Improving the capacity to capture a greater share of mineral rents and manage mineral wealth. This requires better minerals contract negotiation and flexible fiscal regimes that are sensitive to commodity price movements. Ownership, with direct or through joint ventures with the private sector in mining projects, is encouraged. Establishment of sovereign wealth funds to manage windfall revenues, provide infrastructure and set aside income for use by future generations is also encouraged. Minerals contracts should promote: local resource supplies; local processing through beneficiation; investing in training, research and development; and complying with safeguards on good governance, transparency, environmental stewardship, health and safety and affirmative action employment policies for nationals.
- Improving the capacity for mineral sector governance and ensuring compliance with regional and global mineral sector governance frameworks.
- Addressing Africa's infrastructure constraints as infrastructure gaps are the biggest constraint to mineral-based development in Africa.
- Elevating artisanal and small-scale mining as livelihoods of millions of people based on this and it is in the shadow economy, being poorly regulated, beyond the reach of markets, and not integrated into national policies and institutions.

A criticism of the African Mining Vision is that it has not been extensively implemented. Some say it will remain a little more than a vision without sweeping changes made by host governments. However, it is accepted that the vision has had a positive influence on the legislation of many countries.

Influenced by the African Mining Vision, and precursors to this, many African countries have provisions in minerals law that promote gains in human and social capital from exploitation of mineral wealth. The provisions include requirements for each exploitation permit holder to contribute to a research and training fund, a rehabilitation and closure fund, and a local development fund. The research and training fund is generally for financing geological and mining research and supporting training in the earth sciences. In some countries, the local development fund is financed by both the state (from royalties) and exploitation permit holders, administered by the government and monitored by local communities.

In addition to the above, the minerals law often requires that a feasibility study for an exploitation permit application must define a plan for anchoring the activity of the mining company to the local and national economy. This must indicate the economic links upstream and downstream with businesses and economic agents as well as spillover effects.

Furthermore, the minerals law promotes preferential local employment and procurement. Both holders of mineral rights and their suppliers (service providers and subcontractors) must preferentially employ local people and have local people acting in executive roles. Applications for exploitation permits must include plans to support the progression of local managers and staff and to progressively replace expatriates.

The minerals law of many African countries also has overt requirements to respect human rights, engage local communities and deliver fair compensation prior to land disturbance by mining. Exploitation permits cannot be issued without environmental approvals and audits of compliance with permit and environmental-approval conditions are required.

4.6.9.3. Related initiatives: EITI in Africa

Many mineral-rich African countries are members of the EITI and are classified as making meaningful progress towards meeting the EITI Standard by the EITI Board, based on information from regular validation processes [74]. These EITI implementing countries are also progressively making the information required by the EITI Standard available through government and corporate reporting systems (such as databases, websites, annual progress reports, portals), rather than relying on the EITI Report, to bring about transparency. The information being disclosed in the EITI reports for African countries extends beyond information on taxes paid and received by governments, to information on local procurement, local employment, gender diversity of the workforce and investments in community development (including both voluntary and mandatory contributions). Examples are the recent EITI reports for Senegal and Burkina Faso, dated November 2020 and February 2021, respectively.

4.6.9.4. Related initiatives: Mining Local Procurement Reporting Mechanism

Recognising that procurement of goods and services is often the single largest in-country payment type by a mine site, a Mining Local Procurement Reporting Mechanism has recently been proposed to standardise how the global mining industry and host countries measure and talk about local procurement. This mechanism is an initiative of Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH and the Mining Shared Value (MSV) unit of Engineers Without Borders Canada [75]. The mechanism helps mine sites report on local procurement to:

- Improve internal management in mining companies to create more benefits for host countries and to strengthen their social licence to operate.
- Empower suppliers, host governments, and other stakeholders with practical information that helps them to collaborate with mine sites.
- Increase transparency in the procurement process to deter problematic practices such as corruption.

4.6.9.5. Related initiatives; the Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development (IGF)

Leveraging mining for sustainable development to ensure negative impacts are limited and financial benefits are shared is also the focus of the Intergovernmental Forum on Mining, Minerals, Metals and Sustainable Development (IGF). It is a voluntary partnership between countries and any member state of the United Nations may become a member. Currently 79 nations are members, including the UK [76]. The International Institute for Sustainable Development has served as Secretariat for the IGF since October 2015. Core funding is provided by the Government of Canada.

The direction of the IGF is set out in a Mineral Policy Framework [77]. This framework guides members on measures to take in their jurisdictions to ensure mining activities contribute to sustainable development and poverty reduction. The framework covers the following subjects: legal and policy environment; financial benefit optimization; socioeconomic benefit optimization; environmental management; post-mining transition; and artisanal and small-scale mining.

The IGF provides much guidance on the above-mentioned subjects and offers services to its members in the form of assessments and training and capacity building. The assessments involve review of the country's existing mining laws, policies and regulations. The capacity building aims to support with enhancing identified strengths and addressing weaknesses and gaps.

4.6.10. Water Stewardship

The UK has good river basin management practices in place and water use is strictly regulated. However, it is worth considering the notion of water stewardship as it applies to mining further. While many mines do practice water stewardship, past poor water management practices have tarnished the mining industry's reputation and poor practices still prevail in many jurisdictions across the globe.

4.6.10.1. Water Stewardship

The notion of water stewardship extends beyond a mine site to the catchments and river basin occupied by the mine infrastructure and affected by the mine's activities. Ideal mine water stewardship extends beyond preventing significant impacts on water availability and quality to positive participation in river basin management, including understanding the dynamics in the basin and the interests of other water users. Many mining companies are recognising that proactive and holistic water management strategies reduce water-related risk and create opportunities to have positive impacts. In addition, disclosure on water stewardship practices builds trust through improved transparency.

4.6.10.2. Water availability issues

Mines can impact on water availability indirectly through impacts on water quality or directly through abstraction and alteration of flow regimes on surface or underground. Dewatering of mine workings can impact yields in surrounding boreholes.

4.6.10.3. Water quality issues

Of all the negative impacts that mines can have, impacts on the quality of water resources are generally the impacts in need of most attention. Not only is it possible for mine to have far-reaching impacts on water resources and users of these resources, these impacts present significant risks to business stakeholders, such as lenders and shareholders, and government. The risks take the form of high water-treatment costs, fines, enforced suspensions of operations, and high closure liabilities.

Historically regulatory authorities have been hesitant to enforce suspensions of operations, but numerous suspensions have occurred across the globe in the last decade. Suspensions for a few weeks or months can be financially crippling to mine operators. Known suspensions have been due to different causes ranging from sedimentation of watercourses, through spills from polluted water-holding facilities, to ongoing releases of water unfit for discharge.

Water resources have also been extensively polluted when mine-waste facilities fail. The worst water pollution incident was probably the failure of the Fundão tailings-storage facility at the Samarco mine (owned by Vale and BHP), Mariana, Minas Gerais, Brazil, on 5 November 2015. Over 650 km of waterways were affected by the failure [78].

Abandoned mines are a serious source of water pollution in the UK, according to reports published by the Environmental Agency (2008 and 2019 [79]). The polluting mines include coal mines and metal mines. Thousands of mines operational in the eighteenth and nineteenth centuries were abandoned, long before operators became liable for the long-term environmental impacts of their activities. Some of the mines are ancient dating to the Bronze Age. In 2008, it was estimated that 9% of rivers in England and Wales and 2% of the rivers in Scotland were at risk of pollution from these sites.

The main sources of water pollution from abandoned mines in the UK are illustrated in the figure below Figure 4.16. The Environmental Agency (2019) [79] explains that the pollution from abandoned mines comes from:

- Point sources where drainage tunnels or mine entries discharge metal rich mine waters all year round. These cause the most severe river pollution, particularly at lower river flows when there's less dilution.
- Diffuse sources such as mine wastes, metal-contaminated sediments and groundwater contribute increasing amounts of metals as river flows increase, for example after heavy rainfall.
- Impacts on groundwater and aquifers. Where mines were sunk through aquifers, groundwater flowed down from the aquifers into the mine workings below. After a mine closed, the pumps used to control groundwater levels were switched off and the mine workings flooded. As groundwater levels recover within the mines, contaminated mine water can rise into the overlying aquifers, especially where connections like mine shafts were built for access to the mine workings. This type of pollution is generally irreversible and can cover a very large area.

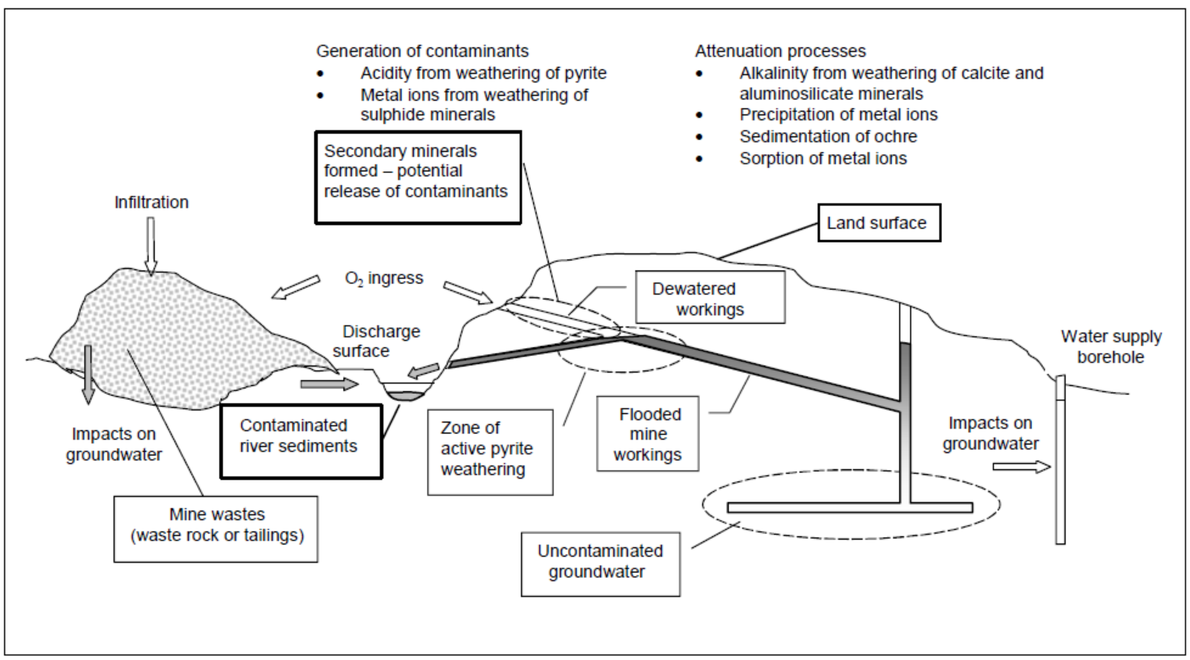

Figure 4.16: Sources and pathways of mine pollution (From Younger et al. 2002 – cited in Environmental Agency, 2008) [80].

Pollutants that are omitted from the above list are suspended solids and residues from ammonium nitrate explosives. These need to be managed in the construction and operational phases. Water in the mine processing circuit and runoff on the mine site is often enriched in nitrogen, particularly nitrates.

Liability for the pollution emanating from abandoned metal and coal mines sits with the government. The taxpayer is the primary source of funds for the government measures to clean up pollution from the mines. Measures are still being implemented as part of programmes under River Basin Management Plans and the UK 25 Year Plan to Improve the Environment [81]. Measures that have been implemented to date include passive water treatment facilities (settlement ponds and wetlands), active water treatment, repair and capping of mine waste facilities and clean-up of metal polluted rivers.

The above discussion highlights the main water pollution risks from mines, particularly coal and metal mines, that can persist for decades or even centuries. Mines can also have shorter term impacts on water quality through alteration of stream beds and flow, through erosion of disturbed land and mine waste facilities, and through releases of runoff from active mine sites.

4.6.10.4. Understanding the geochemistry of exposed rock and mine waste

Geochemical characterisation of rock exposed by mining and of mine waste should be carried out as early as possible in the mine planning process. It is fundamental to the design and operation of mines to prevent water pollution and closure impacts. Unfortunately, many proponents of mining projects still skimp on geochemical testwork during project planning. International standards that apply to the mining industry increasingly emphasise the need for adequate geochemical testwork.

Geochemical studies should predict the acid rock drainage and metal leaching potential (ARDML) and influence the engineering controls such as lining of water-holding facilities and mine-waste facilities and capping of mine-waste facilities at closure. In addition, geochemical studies should inform water and waste management plans and closure plans.

The EU Mine Waste Directive requires characterisation of mine wastes in accordance with the standards listed in Section 4.5.4. In 2016, a review established that the requirements of this directive have been implemented inconsistently across the EU [82,83].

Internationally recognised geochemical characterisation guidelines include the Global Acid Rock Drainage (GARD) Guide (INAP, 2017) and the MEND Report (MEND, 2009). The GARD Guide is recognised as a state-of-the-art summary of the best practices and technology to assist mine operators and regulators to address issues related to sulphide mineral oxidation.

A site-specific geochemical characterisation programme should be implemented. The number of samples required is dependent on the level of study (for example scoping, pre-feasibility and feasibility) and the complexity of the geology surrounding the deposit. The GARD guide recommends that at pre-feasibility level, short term ("static") test work should be carried out on numerous representative samples and longer-term ("kinetic") testing should be undertaken on one or two representative samples of each material type.

The sample selection process for geochemical test work should ensure that there is an appropriate spatial and lithological distribution representative of material that will be generated during operations. Consideration should also be given to any available assay data to ensure that a suitable range of compositions are analysed.

4.6.10.5. Measures for avoiding and mitigating impacts on water quality

The following is a list of basic good practice measures for prevention of water quality impacts:

- Diversion of clean surface water (rain and snowmelt runoff) away from dirty areas on the mine site;
- Interception of runoff from dirty areas of the mine site, including water contaminated by exposure to ore and waste rock, and capture this in dirty-water-holding facilities;

- Installing liners at the sites of ore stockpiles and mine waste facilities to prevent contamination of groundwater, where geochemical characterisation indicates this is necessary;
- Recycling of water as much as possible to minimise fresh-water intakes, using detailed water balances to optimise recycling and help prevent uncontrolled discharges;
- Treatment of water discharged from the site to ensure it conforms with the relevant water quality criteria in the downstream water body, which will depend on standards defined in legislation, catchment water quality objectives defined by the river basin authority and sensitive downstream water uses;
- Capping of mine waste disposal facilities at closure, to prevent contamination of groundwater, where geochemical characterisation indicates this is necessary.

The selection and combination of measures is site-specific, based on robust EIA and risk assessment. Monitoring is used to prove that the technology implemented is adequate and to determine if additional measures are required.

In Northern Ireland, the 2015 MWR and associated standards and guidelines (Section 4.5.4) influence the identification of appropriate measures for water quality management.

4.6.11. Mine Waste Management

Northern Ireland has mine waste regulations based on the EU Mine Waste Directive. The regulations and corresponding EU standards and guidance are robust, if implemented properly. There is a high level of interest in mine waste management internationally and it is important to be aware of the new trends and new standards outlined below.

4.6.11.1. General background

Mines usually produce large quantities of mine waste in the form of waste rock and tailings. The footprint required for the waste storage facilities can be very large. Poor performance of a mine waste facility can have substantial negative impacts on surrounding communities and the environment. In addition, tailings storage facilities can be a major hazard. They pose serious threats to humans and the environment in the event of improper design, handling or management. A failure may result in uncontrolled spills of tailings, dangerous flow-slides or the release of hazardous substances.

The main reported causes of tailings incidents are a lack of control of the water balance, inadequate adherence to design, poor construction control, insufficient understanding of the key features that control safe operations, and governance. This is according to the Australian Leading Practice Sustainable Development Program for the Mining Industry – Tailings Management Guideline [84]. Linked to this dewatering of tailings and dry tailings transport and storage are increasingly being considered as alternatives to conventional slurried tailings transport and storage, if practical and sustainable. In addition, a new suite of new tailings management standards and guides has emerged.

4.6.11.2. Public reporting on management of mine waste

Active investor interest in mine waste management was reflected in the Investor Mining & Tailings Safety Initiative that was established in response to the Brumadinho tailings storage facility (TSF) failure in Brazil in January 2019. The failure occurred at the Córrego do Feijão iron ore mine and caused the death of 270 people and extensive damage to land and rivers. The investor initiative linked up many institutional investors active in extractive industries, co-led by the Church of England Pensions Board and the Council of Ethics of the Swedish National Pension Funds. The investors leveraged disclosure of specific TSF information on mining company websites and called for further action that led to the Global Tailings Review, the subsequent development of the Global Industry Standard on Tailings Management (see below) and establishment of a global public database of over 1,800 TSF at over 750 mine sites. The database is called the Global Tailings Portal [85] and presents information on the capacity of and hazard categorization of the TSFs. It also includes information on whether the design of the facility has been approved by a competent person and whether assessments of extreme weather effects and downstream impacts have been undertaken.

4.6.11.3. Globally recognised standards

A new Global Industry Standard on Tailings Management (GISTM) was published in August 2020. This is a governance standard that adds to, rather than replaces, existing technical tailings management standards. Broad-based uptake of the GISTM across the global mining industry is expected.

The GISTM was developed by international experts using a multi-stakeholder approach. Its development was supported by the International Council on Mining and Metals (ICMM) partnered with the UN Environment Programme (UNEP) and the Principles for Responsible Investment (PRI).

Downstream manufacturers' ESG expectations of mining companies are reflected in responsible sourcing standards, which include the ICMM Principles, the CopperMark, the Aluminium Stewardship Initiative and the Responsible Gold Mining Principles. These standards all require that tailings management lines up with widely supported good practice guidelines.

To facilitate the implementation of the GISTM, ICMM published Conformance Protocols for the GISTM and a Tailings Management Good Practice Guide in May 2021. The two recent documents were presented in a webinar on 6th May 2021 [86].

The Good Practice Guide focuses on good engineering practice, including good governance, that will support continual improvement in the management of tailings facilities and strengthen a 'corporate safety culture'. It sits under the GISTM and is complementary to jurisdictional legal requirements and technical standards, such as the standards of International Commission on Large Dams (ICOLD), Australian National Committee on Large Dams (ANCOLD), the Canadian Dam Association (CDA); and South African National Committee on Large Dams (SANCOLD).

The Good Practice Guide was developed with input from the Mining Association of Canada (MAC), thus promoting alignment of the ICMM guide with MAC tailings guidance documents, and Professor Morgenstern, a world leading expert on tailings management.

The MAC tailings guidance documents are widely recognised, particularly the following:

* A Guide to the Management of Tailings Facilities (Version 3.1, 2019), which is referred to as the “Tailings Guide”;
* Developing an Operation, Maintenance and Surveillance Manual for Tailings and Water Management Facilities, which is referred to as the “OMS Guide”.

The MAC Tailings Guide presents a framework for a tailings management system, based on the ISO 14001 environmental management system framework and promotes a risk-based approach to management throughout the life of the tailings facility to closure. Much emphasis is placed on the need to use risk assessment and transparent decision-making in the selection of the location of the facility and best available techniques. The guide also requires that critical controls to manage high-consequence risks are determined (ideally by means of failure-mode-and-causes analysis).

The Conformance Protocols for the GISTM intend to help operators and independent third parties assess conformance with, and implementation of, the GISTM’s 77 requirements using 219 criteria. The Conformance Protocols note that a multidisciplinary team is needed to provide the experience, skills and knowledge to adequately assess conformance with the GISTM. The audit team in all of the following areas; geography and climate; lifecycle specifically including closure; risk/emergency management; tailings engineering (including geotechnical, hydrotechnical, civil and mining engineering); environmental management; socio-economics and communities; and water.

ICMM members have committed to conformance with the GISTM, specifically that facilities with ‘Extreme’ or ‘Very high’ potential consequences will be in conformance by August 2023 and all other facilities by August 2025.

Mining companies that are not members of ICMM are also likely to align their tailings management practices with this standard for many reasons. The obvious driver is corporate responsibility, other drivers are investor pressure and expectations of downstream manufacturers.

4.6.12. Decarbonisation of Mining

4.6.12.1. General overview

The impact of mining on greenhouse gas emissions is low compared to heavy industries, although it is highly dependent on the location of the mine and the mineral being extracted.

On average, approximately 40% of a mine's Scope 1 (direct) and Scope 2 (indirect) emissions are a result of haulage, the combustion of diesel in haul trucks to raise minerals to the surface [87]. A further 40% on average results from comminution, the crushing and milling of the rock into smaller particles for further refinement (Figure 4.17).

Crushers and mills are typically driven by electricity, so if a mine is connected to the local or national grid, these will likely result in some Scope 2 emissions from fossil fuel combustion for power generation. In some jurisdictions such as Norway, the proportion of renewables within the national grid is high, for Norway in 2007 this proportion was 98% [88].

In more remote areas the operator may have to use onsite power generation. The fuel used in these onsite power facilities also depends on the mine location. In regions such as Kazakhstan, coal is the primary source as there is significant domestic access to coal. In remote areas of Africa, it is frequently diesel, but solar panels are increasingly being used.

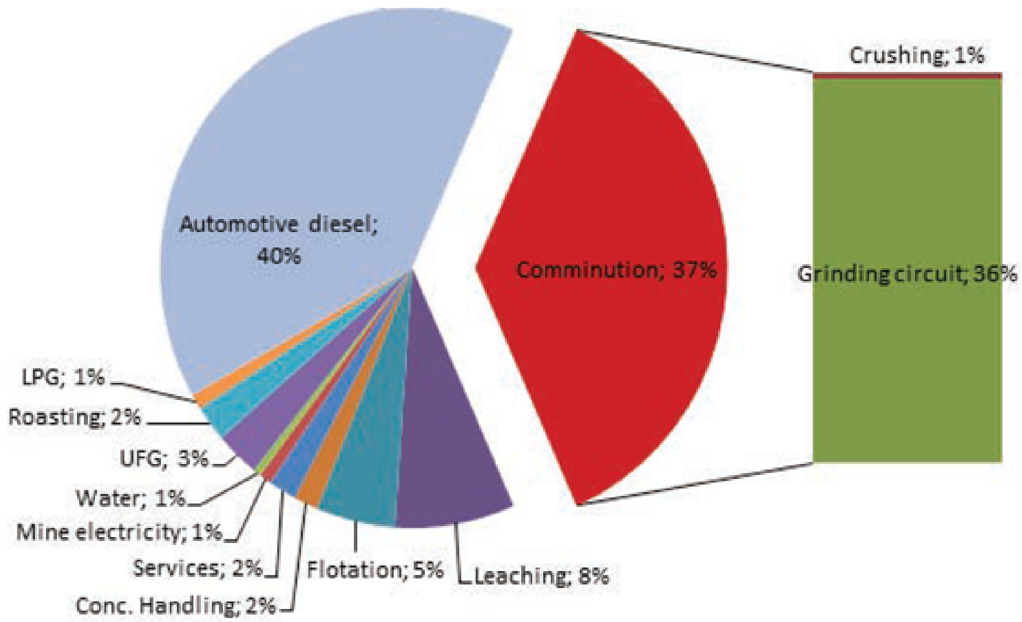

Figure 4.17 - Break-down of sources of GHG emissions from Kalgoorlie Mine, Australia [87].

Within Northern Ireland the grid mix contains a high proportion of renewable electricity, over 58% according to SONI, the System Operator for Northern Ireland [89]. A mine operator connecting to this grid to power comminution could therefore claim half the Scope 2 emissions of a similar mine in a fossil fuel dependent jurisdiction. Through the further purchase of Renewables Obligation Certificates (ROCs) from the energy provider could result in the Operator being able to claim negligible, or even zero Scope 2 emissions.

Electrification of mining equipment in a jurisdiction like the UK would also therefore be of benefit. Through careful planning, conveyors can be used for haulage rather than trucks, continuous remote-controlled mining machines can be used instead of explosives with the right rock conditions and high recovery hydrometallurgical processes can reduce the need for carbon intensive pyrometallurgy.

However, a new mining project, particularly on greenfield sites will likely involve the removal of vegetation disturbing carbon stores such as peatland or forest. Subsequent destruction of vegetation through burning or decomposition releases the carbon dioxide back into the atmosphere. These emissions should be considered as Scope 1 emissions from the project during construction. Open pit mining inevitably has a greater impact in this regard, not only from the footprint of the pit itself, but the areas of land on which the overburden and tailings must be stored.

There is no “typical” greenhouse gas profile for a mine. The method of extraction, processing, refinement and shipping are all very specific to the mine setting and the commodity being exploited. However, globally it can be said that approximately 40% of Scope 1 (direct) and Scope 2 (indirect) emissions result from mineral haulage, and a further 40% result from comminution - crushing and grinding of ore to improve the recovery from tailings. Should ores require roasting, sintering or smelting to further refine them, significant additional GHGs can result from fossil fuel combustion, typically coal as a reductant in steelmaking or natural gas for furnace heating.

The most effective approach for decarbonisation in mining is to design out emissions before construction. The avoidance of emissions through design meets the highest tier on the GHG Mitigation Hierarchy (Figure 4.18). The hierarchy ranks mechanisms for GHG minimisation as follows:

- Avoid – operations are designed to not emit GHGs;
- Eliminate – GHG emitting equipment or processes are replaced with non-emitting processes;
- Reduce – energy efficiency or process optimisation measures are implemented to reduce emissions often without the significant capital investment of retrofitting; and
- Offset – ultimately residual emissions can be reduced by purchasing carbon credits from e.g. tree planting or peat bog restoration schemes.

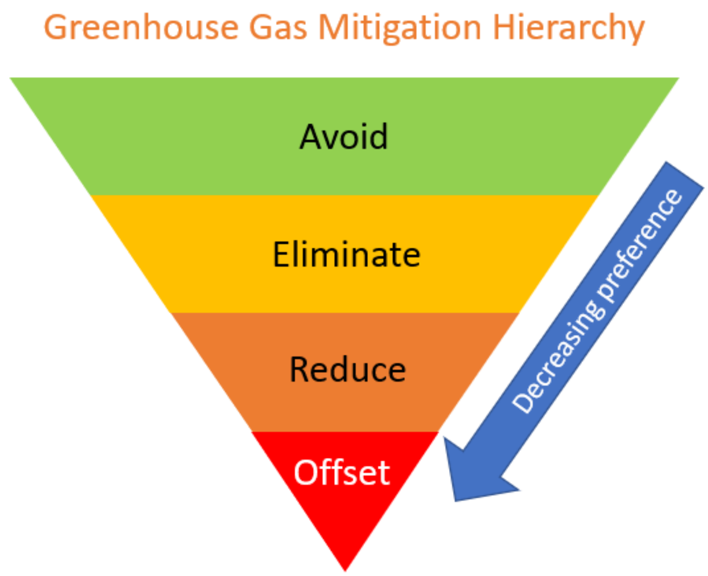

Figure 4.18 - The GHG mitigation hierarchy

The most common approach being taken by mining operators worldwide to decarbonisation is to electrify operations, and then to attempt to increase the proportion of electricity from renewable sources.

4.6.12.2. Mining and materials handling

Across the sector, the haulage of material around and out of a pit or out of an underground mine can be said to be the largest source of direct emissions at an operation. This statement may be challenged at sites that are large enough to also have pyrometallurgy onsite.

The mining of hard rock ores itself commonly means the drilling of holes for charges and then the blasting of sections of rock. This rock is then collected by shovels and dozers. Drill rigs are commonly powered by diesel, as are shovels and dozers, although equipment manufacturers such as Sandvik have developed electricity and battery versions of these equipment [90]. Once mined, bulk ore sorting in the mine or pit can improve the efficiency of downstream beneficiation compared to conventional “blast and haul”.

Once the rock can be moved, there are two primary solutions for its haulage, either by truck or by conveyor. Conveyors are commonly powered by electricity but only recently have entirely electric powered trucks been trialled, although pantograph assistance for hybrid haulage vehicles has been used for many years.

A key constraint in the use of electric haul trucks is the weight of the battery packs. Due to their density, batteries considerably increase the weight of the unladen truck, meaning once at the bottom of a mine and laden, comparatively more energy than a diesel truck is required to get the vehicle back out. One solution currently being trialled by AngloAmerican is a hydrogen powered haul truck [91]. Hydrogen when combusted does not produce carbon dioxide, and its production from cracking water can be decarbonised by using renewable electricity.

4.6.12.3. Comminution

Crushing, grinding and milling ore into the small particles required to liberate target minerals is an energy intensive process. Although globally approximately 40% of mining related GHG emissions can be attributed to comminution, the energy used for the process is electricity, hence GHG emissions are produced from combustion for energy generation. As a result the sourcing of renewable electricity can decarbonise much of this process, although ball mills and semi-autogenous mills use steel balls within the rotating drum that wear down over time requiring replacement. In this respect Scope 3 emissions from the production of the balls may be material unless they are also produced using low carbon technology such as electric arc steelmaking.

Operators can also meet the “reduction” stage of the GHG hierarchy through process optimisation, variable speed drives and efficient pumps and technologies such as high-pressure grinding rollers.

4.6.12.4. Hydrometallurgy

Hydrometallurgical processes refer to mineral concentration processes involving water. By suspending fine particles of ore in water various techniques can be used to refine them. For example rolling magnets through a water column extracts magnetic particles, gravity separation such as cyclones can be used to differentiate minerals based on their mass, and froth flotation which is the separation of minerals from unwanted fractions by taking advantage of differences in hydrophobicity. Reagents are often used to enhance or selectively impart hydrophobic properties on a valuable mineral that can then either be floated to the surface of the solution or extracted from the bottom and scraped from the unwanted gangue material.

Hydrometallurgy produces both concentrated minerals and tailings, the gangue material often still suspended in water. Historically these tailings were then pumped for deposition behind dams or dykes. Over time the water evaporates or drains out through the lined bottom of the dam, leaving dry tails behind, although increasingly tails are being dewatered prior to storage using filter presses for example. Dry-stack tailings facilities can often be safer than wet tailings facilities although the additional process stages increase energy use, and therefore can increase GHG emissions if the energy is not sourced from renewable technologies.

4.6.12.5. Pyrometallurgy

Metals have historically widely been extracted from gangue through heating, and often melting the ore, and skimming off gangue and extracting molten metal from beneath. Pyrometallurgical processes include sintering, roasting, pelletising and smelting, the most common example of which is the reduction of iron ore to pig iron in a blast furnace, before refinement into steel in a Basic Oxygen Steelmaking (BOS) plant.

Pyrometallurgy is used for the refinement of metals such as gold, silver, iron, zinc, copper and Aluminium and other base metals. Roasting, sintering and pelletising can be said to be treatments methodologies whereby an ore is heated and often combined with other ingredients to give the product favourable chemistry for the next stage of refinement. In terms of GHGs these pyrometallurgical processes involve heat produced through combustion of a fuel such as natural gas.

Smelting is often a subsequent stage involving the melting of the feedstock to separate out the valuable mineral, mostly metals, and gangue material. Although some smelting reactions such as the extraction of iron from iron ore involve exothermic reactions, energy in the form of heat or electricity always needs to be added. Typically heat is provided by natural gas combustion. Electrolysis based smelting such as the extraction of aluminium requires electricity, as does Electric Arc Furnace steelmaking, whereby an electrical arc is used to melt scrap steel. Again, as long as the electricity in the process can be obtained from renewable sources there is potential for decarbonisation, although the large power requirements pose a challenge.

Steel production from iron ore is a particularly carbon intensive process, both because of the quantities in which steel must be produced in order to be economical, as well as due to the chemistry of the process. The decarbonisation of steelmaking would involve replacing the reductant in the process, currently metallurgical coal, refined to coke. Although metallurgical coal and iron ore have been identified in other parts of the UK (and indeed have been exploited for several hundred years) none has been identified in Northern Ireland.

Although the smelting of aluminium is via electrolysis, the reaction requires the use of carbon electrodes that oxidise in the process resulting in carbon dioxide generation. Aluminium smelting is also very energy intensive and, as a result, smelters are often associated with reliable sources of power such as coal fired power plants. However, the aluminium producer Alcoa has constructed the Fjarðaál smelter in Reyðarfjörður in Iceland, that uses reliable and renewable geothermal power.

Although the environmental impact of exploration activities is localised, the social impacts can be widespread if not managed, potentially leading to the loss of a social license to operate.

4.6.13. ESG in Exploration

The principal reference for good environment and social practise in exploration is the Prospectors and Developers Association of Canada (PDAC) "e3 plus" framework for responsible exploration [92]. The principles outlined in e3 plus are not standards or requirements, but rather suggestions for exploration companies to improve their corporate social responsibility in three key areas, social responsibility, environmental stewardship and health and safety. These three areas have internet-based toolkits to aid in implementation [93].

PDAC's eight principles for corporate social responsibility include:

1. Adopt responsible governance and management;
2. Apply ethical business practises;
3. Respect human rights;
4. Commit to project due diligence and risk assessment;
5. Engage host communities and other affected and interested parties;
6. Contribute to community development and social wellbeing;
7. Protect the environment; and
8. Safeguard the health and safety of workers and the local population.

PDAC has further tools and guidance for exploration companies in the field of climate change, diversity and inclusion, human rights, engagement with indigenous communities, and disclosure. PDAC's work on disclosure contributed to the adoption by the Government of Canada of the 2015 Extractive Sector Transparency Measures Act (ESTMA) [94], an Act that implements public disclosure requirements for extractives entities active in Canada relating to certain types of payments made to governments in Canada and abroad.

Environment and social barriers to exploration have also been researched in Europe, through the EU Innovative, Non-invasive and Fully Acceptable Exploration Technologies (INFACT) project, funded by the European Union Horizon 2020 programme. The project aims to engage society, to invigorate exploration, and to impact innovation, knowledge and growth in the minerals exploration sector [95].

The objectives of the INFACT Project include building trust in the raw materials cycle, to raise awareness of environmentally friendly, safe and effective exploration techniques as well as establishing a view of best practise for exploration. The INFACT Project has yet to produce documentation outlining best practice as PDAC has, although it has identified several key barriers to exploration in Europe (Williams and MacCallum 2018). The barriers identified in the research included:

- Concern over compromising existing land use;
- Cost of exploration activities;
- A lack of public awareness and prevailing negative attitudes towards the extraction industry in general;
- Negative environmental impacts; and
- Inconsistent or varied regulations.

Project Conclusions

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